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Message From Editor-in-Chief



Dear readers,

I hope this message finds you well. It is my great pleasure, as Editor-in-Chief of the International Journal of Oral Health (IJOH), to welcome you to this issue. I would like to extend my deepest gratitude to our authors, reviewers, editors, and dedicated support team whose expertise, commitment, and passion have made this publication possible.

The strength of IJOH lies in its mission to serve as a trusted platform for the exchange of knowledge in dentistry—where research innovation meets clinical relevance. Each contribution represents not only scientific advancement, but also a shared vision to improve oral health and reduce disparities across communities.

This issue is published in conjunction with the Asian Academy of Preventive Dentistry (AAPD) Conference 2026 in Kota Kinabalu, Malaysia, a gathering that brings together leading experts, young researchers, and practitioners across Asia and beyond. In these pages, you will find a diverse selection of topics ranging from child oral health, epidemiology, and preventive strategies to advancements in innovative dental materials. We believe these articles will stimulate reflection, inspire dialogue, and enrich clinical and public health practice worldwide.

As IJOH continues to grow, I am confident that AAPD will play an even greater role in shaping the future of preventive dentistry and making tangible differences in oral health. Thank you for your continued trust and support. I wish you an insightful reading.

Sincerely,

Professor Dr Rahimah Abdul Kadir

Editor-in-Chief

International Journal of Oral Health

**Proceedings of the
18th International
Conference of the Asian
Academy of Preventive
Dentistry**

**30th September 2026
to
2nd October 2026**

Sabah, Malaysia

Empowering Communities Through Preventive Dentistry

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Keynote & Invited Lecture Abstracts

Oral Health for Older People Workshop

Title: Optimal Oral Health for Older People — A Common Protocol for Oral Health Survey of the Aged Population in Asia

Speaker: Prof. Dr Hiroshi Ogawa

Affiliation: Niigata University, Japan

Abstract

As Asia's population ages rapidly, understanding and safeguarding oral function has become central to achieving healthy ageing. Drawing on Japan's extensive expertise and decades of experience in geriatric oral health, this workshop brings together leading Japanese specialists to define optimal oral health for older people and to build regional capacity for assessing oral functional status in ageing populations. The session opens with an overview of what constitutes optimal oral health for healthy ageing, before turning to practical approaches for the assessment and management of oral function, including chewing, swallowing, saliva and other functional parameters relevant to older adults. Findings from a regional questionnaire survey of oral functional status across Asia will be presented, offering a comparative picture of current practice and gaps across countries. A key highlight of the workshop is the introduction of a Common Protocol for Oral Health Survey for the Aged Population, developed by the Japanese facilitator team to standardise data collection and enable meaningful comparison of oral function across different Asian settings. Participants will take part in a hands-on demonstration of oral functional assessment techniques, followed by a Japanese industry perspective on ongoing initiatives supporting oral health care for older populations. The workshop then turns to application, guiding participants through the design of data collection using the common protocol for both research and clinical work. The session concludes with a forward-looking discussion on future collaboration, including how participating countries can share outcomes and sustain regional cooperation beyond the workshop. Combining lectures, survey findings, live demonstration and structured discussion, this workshop draws on Japan's pioneering experience to equip dental and healthcare professionals, researchers and policymakers with practical tools and a shared framework to advance the assessment and promotion of oral health among older populations across Asia.

Oral Health Literacy Workshop

Title: Leveraging Oral Health Literacy to Advance Oral Health Promotion and Disease Prevention

Speaker: Assoc. Prof. Wong Mun Loke

Affiliation: National University of Singapore

Abstract

Oral diseases continue to affect populations worldwide, with dental caries and periodontal disease remaining the two most common oral conditions encountered across the lifespan. Both diseases have well-established aetiologies and pathogenesis that render them highly amenable to early detection and prevention. Health education has conventionally been used to impart information and encourage the adoption of preventive practices to help individuals and populations avoid the onset of these diseases or halt their progression. Beyond health education alone, however, there is a pressing need to view oral health promotion and disease prevention through the broader lens of health literacy, which extends beyond the simple provision of preventive advice to encompass how individuals access, understand, appraise and apply health information in their daily lives. In an age where information is more readily available to the public than ever before, yet not always accurate or easy to navigate, there is an even stronger imperative to build the oral health literacy of our populations as part of a comprehensive strategy to prevent oral disease and reduce inequalities in oral health outcomes. Through this workshop, participants will review the barriers to, and drivers of, prevention at both the population and individual level; develop a deeper understanding of oral health literacy, its scope and its remit; understand the significance of building oral health literacy within populations; identify practical ways to enhance the oral health literacy of diverse populations; and discuss approaches to effectively measure oral health literacy in research and practice. The workshop comprises short presentations as well as interactive group discussions, equipping participants with practical strategies to advance oral health promotion in their own settings.

Smoking & Vaping Cessation Workshop

Title: Smoking Cessation and Vaping

Facilitator(s):

- Prof. Chris Bullen, University of Auckland, New Zealand
- Dr Amer Siddiq Amer Nordin, Universiti Malaya Centre of Addiction Sciences , Malaysia

Abstract

Tobacco use remains one of the leading preventable causes of death worldwide, claiming more than 8 million lives annually, including approximately 1.3 million deaths from second-hand smoke exposure. While cigarette smoking continues to pose a substantial public health burden, the rapid rise of electronic nicotine delivery systems (ENDS), particularly among adolescents and young adults, has introduced new challenges for clinicians, healthcare systems and tobacco control efforts. Increasing dual use of cigarettes and vaping products further complicates nicotine dependence, cessation strategies and clinical decision-making. This interactive workshop provides participants with practical, evidence-based approaches to managing tobacco and nicotine dependence in everyday clinical practice. Drawing upon contemporary international guidelines and implementation experience in Malaysia, participants will develop core competencies in assessing nicotine dependence, applying behavioural counselling techniques, utilising brief interventions and motivational interviewing, selecting appropriate pharmacotherapy, and managing relapse. The workshop will also examine current evidence surrounding vaping, including its role in harm reduction, vaping cessation, management of dual users, common misconceptions, and emerging clinical recommendations. Designed for frontline healthcare professionals, the workshop combines concise lectures, case discussions, demonstrations and small-group practical exercises to maximise skills acquisition and confidence. By the end of the programme, participants will be equipped with practical tools that can be immediately incorporated into routine clinical practice to improve smoking cessation outcomes while addressing the evolving challenges posed by vaping and emerging nicotine products.

School-Based NaF Varnish Workshop

Title: Use of NaF Varnish in School-Based Programs in Asia

Convener: Dr Ivy Guofang Sun, China Hong Kong SAR

International Expert Panel:

- Prof. Dr Shuguo Zheng, China
- Prof. Dr Chun-Hung Chu, China Hong Kong SAR
- Prof. Dr Michelle Sunico Segarra, Philippines
- Prof. Dr Waranuch Pitiphat, Thailand
- Prof. Dr Rahimah Abdul Kadir, Malaysia
- Prof. Dr Tammy Duangthip, USA
- Associate Prof. Dr Hoang Trong Hung, Vietnam
- Prof. Dr Youn-Hee Choi, Korea
- Prof. Dr Diah Ayu Maharani, Indonesia
- Prof. Dr Hiroshi Ogawa, Japan
- Prof. Dr Karen Glazer De Anselmo Peres, Singapore

Abstract

Dental caries remains one of the most prevalent non-communicable diseases affecting children across Asia, with school-based fluoride varnish (FV) programmes offering a proven, cost-effective strategy for prevention at population scale. Despite strong evidence supporting FV application, considerable variation exists across the region in caries prevalence, programme design, workforce capacity and implementation models, making regional consensus and shared learning essential to guide policy and practice going forward. This closed workshop brings together invited experts from across Asia to review the latest national and regional dental caries prevalence data for five- and twelve-year-old children, alongside existing school-based oral health programmes and service delivery models in their respective countries. Building on this foundation, participants will critically review and refine a systematic review of fluoride varnish application in school-based programmes, contributing their expertise, comments and corrections to strengthen it as a core evidence base for the region. Drawing on this shared evidence and regional data, the workshop aims to develop consensus recommendations for the implementation of school-based fluoride varnish programmes across Asia, addressing variations in population needs, health system capacity and available resources between countries. The session will also lay the groundwork for at least two post-workshop academic publications, translating the workshop's discussions and consensus outcomes into resources that can inform policy and practice across the region. Structured around focused document review and discussion sessions, the workshop is designed to harness the collective expertise of its invited international panel to strengthen the evidence base and regional coordination for school-based caries prevention among children throughout Asia.

Plenary 1

Title: From Bench to Bedside: Innovative Approaches to Radiation-Induced Oral Mucositis and Taste Dysfunction

Speaker: Prof. Dr Xin Xu

Affiliation: West China School of Stomatology, Sichuan University, China

Abstract

Radiotherapy (RT) is a cornerstone treatment for head and neck cancer (HNC), but acute oral complications such as oral mucositis (OM), xerostomia, taste dysfunction, and infections remain common. OM is the most frequent acute toxicity, with 50%–70% of patients developing severe OM (WHO grade 3–4), leading to pain, impaired eating, swallowing, and speaking, reduced treatment tolerance, decreased quality of life (QoL), and treatment interruptions. Patients also frequently experience hypogeusia, ageusia, and dysgeusia, which further increase the risk of malnutrition and psychological distress. Although MASCC/ISOO has issued recommendations for radiation-induced OM, their efficacy and safety require further validation. Our group has long focused on the shared mechanisms and clinical management of radiation-induced oral mucositis (RIOM) and taste dysfunction. We identified a critical role of oral microbial dysbiosis in RIOM and showed in a clinical trial that oral probiotics, particularly *Streptococcus salivarius* K12, can delay onset and shorten the duration of severe OM in HNC patients receiving RT, likely via modulation of the oral microbiota. We also demonstrated that Wnt/ β -catenin signaling is essential for the repair and regeneration of oral mucosa and taste buds, and that LiCl can promote recovery from RIOM and dysgeusia; these findings were highlighted by JDR on the IADR official website. To advance clinical translation, we have initiated two registered clinical trials evaluating the efficacy and safety of lithium-containing mouthrinse and hydrogel for the prevention and treatment of RIOM and taste dysfunction. Collectively, our work proposes promising, mechanism-based strategies for improving the management of radiation-induced oral complications, particularly OM and taste dysfunction, with the goal of enhancing QoL and optimizing oncologic outcomes in HNC patients.

Keynote

Title: Oral Health for All: From Primary Care to Universal Coverage in China and Asia

Speaker: Prof. Dr. Shuguo Zheng

Affiliation: Peking University School of Stomatology, China

Abstract

Globally, oral diseases affect 3.5 billion people, with dental caries remaining the most prevalent non-communicable disease (NCD). The WHO Western Pacific Region bears a disproportionate burden with high caries prevalence in young population and concentrated in underserved regions. Recognizing that "No health without oral health," WHO 2023–2030 Global Oral Health Strategy and Action Plan strongly recommend the integration of essential oral care into primary health care (PHC) to increase universal health coverage, emphasizing task-shifting to train more community dental work force and prioritizing cost-effective interventions. China has responded these with a series of policy framework, mandating oral departments in township health centers, standardizing basic oral disease codes, and incorporating children's oral services into family doctor contracts. However, disparities persist: the national dentist-to-population ratio masks severe shortages at the grassroots—western rural areas report ratios. Early treatment-seeking and service utilization in communities remains low. For Asia-Pacific nations, China's experience underscores three imperatives: embedding oral health within PHC infrastructure rather than siloed specialist care; empowering nurses and community health workers through competency-based training; and addressing sugar as a shared risk factor across NCDs. Critically, strengthening China's grassroots oral health system would effectively cover 60–80% of the WHO regional disease burden, given China's demographic weight in the Western Pacific. China bears profound responsibility and seeks mutual learning with regional partners to advance oral health equity through collaborative PHC innovation.

Symposium 1

Title: The Smoking-Vaping Epidemic: Trends, Dual Use and the New Public Health Risks

Speaker: Assoc. Prof. Dr Amer Siddiq Amer Nordin

Affiliation: Universiti Malaya, Malaysia

Abstract

The introduction and presence of electronic nicotine delivery devices has transformed the tobacco epidemic, creating a complex landscape characterised by increasing vaping, persistent combustible tobacco use, and the emergence of dual-use behaviours. While cigarette smoking has declined in many populations, nicotine dependence remains prevalent as youth uptake of vaping, product diversification, and aggressive marketing continue to reshape patterns of use. Evidence from Malaysia and other low- and middle-income countries demonstrates that dual use is increasingly common, challenging assumptions that vaping is simply replacing smoking. Dual use is associated with sustained nicotine addiction, delayed smoking cessation, and continued exposure to toxicants from combustible tobacco. Among adolescents, early exposure to nicotine through vaping raises concerns regarding addiction, progression to combustible tobacco use, and long-term neurodevelopmental effects. At the population level, the coexistence of cigarettes, vaping products and emerging nicotine products complicates surveillance, regulation, clinical management and public health messaging. This presentation synthesises current epidemiological trends, implementation research, and tobacco control evidence to examine the evolving smoking–vaping epidemic, with particular emphasis on Malaysia and the broader regional context. It discusses the implications of dual use for healthcare systems, highlights gaps in prevention and cessation services, and explores evidence-based strategies integrating behavioural interventions, pharmacotherapy, regulatory approaches and health system strengthening. As the nicotine landscape continues to evolve, effective tobacco control must move beyond product-specific policies towards comprehensive nicotine dependence management that prioritises prevention, early intervention, cessation support and implementation of evidence-based public health strategies. This integrated approach is essential to reducing the future burden of nicotine addiction and tobacco-related disease.

Symposium 1

Title: Vapor Trails and Policy Walls: Dual use and Regulatory Gaps

Speaker: Prof. Dr. Chris Bullen

Affiliation: The University of Auckland, New Zealand

Abstract

The rapid growth in e-cigarette use (vaping) in Malaysia and most other countries around the world over the past decade is an issue of public health concern. While smoking remains the greater public health challenge, vaping, often in conjunction with smoking (“dual use”) and the health effects that arise, should not be downplayed. The impacts from vaping on individual and population health are what I refer to as ‘vapor trails’; whereas ‘policy walls’ are the laws, regulations, enforcement and clinical care required to tackle the problem at multiple levels. In this presentation I will outline the epidemiological and clinical evidence of harm from vaping and especially dual use, and provide evidence in support of my contention that it is the misalignment between the vapor trails and policy walls that has led to the many gaps, asymmetries, incoherence, and barriers to support quitting. Oral health is often regarded as the ‘canary in the coalmine’ of emerging health issues, and vaping is no exception. Oral health professionals therefore have a vital role to play, not only in offering the best clinical care possible but also in bearing witness to the health impacts of vaping and smoking (the ‘vapor trails’) and advocating for the erection of effective ‘policy walls’ that will prevent harm and ensure evidence-based support is available for people to quit both smoking and vaping.

Symposium 1

Title: A Decade of the KOTAK Program: Strengthening Tobacco Control through School Dental Services in Malaysia (2016–2026)

Speaker: Dr. Muhammad Zulkefli Ramlay

Affiliation: Oral Health Programme, Ministry of Health, Malaysia

Abstract

Smoking remains a significant public health challenge, with initiation often occurring during adolescence. In alignment with the WHO Framework Convention on Tobacco Control (FCTC) and the National Strategic Plan for Tobacco Control 2015–2020, the Ministry of Health Malaysia launched a smoking prevention, screening, and intervention program for schoolchildren, namely the KOTAK Program (Kesihatan Oral Tanpa Amalan Merokok), in 2016. This initiative integrates tobacco screening and intervention into the existing School Dental Service, targeting students from Primary 1 to Secondary 5 (spanning an 11-year schooling period). Over the past decade, the program has evolved from a 28-school pilot to a nationwide implementation involving collaboration between the Ministry of Health and the Ministry of Education. The KOTAK framework encompasses three core components: integrated health education, smoking screening (via clinical oral signs or student self-disclosure), and behavioral interventions. As of 2025, the program has achieved a massive reach, screening over 4.7 million students across 7,875 primary schools and 2,520 secondary schools. While smoking prevalence remains low in primary schools (<0.1%), screenings in secondary schools consistently identify current smokers at rates of 1.92%–4.07%. Intervention outcomes indicate variable success, with quit rates reaching up to 10.5% in secondary school cohorts six months post-intervention. Despite its success, the program faces challenges such as intervention time constraints and the rising threat of e-cigarettes. Moving forward, the program aims to re-orient guidelines to include vaping, explore rapid cotinine testing, and strengthen collaboration with school counselors. A decade of KOTAK demonstrates that dental practitioners are vital frontline facilitators in the mission to achieve a tobacco-free generation by 2045.

Symposium 2

Title: Strengthening Public Health Impact through Education and Training: The SEAAD Perspective

Speaker: Prof. Dr. Waranuch Pitiphat

Affiliation: Khon Kaen University, Thailand – President, SEAAD

Abstract

The global oral health agenda is undergoing a major shift, guided by the WHO Global Oral Health Action Plan (2023–2030) and reaffirmed by the 2024 Bangkok Declaration. Translating these policy mandates into tangible population health outcomes requires a socially accountable workforce embedded within primary healthcare. As the regional authority on dental education, the South East Asia Association for Dental Education (SEAAD) plays a pivotal role in aligning regional training with these public health needed. This presentation outlines how SEAAD has revised regional core competencies for graduating dentists, providing a practical framework that member institutions can use for curriculum reform. Key strategies focus on transitioning from restorative-heavy models to preventive-centric competencies, integrating the Common Risk Factor Approach, and fostering interprofessional collaboration across the dental team. Shifting clinical training into primary care environments ensures students gain direct experience delivering equitable, community-focused care to underserved populations. Ultimately, strengthening public health impact begins in the university setting. By uniting global partnerships with international bodies like ADEA and ADEE with regional competency harmonization, SEAAD provides a framework to ensure that tomorrow's dental graduates are equipped not only as skilled clinicians, but as active public health leaders driving sustainable, preventive oral health across Asia.

Symposium 2

Title: Community-Based Dental Education Lessons from the United States

Speaker: Assoc. Prof. Dr. Tammy Duangporn Duangthip

Affiliation: The Ohio State University, USA

Abstract

Community-based dental education (CBDE) has become an important approach in dental training in the United States, aiming to prepare future dentists to address population oral health needs and improve access to care. By integrating education with real-world service settings, CBDE exposes students to diverse communities and strengthens their understanding of social determinants of oral health. This symposium presentation will discuss key lessons learned from CBDE models implemented in U.S. dental schools. The presentation will highlight how partnerships between academic institutions and community stakeholders create meaningful learning environments while expanding preventive oral health services for underserved populations. Examples from community-based dental programs will illustrate how CBDE enhances clinical competence, professional responsibility, and community engagement. In addition, the talk will address practical considerations for implementing CBDE, including curriculum integration, logistics, and sustainable community partnerships. Challenges such as resource limitations, and alignment with accreditation expectations will be explored. The session will also discuss opportunities and future research directions to strengthen CBDE globally as a pathway toward socially accountable dental education and improved oral health equity.

Symposium 2

Title: Building Public Health Leaders through DPH Specialist Training: The Malaysian Experience

Speaker: Prof. Dr. Tuti Ningseh Mohd Dom

Affiliation: Universiti Kebangsaan Malaysia

Abstract

Dental Public Health (DPH) specialist training should contribute not only to specialist qualification, but also to the development of leaders who can strengthen oral health systems. In Malaysia, DPH specialist training follows a 1+3 pathway comprising a one-year Master in Dental Public Health followed by a three-year Doctorate in Dental Public Health within the nationally recognised postgraduate dental specialty framework. This presentation reflects on the Malaysian experience and considers how DPH specialist training can be positioned as a pathway for leadership development in policy, services, research, and education. The Master phase provides grounding in public health principles, epidemiology, and research methods, while the doctoral phase strengthens more advanced competencies in programme planning, evaluation, scholarship, and service leadership. The Malaysian experience suggests that the public health value of DPH specialist training lies not only in producing specialists, but also in preparing professionals who can respond to priorities such as inequity, population ageing, and stronger health system integration.

Symposium 3

Title: Advancing Oral Health among Adolescents: Impact of the MySmile® App in Community School Settings

Speaker: Prof. Dr. Zamros Yuzadi Mohd Yusof

Affiliation: Universiti Malaya, Malaysia

Abstract

Children's oral health remains a fundamental aspect of overall well-being, affecting not only physical health but also psychosocial health, learning, quality of life, and future success. Over the past twenty years, evidence-based research has provided valuable insights into how early prevention, behavioural modification, and community engagement can help reduce caries and gum disease, and improve children's oral health. In this lecture, three evidence-based interventions for preschool, primary school, and adolescent populations will be briefly mentioned, followed by detail presentation of the development and evaluation of The MySmile® app for secondary schoolchildren. The app was developed using a user-centred approach and co-designed with input from adolescents to complement the School Dental Service (SDS). We evaluated its effectiveness through a cluster-randomised controlled trial involving secondary schoolchildren in Perak, with follow-ups at 4 and 12 weeks. The findings indicate that co-designed digital behavioural interventions can improve adolescents' preventive practices, oral hygiene, gum health, and caries-prevention efforts at the population level. Policy implications for scaling digital health within national programmes will be highlighted.

Symposium 3

Title: From Research to Routine: The Evolution of Outreach Dental Care for Caries Prevention in Hong Kong Preschool Children and the Transition to a Government-Integrated Programme

Speaker: Dr. Flora Chan Wai Ling

Affiliation: Society of Preventive Dentistry of Hong Kong

Abstract

According to the territory-wide Oral Health Survey conducted in Hong Kong in 2021, the prevalence of dental caries among 5-year-old children was 41.6%, representing a marked improvement compared with the surveys conducted in 2011 and 2001, in which prevalences of 55% and 60% were recorded, respectively. This improvement is in large part attributable to oral health policies implemented by the Hong Kong Government over the past decades, including water fluoridation, oral health education in kindergartens, and the provision of the School Dental Care Service. In addition to these measures, the development of outreach programmes utilising Silver Diamine Fluoride (SDF) for the arrest of Early Childhood Caries (ECC) has made a significant contribution to the reduction of dental caries burden among preschool children in Hong Kong. Beginning as a small-scale university research project funded by government grants in 2008, the outreach service progressively expanded to 50 kindergartens (2010–2013) and subsequently to 100 kindergartens (2013–2016), sustained through competitive research funding and industry support. The initiative became territory-wide in 2019 when the Hong Kong Jockey Club Charities Trust provided funding for the establishment of the Jockey Club Children Oral Health Project (JCCOHP), extending the service to over 900 kindergartens across Hong Kong. Beyond clinical screening and intervention for the prevention and management of dental caries in an outreach setting, the programme incorporates a robust oral health education component aimed at achieving long-term reductions in caries prevalence among young children. The effectiveness and efficiency of this community oral health care model in managing dental caries among preschool children has been well demonstrated through the JCCOHP. Accordingly, upon the completion of the programme at the end of the 2025–2026 school year, the Department of Health launched the Outreach Oral Health Programme for Pre-school Children (OOHP) in September 2026, adopting the same service delivery model as the JCCOHP. This marks the transition of outreach dental services for preschool children — complemented by oral health education — into a recurrent public health measure provided by the Government. The programme is overseen by the Department of Health and delivered by a panel of recognised non-governmental organisations, each operating at least one dental clinic in Hong Kong. The presenter has participated in this journey as an outreach dentist, a researcher on the effectiveness of SDF in managing ECC, the CEO of an OOHP-participating NGO, and the President of the Society of Preventive Dentistry of Hong Kong. The Hong Kong experience demonstrates how rigorous clinical evidence and philanthropic investment can catalyse lasting policy change, embedding preventive preschool dental care into the public health system and establishing a model for sustainable community-based caries prevention. Finally, the next steps, emerging challenges, and new opportunities for the prevention of oral diseases in children and adolescents in Hong Kong — including those with special needs — will be discussed.

Symposium 3

Title: The Next Challenge in Caries Prevention: From Evidence to Implementation Operationalizing School-Based Caries Prevention in Indonesia

Speaker: Dr. Irene Adyatmaka

Affiliation: Christian Maranatha University, Indonesia

Abstract

Over the past decades, substantial progress has been made in identifying effective strategies for preventing dental caries in children and adolescents. Many countries have translated this evidence into national policies and school oral health programmes. However, successful implementation at scale remains a major challenge. Using Indonesia as a case study, this presentation explores the transition from evidence and policy to routine implementation of school-based caries prevention. Indonesia has established national policies supporting preventive oral health in schools, yet translating policy into consistent daily practice across diverse settings requires more than clinical evidence alone. The presentation will discuss key implementation challenges, including workforce readiness, stakeholder engagement, programme standardization, monitoring, and sustainability. It will also share practical lessons on how preventive interventions can be integrated into existing school health systems while remaining adaptable to local contexts. The future of caries prevention depends not only on discovering effective interventions but also on implementing them consistently, sustainably, and at scale. Indonesia's experience highlights the importance of implementation science as the next frontier in preventive dentistry and offers practical insights that may be applicable to other countries in the Asia-Pacific region.

Symposium 4

Title: How AI is Shaping the Future of Primary Dental Disease Prevention

Speaker: Prof. Dr. Ollie Yiru Yu

Affiliation: The University of Hong Kong

Abstract

The high prevalence of preventable oral diseases highlights the limitations of traditional, treatment-oriented dental care. Artificial intelligence (AI) offers an opportunity for paradigm shift, moving the focus from reactive intervention to proactive prevention. By leveraging machine learning, AI analyzes vast datasets such as clinical images, health records, behavioral patterns, and lifestyle factors to generate highly accurate, personalized data for risk assessments. The data-driven approach enables targeted preventive interventions tailored to individual needs. This evolution promises not only to improve oral health outcomes but also to reduce long-term healthcare costs by prioritizing health maintenance. This lecture will explore AI's key application and impact in primary dental disease prevention. Additionally, the current barriers of clinical implementation, including ethical and integration challenges, and future perspectives for AI-driven preventive oral healthcare will be discussed.

Symposium 4

Title: Digital Tech & AI in Preventive Dentistry: Perspectives from Contemporary Dental Education Curriculum

Speaker: Assoc. Prof. Dr. Hoang Trong Hung

Affiliation: Pham Ngoc Thach University of Medicine, Vietnam

Abstract

Digital technology and artificial intelligence (AI) are redefining preventive dentistry and transforming how future dentists must be trained. This lecture discusses how contemporary dental curricula should integrate digital tools and AI-driven approaches to strengthen risk-based, early, and personalized prevention. From AI-assisted caries detection and predictive analytics to digital epidemiology and teledentistry, emerging technologies enable more accurate diagnosis, better risk stratification, and data-informed decision-making. However, meaningful impact depends on structured curricular integration—not merely technological exposure. Drawing on regional curriculum innovation experiences, the lecture outlines key competency domains: digital literacy, AI-supported clinical reasoning, preventive care planning frameworks (e.g., risk-based models), data ethics, and interprofessional collaboration—especially in managing non-communicable diseases. Aligned with the WHO Oral Health Action Plan 2030, this session emphasizes that AI in preventive dentistry represents a paradigm shift in education. The future preventive dentist must be digitally competent, ethically grounded, and system-oriented to lead sustainable oral health transformation.

Symposium 4

Title: Digital Health Technologies for Early Detection of Oral Cancer: Opportunities, Challenges and Future Directions

Speaker: Prof. Emeritus Dr. Rosnah Md Zain

Affiliation: MAHSA University, Malaysia

Abstract

Oral cancer remains a major global health challenge, with high morbidity and mortality largely resulting from delayed diagnosis. Although the oral cavity is readily accessible for examination, many patients continue to present with advanced disease, particularly in low- and middle-income countries where access to specialist care is limited. Early recognition of oral potentially malignant disorders (OPMDs) and early oral cancer remains the key strategy for improving patient outcomes. Advances in digital health technologies are creating new opportunities to support prevention, facilitate earlier detection, and improve access to timely care. This presentation provides an overview of how digital technologies are transforming oral cancer prevention and early detection. Smartphone-based intraoral imaging, tele-dentistry, artificial intelligence (AI), digital pathology, cloud-based referral systems, and mobile health (mHealth) applications are improving screening, referral, diagnosis, and follow-up care. These technologies can support more consistent clinical decision-making, strengthen referral pathways, and extend specialist expertise to underserved communities. Using examples from Malaysia and the ASEAN region, the presentation will further highlight how digital innovations are being applied to community screening and clinical practice. It will also discuss key challenges, including technology validation, ethical and regulatory considerations, digital inequity, and workforce readiness. Finally, future developments such as integrated digital health systems, predictive analytics, digital biomarkers, and precision prevention will be explored. Used responsibly alongside evidence-based clinical practice, digital health technologies have the potential to strengthen preventive dentistry, promote earlier detection, and improve outcomes for patients with oral cancer.

Plenary 2

Title: High Molecular Weight Hyaluronic Acid in Dentistry: Evidence-Based Applications for Oral Soft Tissue Healing and Regeneration

Speaker: Assoc. Prof. Dr. Cheah Chia Wei

Affiliation: Universiti Malaya, Malaysia

Abstract

High molecular weight hyaluronic acid (HMW-HA) has gained increasing attention in dentistry due to its anti-inflammatory and wound-healing properties. This lecture will provide an evidence-based overview of the biological mechanisms and clinical applications of HMW-HA in promoting oral soft tissue healing and regeneration. Current literature on its use in periodontal therapy, oral surgery, implant dentistry and post-operative care will be discussed. Emphasis will be placed on clinical effectiveness, indications, delivery methods in contemporary dental practice. Participants will gain practical insights into integrating HMW-HA into daily patient management for improved healing outcomes.

Symposium 5

Title: Oral Health Literacy and its Value Proposition for Oral Health Promotion

Speaker: Assoc. Prof. Dr. Wong Mun Loke

Affiliation: National University of Singapore

Abstract

Dental caries and periodontal disease remain the two most common oral diseases which affect the population. Poor oral health can exert a significant impact on the overall wellbeing of individuals. The promotion of oral health and prevention of oral diseases is therefore pivotal in helping individuals and populations embrace holistic health. Oral health literacy is an important part of overarching public health strategies to promote oral health. This presentation revisits the concept of oral health literacy, highlights its value in the context of oral health promotion and key considerations in advancing oral health literacy among individuals and populations.

Symposium 5

Title: From Information to Action: Oral Health Literacy for Healthier Populations in the Digital Age

Speaker: Dr Melissa Adiatman

Affiliation: Universitas Indonesia

Abstract

Oral health literacy is increasingly recognized as an important determinant of oral health and a foundation for effective prevention. Beyond the ability to read or understand health information, contemporary health literacy encompasses the capacity to access, understand, critically appraise, and use information and services to make appropriate health decisions. Importantly, these abilities are shaped not only by individuals, but also by the responsiveness of healthcare organizations, communities, and information environments. Evidence increasingly links oral health literacy with preventive behaviours, healthcare utilization, and oral health outcomes across the life course. Recent findings among independent older adults in Indonesia using the Health Literacy in Dentistry-29 (HeLD-29) similarly demonstrate associations between oral health literacy, sociodemographic characteristics, oral health behaviours, and clinical oral health status. Such findings reinforce the need to move beyond conventional health education towards communication and services that enable people to translate knowledge into action. The digital environment adds a new dimension to this challenge. Social media provides unprecedented access to oral health information while simultaneously exposing populations to misinformation, commercial influence, and content of variable quality. Emerging concepts such as social media oral health literacy, and instruments including the Social Media Oral Health Literacy Questionnaire (SMOHLQ), emphasize the ability to access and understand digital information, critically assess its credibility, and determine whether it should influence behaviour. This presentation proposes an adapted translational framework — Access, Understand, Appraise, Act and Enable — for translating oral health literacy into population health. It will explore practical opportunities through professional communication, community-based prevention, digital platforms, and health-literate services. In alignment with the WHO Global Oral Health Action Plan 2023–2030, strengthening oral health literacy should be viewed not merely as patient education, but as a strategic component of prevention, equity, primary healthcare, and healthier populations.

Symposium 5

Title: Advancing Oral Health Literacy and Population Health

Speaker: Dr. Habibah Yacob

Affiliation: Oral Health Programme, Ministry of Health Malaysia

Abstract

Oral health literacy plays a critical role in improving population health by empowering individuals to understand, access, and utilize dental care effectively. This paper explores strategies to advance oral health literacy through education, community engagement, and healthcare integration. Emphasizing the importance of clear communication and culturally sensitive approaches, it highlights how enhanced literacy can lead to better prevention, early detection, and management of oral diseases. The paper also discusses policy initiatives and collaborative efforts needed to reduce disparities in oral health outcomes across diverse populations. By promoting oral health literacy, communities can achieve improved overall health, reduced healthcare costs, and enhanced quality of life.

Symposium 6

Title: Preventive Dentistry for Older Adults: Aligning Oral Health with Indonesia's Health Transformation

Speaker: Prof. Dr. Sri Susilawati

Affiliation: Universitas Padjadjaran, Indonesia

Abstract

Indonesia is experiencing a rapid demographic transition, with a steadily growing older adult population that has increased the demand for comprehensive oral healthcare. Recent findings from the Indonesian Health Survey and the National Free Health Check /Screening Program reveal a substantial burden of oral diseases among older adults, particularly dental caries, periodontal disease, tooth loss, and unmet treatment needs, all of which significantly affect nutrition, systemic health, functional ability, and quality of life. Recognizing these challenges, Indonesia's National Oral Health Action Plan 2026–2030 has established a target to ensure that at least 50% of the population, including older adults, has access to essential oral healthcare services. Aligned with Indonesia's Health Transformation Agenda, achieving optimal oral health among older adults is guided by four strategic pillars, including health promotion and disease prevention, strengthening oral healthcare services, governance, workforce development and sustainable financing, as well as health information systems and research. Implementation of this strategy has been driven by strong collaboration among government agencies, professional organizations, universities, healthcare providers, schools, communities, and other stakeholders. This collaborative approach has strengthened the integration of preventive oral healthcare into primary healthcare services, community-based oral health programmes, fostering shared responsibility for improving oral health outcomes among older adults. Strengthening preventive dentistry through integrated health system transformation represents a sustainable pathway towards improving oral health among older adults. Indonesia's experience offers valuable insights into how prevention-oriented policies and multisectoral collaboration can strengthen oral health systems and support healthy ageing.

Symposium 6

Title: Caring for Oral Health and Improving the Quality of Life Among the Elderly in the Philippines

Speaker: Prof. Dr. Michelle S. Segarra

Affiliation: University of the Philippines College of Dentistry

Abstract

2025 data shows that there are 10.9 million senior citizens (60 years old and above) in the Philippines. This is 9.6% of the country's 113.7 million population. The Philippines is approaching an aging society and is predicted that there will be 23.7 million elderly by 2050. It is thus important that the Philippines set up programs for the oral health of the aging population. Common health problems among the elderly include periodontal diseases, root caries, dry mouth, tooth loss and dentures, oral cancer among others which are compounded by limited access to dental care and financial constraints. Data from the Economic Research Institute for ASEAN and East Asia in 2019 shows that 32% of older Filipinos suffer from complete tooth loss, with the rate climbing higher among older individuals; only 60% of those with nor remaining teeth have dentures; and many elderly, especially those who are institutionalized have high rates of untreated caries. A published study in 2024 shows that periodontal disease affects 83.58% of Filipinos aged 65 to 74. As poor oral health affects quality of life, the government and private sectors including dental academic institutions are setting up programs and interventions to improve the elderly population's quality of life. The inclusion of oral health services in the Universal Health Care (UHC) Act (Republic Act 11223) and alliances between organizations such as the National Commission of Senior Citizens (NCSC) and the Philippine Dental Organizations will ensure that oral healthcare is integrated into over-all health towards elderly well-being. Accessibility and awareness programs are being advocated. In the academic setting, geriatric dentistry and special care dentistry are integrated in the undergraduate dental curriculum since 2018. This brief presentation will discuss in detail the oral health status of the elderly in the Philippines and the actions being done to improve their oral health and their quality of life.

Symposium 6

Title: Oral Frailty and Hypofunction in Korea's Super-Aging Society: Current Status and Perspective

Speaker: Prof. Dr. Youn-Hee Choi

Affiliation: Kyungpook National University, Korea — President, KAPDOH

Abstract

Korea, like many countries worldwide, has entered a super-aging society, with a rapidly growing older population and rising socioeconomic burden. Remarkably, Korea reached an aging society (7% of the population aged 65 or older) in 2000, and is projected to transition from an aged society (14%) to a super-aged society in just 7–8 years—an unprecedented pace in human history. This rapid demographic shift underscores the urgent need to address factors that influence health and functional independence in later life. Oral frailty and oral hypofunction have recently been defined as important clinical conditions reflecting declines in oral function. These conditions are closely associated with general frailty and systemic health, and have become key determinants of healthy aging and quality of life. Maintaining oral function is therefore essential for preventing frailty and supporting active and independent living among older adults. From a broader perspective, concepts such as anti-aging and healthy longevity have gained increasing attention. Preventing oral frailty is emerging as a key component of these strategies, contributing not only to maintaining oral function but also to promoting overall functional health and longevity. Effective prevention requires not only individual efforts but also comprehensive approaches at the community and national levels. In particular, integrated strategies within public oral health, along with active involvement of dental professionals, including dentists and dental hygienists, are essential to manage and prevent oral functional decline in a super-aging society. Addressing oral frailty and hypofunction through multidisciplinary and population-based strategies is therefore crucial for supporting healthy aging, preserving functional independence, and improving quality of life among older adults in Korea's rapidly aging society.

Plenary 3

Title: Beyond the Dental Chair: The Unsung Heroes in Mass Disaster Management

Speaker: Brigadier Jeneral (Dr) Hazudin Hassan

Affiliation: Malaysian Armed Forces Health Services

Abstract

The Malaysian Armed Forces Dental Services (MAFDS) is part of a military healthcare system known as the Malaysian Armed Forces Health Services (MAFHS). MAFHS's mission is centered on maintaining the combat readiness of the MAF by providing optimal health services and support. Apart from its core function, MAFHS remains committed in contributing towards national resilience through its role in the Whole-of-Government and Whole-of-Society (WOGOS) approaches, especially in crises, humanitarian and disaster relief (HADR) missions, and civil–military cooperation. Field and mobile capabilities are a crucial component of MAFHS, with dedicated medical battalions ready to be rapidly deployed to conflict zones and HADR missions. These battalions are trained to set up mobile field hospitals to provide immediate care, including resuscitative treatment, advanced trauma surgery, and other health services. The distinctive ability to be deployed at a moment's notice is the main thrust of MAFHS's involvements in various international missions. It has participated in multiple United Nations (UN) operations in conflict areas such as Namibia, Bosnia, Cambodia, East Timor, Western Sahara and Lebanon. During these peacekeeping missions, the medical and dental elements also provide care for the displaced populations and refugees. In HADR, MAFHS has served globally in disaster areas such as during tsunami in Aceh, earthquake in Pakistan, Padang, Türkiye and Myanmar, post-conflict support in Afghanistan and typhoon in the Philippines. Domestically, MAFHS has been deployed to assist local authorities during landslides, floods and the COVID-19 pandemic. The MAFHS forensic odontologists team has been deployed to mass disaster areas aiding in disaster victim identification (DVI) during the MH17 tragedy, civilian and military aircraft crashes and the Lahad Datu incident. In executing its duty, the MAFHS receives full logistic support from the Malaysian Armed Forces; the Army provides security protection, land transport, engineers, and other services through its various Corps, the Air Force provide air transport and CASEVAC support and the Navy provides maritime transport.

3-Minutes Presentation Abstracts

Presenter ID: 3M1

Lithium Mouthwash for Preventing Radiotherapy-induced Oral Mucositis and Taste Dysfunction in HNC: Phase I/II Trial

Zixia Li¹, Yiyang Pei², Zhiying Zhang¹, Shuhao Zheng¹, Yixuan Li¹, Qinrui Wu¹, Jingjing Wang², Huaju Yang², Ruidan Li², Liwei Zheng³, Xingchen Peng², Xin Xu⁴

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Background: Oral mucositis (OM) and taste dysfunction (TD) are common oral complications of radiotherapy for head and neck cancer (HNC) patients. The goal of this phase I/II trial (NCT06251050) was to evaluate the efficacy and safety of a lithium-containing mouthwash in the prevention of OM, and explore its potential protection against TD. **Methods:** HNC patients planned for definitive or postoperative adjuvant radiotherapy received lithium-containing mouthwash or placebo thrice daily during radiotherapy. OM was assessed twice a week during radiotherapy and weekly thereafter for up to 8 weeks. The primary endpoint in phase II was the incidence of severe OM (SOM). Second endpoints included the time to onset and duration of SOM, taste alteration, the incidence of dry mouth, quality of life and safety. **Results:** 175 patients were enrolled (n=15 for phase I and n=160 for phase II, respectively). A dose of 0.15 mol/L lithium was recommended for phase II study based on phase I. Compared with placebo, the lithium-containing mouthwash significantly reduced the incidence of SOM (34.2% vs. 52.1%; P= .0304). The duration (mean, 9.0 vs. 17.3 days, P= .0101) and time to develop SOM (median, NE vs 35.0 days; P= .0262) were also improved in the lithium group. Additionally, lithium-containing mouthwash alleviated TD during RT. Adverse events (AEs) were similar between groups, and no severe AE related to lithium was observed. **Conclusion:** Lithium-containing mouthwash at 0.15 mol/L significantly reduced the incidence, duration and severity of SOM, with an acceptable safety profile. It also showed potential for ameliorating radiotherapy-induced TD.

Keywords: Radiotherapy, Oral Mucositis, Taste Dysfunction, Head and Neck Cancer, Lithium

Presenter ID: 3M2

A Deal Worth Making? Stakeholder Perspectives on Publicly Funded Oral Healthcare in Malaysia

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Background: In Malaysia's dual-tier healthcare system, lower-income adults encounter limited access in the public sector and prohibitive costs in the private sector. Contracting private dentists to deliver publicly funded oral healthcare through Strategic Purchasing and Public-Private Partnerships (PPPs) offers a promising solution. Yet evidence on the operational requirements of such arrangements, from the perspectives of those who would design and deliver them, remains scarce. This study examined stakeholders' perspectives on the feasibility and acceptability of participating in a publicly funded oral healthcare. **Methods:** This qualitative exploratory case study was conducted through two in-depth interviews with purchasers and three focus group discussions with 14 private general dental practitioners (GDPs). Transcripts underwent inductive thematic analysis. Study rigour was ensured through investigator triangulation via collaborative team discussions, reflexive documentation, and comprehensive audit trail. **Results:** Divergent priorities emerged between the two parties to any potential deal. Purchasers framed PPPs as institutional mechanisms for cost-containment, equitable resource distribution, and outcome monitoring through centralized data systems. Private GDPs expressed conditional willingness to participate but cited heavy administrative burdens and reimbursement delays as critical barriers, particularly for solo practices. Their preferred terms centered on simple digital claims, rapid payments, and a clearly defined health benefit package. These perspectives were synthesized into a preliminary framework of contract attributes spanning service scope, financial structures, digital infrastructure, and audit intensity. **Conclusion:** Strategic purchasing through PPPs can work, but its success is predicated on contract arrangements that honour both institutional accountability and the operational realities of frontline providers. This framework offers policymakers a practical blueprint for purchasing arrangements that are both equitable and sustainable

Keywords: Strategic Purchasing, Public-Private Partnerships, Stakeholders, Low-income, Contract Attributes

Presenter ID: 3M3

Supportive Care and Survivorship Policy Gaps for Malaysian Head and Neck Cancer Survivors: Asian Lens

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Background: Head and neck cancer (HNC) survivorship extends beyond disease control. Survivors often face long-term physical, psychosocial and financial challenges that require coordinated supportive care. In Malaysia, supportive care is addressed across cancer, palliative care, rehabilitation, and welfare-related policies, however, the extent to which these policies provide comprehensive survivorship support remains unclear. This policy review aimed to identify supportive care policy gaps for HNC survivors in Malaysia through comparison with selected Asian countries. **Methods:** A qualitative document analysis was conducted using the READ (Ready materials, Extract data, Analyse data and Distil findings) approach. Relevant policy documents were reviewed, and key data were extracted into a cross-country comparative table to synthesise findings, identify similarities and differences, and map policy gaps. **Results:** Of 121 policy documents reviewed, 82 met inclusion criteria. Comparative analysis showed 86.7% of countries had cancer control or survivorship policies. Four themes emerged: financial protection, legal and community support, palliative care, and rehabilitation. Most countries (80.6%) provided universal health insurance. Malaysia offers subsidised healthcare, while Japan and Singapore extend lifelong coverage. Legal and community support were weak, with the Philippines the only country enacting anti-discrimination laws. Palliative care was mainly hospital-based with limited home services. Rehabilitation was available in 86.7% of countries, but structured survivorship plans were confined to high-income settings. **Conclusion:** Malaysia requires stronger integration between supportive care and survivorship policy. Strengthening follow-up, rehabilitation, nutrition support, appearance support, disability recognition, and post-discharge continuity may extend preventive dentistry beyond treatment by reducing long-term complications and improving quality of life among survivors.

Keywords: Head and Neck Cancer, Cancer Survivorship, Supportive Care, Health Policy, Asia

Presenter ID: 3M4

More Than Just Brushing Teeth, How Caregivers Shape Oral Health in Older Adults

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Background: Poor oral health among dependent older adults contributes to preventable complications including aspiration pneumonia, malnutrition, and reduced quality of life. As many older adults rely on caregivers for daily oral hygiene, caregiver-delivered oral care programmes have become increasingly important. However, evidence on how these programmes are designed, implemented, and evaluated remains fragmented. This study mapped the current evidence to identify key programme characteristics and research gaps. **Methods:** A scoping review was conducted following the Joanna Briggs Institute methodology and reported according to PRISMA-ScR guidelines. Five electronic databases were searched, identifying 816 records. After duplicate removal and two-stage screening, 36 studies published between 2015 and 2025 were included for narrative synthesis. **Results:** Studies from 19 countries were identified, predominantly in long-term care settings. Hands-on practical training and didactic education were the most common training approaches, while oral hygiene and caregiver knowledge were the most frequently evaluated outcomes. Resident-related challenges were the leading implementation barrier, whereas structured training programmes were the main facilitator. Notably, 80.6% of studies did not report using a behavioural or implementation theory, limiting understanding of how caregiver behaviour is influenced and sustained over time. **Conclusion:** This review provides a comprehensive mapping of caregiver-delivered oral care programmes for older adults, highlighting opportunities to strengthen caregiver training and implementation strategies. The findings offer an evidence base for developing standardised, theory-informed programmes and support future integration of oral health into routine aged-care services and community-based care.

Keywords: Caregiver, Older Adults, Oral Care

Presenter ID: 3M5

The Missing Piece in Diabetes Care: Lessons from a Scoping Review of Oral Health Integration

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Background: Despite decades of evidence supporting the bidirectional relationship between diabetes and periodontal disease, oral health has yet to become routine in diabetes care. Although integrated care is widely advocated, evidence on its implementation remains fragmented. This scoping review examined implementation approaches, outcomes, and determinants of oral health integration into primary diabetes care. **Methods:** A scoping review was conducted following Joanna Briggs Institute methodology and PRISMA-ScR guidelines. PubMed, Scopus, Web of Science, and the Cochrane Library were searched for studies published between 2015 and May 2026. Study selection followed the Population–Concept–Context framework, and evidence was synthesised using the Consolidated Framework for Implementation Research (CFIR) and Proctor et al.'s Implementation Outcomes Framework. **Results:** Twenty-one studies met the inclusion criteria. Although providers consistently perceived oral health integration as acceptable and appropriate, adoption remained limited, with referral rates as low as 8%, revealing a substantial evidence-to-practice gap. Three implementation approaches emerged: referral-based strategies, oral health screening and risk assessment, and digitally supported integration. Organisational and system-level barriers including competing clinical priorities, limited oral health training, and role ambiguity outweighed individual-level factors. Facilitators included positive provider attitudes, validated screening tools, and workflow-embedded implementation supports. **Conclusion:** The findings suggest that the greatest barrier to oral health integration is not provider willingness but insufficient implementation infrastructure. Embedding protocolised oral health activities within existing diabetes care pathways, supported by organisational systems and clear referral processes may provide a practical foundation for task shifting and integrated diabetes care in Malaysia.

Keywords: Diabetes Care, Oral Health Integration, Primary Care, Task Shifting

Presenter ID: 3M6

Developing STEP-5S: A Theory-Guided Parent-Targeted Framework for Supervised Toothbrushing

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Background: Despite decades of oral health education, early childhood caries remains highly prevalent, indicating that education alone is insufficient to change daily toothbrushing behaviours. Although parent-supervised toothbrushing is an effective preventive strategy, existing recommendations are often fragmented and difficult to implement consistently at home. This study aimed to develop STEP-5S, a theory-guided framework to support parent-supervised toothbrushing. **Methods:** Phase 1 involved a systematic multi-stage development process. Evidence from a scoping review of 67 eligible studies was synthesised with previous quantitative and qualitative findings and mapped to the Capability, Opportunity, Motivation–Behaviour (COM-B) model to identify implementation strategies for parent-supervised toothbrushing. These findings informed the preliminary development of the STEP-5S framework and accompanying materials, which are currently undergoing expert content validation. **Results:** Evidence synthesis and COM-B mapping informed the preliminary development of the STEP-5S framework, comprising five implementation strategies: Set Up, Show, Support, Schedule, and Signal. The Show component incorporates the STEP-5S toothbrushing technique (See, Stay, Squeeze, Spit, and Skip), providing parents with a simple sequence of supervised toothbrushing actions. Supporting materials include a parent brochure, instructional video, reminder calendar, and monitoring checklist. The framework and materials are currently undergoing expert content validation. **Conclusion:** STEP-5S translates research evidence and behaviour change theory into a practical framework for parent-supervised toothbrushing. Following expert validation, the framework will undergo feasibility and acceptability testing in preschool settings before evaluation of its implementation and potential effectiveness.

Keywords: Early Childhood Caries, Supervised Toothbrushing, Parental Supervision, Implementation Science, Behaviour Change

Presenter ID: 3M7

The Magic Silver Bullet: Does Frequency Matter?

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Background: Silver Fluoride (SF) is a modified formulation derived from Silver Diamine Fluoride (SDF) that is ammonia-free and has improved physicochemical properties and potential for arresting dentine caries. However, evidence regarding the optimal frequency of SF application remains inconclusive. This study protocol aims to compare single versus dual applications of 38% SF for arresting dentine caries in primary teeth over a 6-month follow-up period. **Methods:** This two-arm, parallel, assessor-blinded randomised controlled trial will recruit children aged 4–6 years from selected preschools in Perak using multistage sampling. Participants will be stratified by caries severity (≤ 3 or > 3 active lesions) and randomly allocated 1:1 to receive either a single SF application at baseline or two applications at baseline and three months. Eligible dentine caries lesions will be identified using ICDAS criteria, with caries arrest assessed using visual-tactile criteria. The primary outcome is caries arrest rate, assessed at 3 and 6 months, while changes in oral health-related quality of life (OHRQoL), measured using the Malay-ECOHIS at baseline and 6 months, are the secondary outcome. Data will be analysed according to intention-to-treat principle. **Results:** It is hypothesised that children receiving two applications of 38% SF will exhibit higher caries arrest rates, along with greater improvements in OHRQoL. **Conclusion:** The findings from this study will generate evidence to optimise SF application protocols and support wider adoption of SF within Malaysian preschool oral health programmes.

Keywords: Silver Fluoride, Randomised Controlled Trial, Preschool Children, Caries, Protocol

Presenter ID: 3M8

Factors Associated with Dental Caries among 12-Year-Old Children: 2024 KNCOHS

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Background: Dental caries is one of the most prevalent chronic conditions in childhood and may have long-term effects on oral health. This study investigated factors associated with dental caries experience among 12-year-old children, with particular emphasis on cariogenic snack consumption and dental floss use. **Methods:** Data from 12-year-old participants in the 2024 Korea National Children's Oral Health Survey were analyzed. Dental caries experience was defined as a decayed, missing, and filled permanent teeth index of ≥ 1 . The main explanatory variables were the presence of fissure-sealed permanent teeth, frequency of cariogenic snack consumption on the previous day, and dental floss use during the preceding week. Sex, residential area, perceived household economic status, oral health education, toothbrushing frequency, and mouthrinse use were included as covariates. Complex-sample hierarchical logistic regression analyses were performed. **Results:** Female children had higher odds of dental caries experience than male children (OR=1.274, 95% CI=1.111–1.460). Compared with children living in rural areas, those living in metropolitan districts (OR=0.759, 95% CI=0.599–0.962) and cities (OR=0.741, 95% CI=0.580–0.946) had lower odds. Children without fissure-sealed permanent teeth had higher odds of caries experience (OR=1.362, 95% CI=1.188–1.560). Consuming cariogenic snacks once daily was associated with lower odds than consuming them at least twice daily (OR=0.846, 95% CI=0.743–0.964). Dental floss use was not significantly associated with caries experience. **Conclusion:** Dental caries experience was associated with sex, residential area, fissure sealant status, and cariogenic snack frequency. Population-based strategies should prioritize equitable access to preventive services and reduced frequency of cariogenic snack consumption.

Keywords: Child, Dental Carie, Diet, Cariogenic, Pit and Fissure Sealants

Presenter ID: 3M10

The Digital Dental Advertising Dilemma: Balancing Marketing and Professional Ethics

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Background: Digital dental advertising is increasingly prevalent, however evidence among Malaysian dentists remains limited. This study aimed to assess dentists' perceptions, attitudes and practices towards digital dental advertising. **Methods:** This study employed a sequential exploratory mixed-methods approach. The qualitative phase explored dentists' perceptions and informed item generation for questionnaire development. The quantitative phase involved content and face validation, followed by a pilot survey to assess dentists' attitudes and practices towards digital dental advertising. Qualitative data were analysed using thematic analysis, while quantitative data were analysed descriptively. **Results:** The FGDs (n=21) identified three main themes that informed questionnaire development: perceived benefits, ethical considerations and type of dental advertising practices. Thirty private dentists (29–45 years) participated in the pilot survey. They were mostly clinic owners (40.0%) and associates (30.0%). Most participants agreed that digital dental advertising is acceptable (90.0%), can be conducted professionally and ethically (96.7%), and increases opportunities to reach potential patients (96.7%). Many considered short-form content (83.3%), paid advertisements (80.0%) and influencer marketing (50.0%) acceptable, although these practices requiring careful consideration under current professional guidance. Approximately 60% reported advertising promotional/discounts related content. Instagram (86.7%), TikTok (83.3%) and Facebook (73.3%) were the most commonly used platforms for informational (90.0%) and oral health education (80.0%) content. However, concerns regarding ethical risks remained, with 73.3% acknowledging that digital dental advertising may potentially mislead patients. **Conclusion:** Malaysian dentists generally viewed digital dental advertising positively. However, evolving practices require clearer ethical guidance to ensure responsible implementation.

Keywords: E-professionalism, Dental Advertising, Private Dental Practitioners, Social Media

Presenter ID: 3M11

Factors Influencing Out-of-Pocket Payment Decisions for Private Dental Care in Malaysia: An Exploratory Qualitative Study

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Background: Achieving Universal Health Coverage requires financial protection against healthcare costs. In Malaysia, private dental services are predominantly financed through out-of-pocket (OOP) payments, accounting for 52% of total dental care expenditure. Despite the financial challenges, many patients, including those from low-income households, continue to utilise private dental services. This study aimed to explore factors influencing patients' decision-making regarding OOP payments for care in private dental clinics. **Methods:** This exploratory qualitative study employed in-depth interviews (IDIs) using a validated topic guide. Participants were adults who had received treatment in private dental clinics within the preceding year. Purposive sampling ensured maximum variation among participants. IDIs were conducted online until data saturation was achieved, transcribed verbatim, and analysed using thematic analysis with Atlas.ti software. **Results:** Seven participants from different income groups (B40, n=5; M40, n=1; T20, n=1) were interviewed. Financial considerations emerged as the overarching theme, with affordability identified as the primary determinant of payment decisions. Participants from the low-income group reported financial constraints that delayed or limited care-seeking. Cost transparency enabled participants to anticipate treatment costs, while financial management strategies, such as budgeting and instalment plans, influenced their decisions. Participants also considered whether the services offered good value for money. Beyond financial considerations, perceived service quality, accessibility and convenience, and time-related factors also influenced decision-making. **Conclusion:** Patients' decisions regarding OOP payments for private dental care were influenced by financial, service-related, and personal factors. These findings may inform health financing reforms to reduce barriers associated with OOP payment.

Keywords: Dental Care, Health Financing, Out-of-Pocket Payments

Oral Presentation Abstracts

Presenter ID: OP1

Disrupted Sensory Innervation Impairs Bone Repair in Diabetes-Related Periodontitis

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Background: Diabetes is a major risk factor for periodontitis. This study aimed to identify neural signaling as a potential therapeutic target to limit bone resorption and promote regeneration in diabetes-related periodontitis. **Methods:** Diabetes-related periodontitis was induced by ligating molars in db/db mice. Periodontal denervation was achieved by inferior alveolar nerve (IAN) axotomy. Neural alterations in the periodontium and trigeminal ganglion were assessed by immunofluorescence staining. Single-cell RNA sequencing (scRNA-seq) was used to profile the endothelial cell (EC) subsets with or without sensory innervation and to reveal crosstalk between periodontal ECs and trigeminal neurons. **Results:** Under non-diabetic conditions, sensory fibers and type H vessels increased in parallel during periodontal regeneration and preceded bone formation. Denervation markedly reduced type H vessels and subsequent osteogenesis. scRNA-seq identified Sox17-positive ECs as a pro-osteogenic subset whose proportion also decreased after IAN axotomy. Furthermore, scRNA-seq reconstructed potential nerve-to-EC signaling interactions, involving neuropeptides (Calca, Tac1) and other molecules (Vegfa, Vegfb, and Jag2). In db/db mice, diabetes led to periodontal neuropathy characterized by loss of peripheral calcitonin gene-related peptide (CGRP)+ nerves, which worsened with disease progression and was accompanied by reduced type H vessels and aggravated bone loss. Periodontal denervation in non-diabetic mice phenocopied the diabetic periodontal phenotype. **Conclusion:** Sensory nerves orchestrate angiogenesis and bone regeneration in the periodontium by regulating type H vessel formation. Diabetes-induced sensory neuropathy disrupts this neurovascular axis and accelerates periodontal destruction. These findings suggest that enhancing nerve reinnervation may represent a promising preventive and therapeutic approach in diabetes-related periodontitis.

Keywords: Angiogenesis, Bone Regeneration, Endothelial Cell, Periodontitis, Sensory Nerve

Presenter ID: OP2

Untargeted Metabolomic Analysis Reveals the Mechanisms of Fluoride Application in Early Childhood Caries Prevention

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Background: Early childhood caries (ECC) remains a major global oral health challenge. Fluoride application is one of the most effective preventive strategies, however, the metabolic mechanisms underlying its protective effects are not fully understood. This study aimed to investigate salivary metabolic alterations associated with ECC and fluoride application using untargeted metabolomics. **Methods:** Forty preschool children were recruited, including 20 children with high caries experience and 20 caries-free controls. Unstimulated saliva samples were collected at baseline, 1 day, and 14 days after topical fluoride application, generating a total of 120 saliva samples across six study groups. Untargeted metabolomic profiling was performed using liquid chromatography–tandem mass spectrometry (LC MS/MS). Differential metabolites and enriched metabolic pathways were analyzed using bioinformatics approaches. **Results:** A total of 1,951 and 1,245 molecular features were detected in positive and negative ion modes, respectively. Comparative analyses revealed metabolic alterations following fluoride application. Pathway enrichment analysis identified significant involvement of amino acid metabolism, aminoacyl-tRNA biosynthesis, protein digestion and absorption, and lipid metabolism. Acetophenone and choline were enriched in the ECC group, suggesting associations with microbial metabolism, membrane turnover, and oral inflammation. Following fluoride application, several amino acid-related metabolites changed, potentially reflecting altered bacterial protein metabolism and reduced proteolytic activity in the oral microenvironment. Among caries-free children, abietic acid and sedanolide were upregulated, while gamma-linolenic acid decreased in specific comparisons. In the high-caries group, 3-hydroxybutyric acid increased following fluoride exposure. Palmitic acid and abietic acid showed potential as candidate markers of fluoride-related metabolic responses. **Conclusion:** Fluoride application induces significant alterations in salivary metabolic profiles in preschool children, particularly in pathways related to amino acid and lipid metabolism, providing preliminary insights into the metabolic mechanisms underlying fluoride mediated ECC prevention and may contribute to future biomarker exploration.

Keywords: Fluoride, Early Childhood Caries, Untargeted Metabolomics, Saliva

Presenter ID: OP3

Clinical and Microbiome Effects of Chlorine Dioxide Mouthwash in Gingivitis: A Randomized Controlled Trial

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Background: Chlorhexidine is widely used for gingivitis control but is associated with adverse effects and potential disruption of the oral microbiome. Chlorine dioxide has emerged as a potential alternative; however, its clinical efficacy and ecological impact on supragingival plaque remain unclear. **Methods:** In this randomized controlled trial, 90 patients with gingivitis were assigned to a chlorine dioxide group (n = 44) or a chlorhexidine group (n = 46). Clinical parameters, including gingival index (GI), plaque index (PLI), and sulcus bleeding index (SBI), were recorded at baseline and on Days 1, 3, and 7. Supragingival plaque samples were analyzed using 16S rDNA sequencing to evaluate microbial diversity and compositional changes. **Results:** Both groups showed significant reductions in GI and PLI from Day 1 and in SBI from Day 3 ($P < 0.05$), with no significant intergroup differences ($P > 0.05$). Microbiome analysis demonstrated an early decrease in alpha diversity followed by recovery at Day 7 in the chlorine dioxide group, whereas recovery was less pronounced in the chlorhexidine group. Beta diversity indicated a reversible microbial shift with chlorine dioxide. At the genus level, chlorine dioxide exhibited stronger inhibition of *Neisseria* and *Kingella*, suggesting a selective antimicrobial effect. **Conclusion:** Chlorine dioxide achieved comparable clinical efficacy to chlorhexidine while demonstrating a more favorable microbiome modulation pattern, characterized by selective bacterial suppression and faster ecological recovery. It may represent a microbiome-friendly alternative for gingivitis management.

Keywords: Chlorine Dioxide, Gingivitis, Randomized Controlled Trial, Oral Microbiome

Presenter ID: OP5

Modulatory Role of Magnesium in Fluoride-Induced Enamel Lesion Remineralization

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Background: This study aimed to investigate whether the application of magnesium after fluoride solution could enhance remineralization within the lesion body, which is ordinarily limited when fluoride applied alone. **Methods:** 18 enamel specimens with exposed area of 2 mm x 2 mm were prepared from 18 bovine incisors. Each specimen was immersed in 50 mL demineralization solution for 3 days and subsequently stored in artificial saliva, except during treatment application. The specimens were randomly assigned to 3 groups (n = 6/group) and immersed in solutions each for 15 minutes daily (F+Mg group: first 0.1% sodium fluoride, then 2.5% chloride magnesium; F group: 0.1% sodium fluoride; DW group: deionized water). Micro-computed tomography (micro-CT) was performed before, after demineralization, after 1 week, 2 weeks and 4 weeks of remineralization. Attenuated total reflectance–Fourier transform infrared spectroscopy (ATR-FTIR) was conducted before scanning electron microscope (SEM) and energy-dispersive X-ray spectroscopy (EDS). **Results:** Micro-CT revealed more homogeneous remineralization patterns in F+Mg and DW groups, whereas F group exhibited relatively lower mineral density within lesion body. F+Mg group revealed significantly more mineral regain and more reduction in lesion depth. FTIR implied more amorphous PO₄³⁻ incorporation on surface in F+Mg group. SEM exhibited a more porous surface structure with thinner enamel rods in F+Mg group, while thicker rods were observed in F group. EDS indicated higher fluorine (F) composition in F+Mg group. **Conclusion:** The sequential application of fluoride followed by magnesium solution may enhance remineralization and incorporation of F within lesion body by stabilizing the lesion surface, compared to fluoride applied alone.

Presenter ID: OP6

Effects of Fluoride-Containing Coating Agents Combined with Icon on Enamel White Spot Lesions Treatment

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Background: Non-invasive treatments for white spot lesions (WSLs) aim to inhibit demineralization, despite esthetic improvement is not predictable. Resin infiltration (Icon) is a microinvasive approach improves appearance. This study evaluated the effects of Icon combined with remineralization-promoting coatings, CareDyne Shield (CS) or NanoSeal (N), compared with each material alone, on enamel WSLs using swept-source optical coherence tomography (SS-OCT), transverse microradiography (TMR), and scanning electron microscopy (SEM). **Methods:** Sixty-six bovine enamel specimens were prepared with a 3×3 mm² window and randomly assigned to six groups (n = 11): Icon, Icon + CS, Icon + N, CS, N, and deionized water (control). Artificial WSLs were created by 72 h demineralization, followed by material application and 1-week pH cycling. SS-OCT images were obtained at baseline, after WSL formation, before pH cycling, and after pH cycling. Signal attenuation coefficients (μ t) were calculated from selected regions of interest. After SS-OCT observation, specimens were sectioned and analyzed by TMR to determine integrated mineral loss (ΔZ). **Results:** μ t only showed no significant difference between before and after pH-cycling in Icon + CS and CS groups. $\Delta\mu$ t (After- Before pH cycling) was highest in control and significantly higher than in Icon + CS and CS. TMR showed that Icon + CS had the lowest ΔZ , significantly lower than control. **Conclusion:** Icon + CS showed the strongest inhibition of demineralization, CS-containing treatment showed the most consistent anti-demineralization response. In contrast, control group showed inferior performance.

Keywords: SS-OCT, White spot lesion, Infiltration resin, Remineralization

Presenter ID: OP7

Microbial and Functional Characteristics of Extrinsic Black Stain on Primary Dentition

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Background: Extrinsic black stain (EBS) is characterized by pigment deposition on primary teeth. This study aimed to investigate the microbial and functional features associated with EBS. **Methods:** A case-control study design was employed. Supragingival plaque samples were collected from children aged 3–5 years with and without EBS. Metagenomic sequencing was performed. Microbial community structure was assessed using α - and β -diversity indices. Functional annotations were mapped to the Kyoto Encyclopedia of Genes and Genomes (KEGG) database. LEfSe analysis identified taxonomic and functional differences between groups. The Wilcoxon rank-sum test was used to compare KEGG Orthology (KO) genes. Pathway modules were visualized using KEGG pathway maps. **Results:** A total of 44 children were enrolled: the EBS group (EG, n = 22) and the control group (CG, n = 22, without EBS). The Chao index ($P = 0.018$) and Shannon index ($P = 0.034$) indicated lower species richness and diversity in EG compared to CG. Principal coordinate analysis (PCoA) and ANOSIM further revealed a significant difference in microbial community structure between the two groups ($R = 0.42$, $P = 0.001$). A total of 38 species were significantly enriched in EG (LDA score [$\log_{10}$] > 2, $P < 0.05$). Additionally, 32 KEGG pathways and 967 KOs were significantly elevated in EG. These KOs were summarized into five functional modules: iron acquisition and siderophore-related functions; aromatic/phenolic/indole-related metabolism; sulfur-related metabolism and redox reactions; adhesion/matrix formation/initial attachment; and environmental response and genetic regulation. **Conclusion:** EBS formation may involve increased local iron availability, along with aromatic-derived organic intermediates and sulfur-related redox changes, thereby creating a microenvironment that favors the deposition and retention of pigment-related products on tooth surfaces.

Keywords: Extrinsic Black Stain, Microbiota, Metagenomics, Primary Dentition

Presenter ID: OP8

Association between Dietary Intake and Volatile Sulfur Compound Levels in Diabetic Patients with Periodontitis

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Background: Halitosis is common in patients with diabetes. While clinical and periodontal determinants of halitosis have been widely studied, the role of dietary factors in modulating volatile sulfur compound (VSC) levels remains unclear. This study aimed to evaluate the association between dietary intake and changes in VSC levels over 24 weeks in diabetic patients with periodontitis.

Methods: This study is a secondary exploratory analysis of a randomized clinical trial (RCT) including 37 diabetic patients aged ≥ 35 years treated at Niigata University Hospital. Dietary variables included frequency of vegetable, fruit, and seaweed intake, salt reduction effort, and preference for fish versus meat, assessed by questionnaire. VSC levels (H_2S , CH_3SH , $(\text{CH}_3)_2\text{S}$, and total VSC) were measured using a gas chromatograph. In the RCT, participants were assigned to two periodontal treatment groups (with and without antimicrobials), and both groups were included in the present analysis to reduce bias. Changes in VSC levels were analyzed using analysis of covariance models. Interaction terms between seaweed intake and study group were included to assess effect modification. **Results:** A total of 37 participants were included (mean age 62 years; 45% male). Most reported high intake of vegetables (80.6%) and fruits (47.2%), salt reduction practices (65.7%), and regular legume consumption (91.9%). Only 13.9% reported high seaweed intake. A significant interaction for seaweed intake was observed for H_2S ($p = 0.048$). **Conclusion:** This preliminary finding showed that higher baseline seaweed intake was associated with greater reductions in H_2S , indicating a potential beneficial effect on halitosis.

Keywords: Halitosis, Diabetes Mellitus, Volatile Sulfur Compounds, Seaweed

Presenter ID: OP9

Functional Evaluation of Coral-Derived Biomaterials for Early Root Caries Management

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Background: Coral apatite (CA) is a novel low-crystalline apatite biomaterial derived from fossilized coral (FC) from Yonaguni Island, Japan. Because FC and CA differ in mineral composition and crystal structure, they may exhibit distinct functional properties relevant to root caries management. This study compared the physicochemical characteristics, bacterial adsorption ability, and demineralization-inhibitory effects of these coral-derived materials. **Methods:** FC, CA, and monoclinic hydroxyapatite were characterized by XRD, SEM, and EDS. Cariogenic bacterial adsorption was evaluated by colony-forming unit (CFU) counting and fluorescence microscopy (FM), with bacterial attachment further observed by SEM. For demineralization analysis, bovine root dentin specimens were assigned to five groups and subjected to pH cycling with repeated gel application. Lesion changes were evaluated using swept-source optical coherence tomography (SS-OCT) and transverse microradiography (TMR). **Results:** XRD analysis revealed that CA consisted mainly of low-crystalline apatite with residual calcite. SEM revealed finer particles after apatite conversion, while EDS confirmed phosphate incorporation and the retention of trace elements, including Mg, Si, and Sr. CFU counting, FM, and SEM demonstrated that FC exhibited the greatest bacterial adsorption. CA showed the strongest inhibitory effect against demineralization, while the mixed material exhibited both bacterial adsorption and demineralization-inhibitory properties. **Conclusion:** These findings indicate that FC and CA possess distinct functional strengths, with FC showing superior bacterial adsorption and CA exerting a strong demineralization-inhibitory effect. Together, these findings suggest that combining FC and CA may provide a promising dual-functional strategy for early root caries management while contributing to SDGs 3 and 12.

Keywords: Coral Apatite, Fossilized Coral, Root Caries, Demineralization Inhibition, Bacterial Adsorption

Presenter ID: OP11

Intervention Effect of Risk-Stratified Caries Management in 3-Year-Old Children

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Background: Dental caries in 3-year-old children remains a significant public health concern. This study therefore aimed to explore a risk-based precise prevention and control model to reduce caries incidence. **Methods:** A stratified random cluster sampling method was adopted, and a total of 5737 children aged 3 years were enrolled from 16 districts in Shanghai, including 3361 children in the intervention group and 2376 in the control group. Caries risk assessment classified intervention children into low-, medium-, and high-risk groups. Stratified management was implemented in accordance with risk levels, with differentiated follow-up frequencies and topical fluoridation intervention schedules formulated correspondingly. Meanwhile, on-campus oral examinations and targeted oral health education were carried out. The control group received routine oral health prevention measures including oral examinations, fluoridation treatment once or twice a year and general health education. **Results:** Among the 3-year-old children, the proportions of high-risk, medium-risk and low-risk dental caries were 31.69%, 3.80% and 64.53% respectively. After one year of intervention, the incidence rate of dental caries was 12.69% in the intervention group, which was significantly lower than 16.93% in the control group ($P < 0.001$). The awareness rate of oral health knowledge among parents increased from 88.20% at baseline to 91.00% at the end of the intervention, and the proportion of parents with positive oral health attitudes rose from 87.08% to 88.40% ($P < 0.05$). **Conclusion:** Targeted stratified interventions can effectively reduce the incidence of dental caries, which is worthy of further promotion and application.

Keywords: Children, Dental Caries, Risk Assessment, Stratified Management

Presenter ID: OP12

Comparative Infodemiological Analysis on Public Perceptions of Fluoride: TikTok vs Douyin

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Objective: To explore public perceptions of fluoride in dental/public health contexts through a comparative infodemiological analysis of videos and associated comments on TikTok and Douyin. **Methods:** Video search was conducted on TikTok and Douyin using the term 'fluoride'. Video's metadata and comments were collected via Python. Sentiment analysis, Latent Dirichlet Allocation topic modeling, word co-occurrence and frequency analysis were performed. **Results:** A total of 102 eligible videos (TikTok:49; Douyin:53) and 23,386 comments (TikTok: 7476; Douyin: 15910) were analyzed. Video metadata did not differ significantly between the two platforms ($p > 0.05$), but content varied. On TikTok, more videos mentioned non-professional used fluoride products, while equal proportion of videos of professional and non-professional used fluoride products on Douyin. TikTok videos more often addressed harms to the nervous and endocrine systems, while Douyin videos tended to focus on dental and skeletal fluorosis. Comments on the two platforms differed significantly by upload year, uploader, format, length, and fluoride source ($p < 0.001$). The mean ($\pm$ standard deviation) sentiment scores of comments on TikTok (6.8 ± 39.5) and Douyin (7.3 ± 38.4) were overall positive without significant difference ($p > 0.05$). Nevertheless, distributions of comment sentiment scores differed significantly by fluoride source, uploader, format, video length, and year ($p < 0.001$). On TikTok, comments on dentist-uploaded videos showed a higher sentiment score compared with Douyin (16.1 ± 38.6 vs. 4.4 ± 34.8 , $p < 0.001$). **Conclusion:** An overall positive attitude towards fluoride on both TikTok and Douyin, but sentiment patterns differed. It highlights the sociocultural diversities among populations where customized strategies should be considered in the dissemination of health-related information.

Keywords: Infodemiology, Social Media, Dental Health, Public Health, Fluoride, TikTok, Douyin

Presenter ID: OP13

Oral Hygiene During Fixed Appliance Treatment: A TikTok and YouTube Netnography

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Background: Maintaining good oral hygiene is essential during fixed appliance orthodontic treatment, yet patients may experience practical difficulties, uncertainty, and reduced confidence in effectively cleaning around brackets and archwires. As YouTube and TikTok are increasingly used as sources of orthodontic information, this study aimed to explore viewer comments posted on selected videos relating to oral hygiene practices during fixed appliance orthodontic treatment. **Methods:** An exploratory, focused, non-participant netnographic design was used. Publicly available comments were collected from nine videos, comprising four YouTube and five TikTok videos. The videos were selected independently by two orthodontists based on engagement and relevance to oral hygiene during fixed appliance treatment. A total of 3,061 non-empty comments were screened. Following exclusion of emoji-only, tag-only, non-English, duplicate, and low-information comments, 2,151 usable comments were analysed using inductive thematic analysis. **Results:** Five themes were identified: (1) videos as practical educational support; (2) negotiating braces hygiene routines, tools, and techniques; (3) burden and barriers to maintaining oral hygiene with braces; (4) emotional and bodily responses to braces hygiene; and (5) anticipated consequences of poor hygiene, including peer exchange of coping strategies. TikTok generated higher engagement and shorter, more reactive comments, whereas YouTube comments were generally longer and more explanatory. **Conclusion:** Viewer comments revealed persistent practical, informational, and emotional challenges in maintaining oral hygiene during fixed appliance treatment. Social media may supplement orthodontic education, but clearer, repeated, and patient-centred guidance remains important to support patients beyond the clinical appointment.

Keywords: Orthodontic, Oral Hygiene, Social Media

Presenter ID: OP16

Tobacco Use Patterns, Smoking Behaviour, and Cessation Attempts Among Malaysian Armed Forces Personnel

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Background: Tobacco use is substantially higher among military personnel than in the general population. Military operational demands, peer influence, anxiety, boredom, sleep deprivation, and challenging environments may contribute to smoking behaviour. This survey aimed to determine the characteristics of smoking behaviour among Malaysian Armed Forces (MAF) personnel. **Methods:** A national epidemiological survey was conducted in 2023 among active MAF personnel from the Army, Navy, and Air Force. A multistage sampling method using stratified, proportional, and simple random sampling was applied. Data were collected through a self-administered questionnaire, with a calculated sample size of 2,216. Descriptive statistics were performed. **Results:** A total of 2,625 personnel participated. The prevalence of smokers, non-smokers, and ex-smokers was 51.0%, 33.4%, and 15.6%, respectively. Among smokers, 79.8% used conventional cigarettes, while 18.7% reported e-cigarette use, particularly among younger personnel. The mean number of cigarettes smoked daily was 8.1 sticks, with most smokers showing low nicotine dependence. Half of the smokers had attempted to quit, but only 10.8% received professional advice. Among ex-smokers, 58.8% quit without professional help. Additionally, 80% of non-smokers reported exposure to second-hand smoke. **Conclusion:** Smoking prevalence among MAF personnel remains high, with increasing e-cigarette use and dual tobacco use requiring attention. These findings provide evidence to support tobacco cessation policies and risk control strategies within the Malaysian Armed Forces.

Keywords: Tobacco Use, Cessation Attempts, Military Personnel, National Survey

Presenter ID: OP17

Patient Perception towards Dental Visits during COVID I-19 Pandemic: A Cross-Sectional Study

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Background: This study evaluated the perception of patients or caregivers towards dental visits at the Special Care Dentistry clinic, Queen Elizabeth Hospital during the COVID-19 pandemic, assessed their acceptance level towards postponement of dental appointments, and preferences towards alternative dental services. **Methods:** This cross-sectional study collected data via telephone interview using a validated 5-point Likert scale questionnaire. Information collected included patients' sociodemographic profiles and their perceptions towards dental visits during the pandemic and analysed using descriptive statistics. **Results:** A total of 190 respondents (98 patients, 92 caregivers; 33.2% male, 66.8% female; mean age 52.62) with half had education up to secondary school. Most of them did not have children aged 0-4years old (74.4%) and had at least one comorbidity (67.4%). Majority of the participants perceived that COVID-19 is infectious and were afraid of being infected during dental visits (97.3%). However, up to 81.1% of participants are confident in the infection control/standard operating procedure of dental clinics. Many do not think that their dental problem will worsen if a dental visit is postponed (61.6%). Regarding alternative care, participants preferred teledentistry (95.8%), domiciliary care (61.6%), and rescheduling appointments (55.8%), but rejected transferring to other private (84.7%) or government (86.3%) clinics. **Conclusion:** Our survey has shown the majority of the participants were afraid of being infected with SARS-CoV-2 at dental clinics during the pandemic. Most participants preferred to postpone their appointments and did not perceive that their oral health would worsen with delayed dental visits.

Keywords: Pandemic, Dental response, Special Care Dentistry

Presenter ID: OP18

Dietary Pattern as a Risk Factor for Dental Caries among Primary School Children

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Background: Dental caries remains a major oral health problem among school-aged children. Twelve years old children as a preadolescent children is the global indicator age for assessing dental caries. The dietary habit related to dietary patterns of children, such as consuming sugary foods and fermented carbohydrates increase the risk of dental caries. The aim of this study is to analyze the dietary habit related to dental caries prevalence among preadolescents in elementary school children in Selaparang District, Mataram City, Indonesia. **Methods:** The cross-sectional study included 373 preadolescent children from eight elementary schools in Selaparang District Cariogenic. Dietary patterns were assessed using a semi-quantitative food frequency questionnaire (SQ-FFQ) with high inter-enumerator reliability ICC = 0.97, and test-retest reliability (ICC = 0.984; 95% CI: 0.959–0.994), and dental caries status was examined by two calibrated dentists with an inter-examiner kappa coefficient of 0.828, and evaluated using the DMFT index. The association between cariogenic dietary patterns and dental caries status was analyzed using Chi-square test. **Results:** The prevalence of dental caries was 83.9%, with 187 children (50.1%) frequently consuming cariogenic foods and beverages, 186 (49.9%) in frequent. Children who frequently consumed cariogenic foods and beverages had significantly higher odds of dental caries than those who consumed them in-frequently (odds ratio = 2.74; 95% CI: 1.51- 4.97; p < 0.001). **Conclusion:** These results highlight that frequent consumption of cariogenic foods and beverages was significantly associated with dental caries among preadolescent children age 12-year-old in elementary school Selaparang District, Mataram City.

Keywords: Diet Cariogenic, Dietary Patterns, Dental Caries, Preadolescent Children

Presenter ID: OP53

Teacher Empowerment Training for Dental Health Education Using Pop-Up Books and Snakes-Ladders at SRMP Cimahi

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Background: The prevalence of dental caries in children is 60–80% globally (WHO) and 82.8% in Indonesia (2023 Indonesian Health Survey). At SRMP 8 Cimahi, the DMFT index was 4.47 (bad) and 60.3% of students had a moderate OHI-S. Dental caries is a multifactorial disease influenced by knowledge, behavior, oral hygiene practices, and environmental factors. Low socioeconomic status is often influenced by a person's education regarding oral health and contributes to unhealthy eating and toothbrushing habits, thereby increasing the risk of dental caries. **Methods:** A cross-sectional study was conducted at SRMP 8 Cimahi involving 68 participants using total sampling. Teachers received a Training of Trainers (ToT) in oral health education using pop-up books and snakes and ladders. Oral health education was delivered to students by trained teachers. Primary data on participants' knowledge were collected using SPSS. **Results:** Statistical tests showed that 58 students had better knowledge after the intervention, while only 10 students had the same level of knowledge before and after. The Wilcoxon test revealed a significant difference before and after receiving education using a pop-up book and playing Snakes and Ladders ($p=0.000$, $p<0.05$). OHI-S also improved significantly ($p=0.000$, $p<0.05$), with the poor category decreasing from 20 to 6. **Conclusion:** The Snakes and Ladders game with education from the teacher using pop-up book media positively improved students' knowledge regarding dental and oral health.

Keywords: Training on Trainer, Pop-up Book, Snakes and Ladders

Presenter ID: OP70

A Survey on the Competency of Dental Officers in Orthodontic Referral

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Background: Index of Orthodontic Treatment Need (IOTN) is an important screening tool for dental officers to prioritize patients for public care. This study aimed to assess dental officers' competency in using the IOTN and its correlation to training status, types of training, gender and years of practice. **Methods:** This cross-sectional study involved the distribution of validated questionnaires to all clinical dental officers under Kedah State Health Department. Mean IOTN knowledge score was calculated based on responses to IOTN related questions. Linear regression was used to examine the association between knowledge scores with study variables. Challenges and suggestions for improving future training were recorded and analysed using qualitative methods. **Results:** A total of 298 respondents participated in this study, the majority of whom were females (78.9%). More than half of the respondents had less than 5 years of clinical experience (57.7%). Mean IOTN knowledge score was 0.611 (0.582, 0.640). Most of the respondents utilised IOTN (99.7%) and 54.5% used it to grade potential case complexity. Majority of respondents received some training (81.5%) during their undergraduate education (59.7%). Most respondents feel there should be more IOTN training (98.2%). Type of training and gender were found to be associated with the knowledge score of IOTN. Challenges faced include knowledge and training gaps, diagnostic variability, operational and communication barriers. Suggested strategies to overcome the challenges included continuing education and standardization. **Conclusion:** Dental officers demonstrated moderate competency in the use of the IOTN for orthodontic referrals. Strengthening continuing education may help address existing knowledge gaps, reduce diagnostic variability and improve referral accuracy.

Keywords: Orthodontic Referral, Index of Orthodontic Treatment Need (IOTN), Dental Officers, Competency, Orthodontic Screening, Training

Presenter ID: OP71

Access as Prevention: PWD Accessibility Audits at the University of the Philippines College of Dentistry

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Background: Preventive oral health relies on inclusive systems, yet people with disabilities continue to face barriers that result in poorer oral health outcomes compared to the general population. This study assessed the compliance of the University of the Philippines College of Dentistry's (UPCD) facilities with the Accessibility Law of the Philippines; and assessed the knowledge of its clinicians, clinical instructors, and staff on the use of Filipino Sign Language (FSL). **Methods:** This cross-sectional study employed a dual-audit methodology. To assess physical access, researchers navigated the UPCD facilities in a wheelchair alongside experts from the United Architects of the Philippines - Architects for Accessibility. Features such as ramps, entrances, and toilets were scored based on national regulatory standards. Furthermore, an online questionnaire developed with Deaf and Terp Corporation was used to measure the knowledge on the use of FSL of 99 clinicians, clinical instructors, and staff. **Results:** The UPCD building attained 55 out of 100 points, resulting in a 3-star accessibility rating and identification of significant gaps in facilities. Additionally, 65 respondents received low scores in the FSL knowledge assessment, indicating a possible communication barrier within the clinical environment. **Conclusion:** The UPCD requires enhancement in both physical aspects and the knowledge on the use of FSL among its constituents. To help empower marginalized communities, dental education and training must bridge the gap between policy and practice by integrating disability awareness programs and FSL training into the curriculum. Establishing inclusive environments is a fundamental prerequisite for delivering equitable preventive oral health care.

Keywords: Oral Health Services Accessibility, Health Service Delivery, Persons with Disabilities, Physical Accessibility, Filipino Sign Language

Presenter ID: OP20

Characterizing Respiratory Pathogen Profiles on Removable Prostheses with Different Hygiene Status in Long-Term Care Facilities

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Background: Biofilms on removable dental prostheses are potential reservoirs for respiratory pathogens. In frail long-term care facility (LTCF) residents, the relationship between prosthesis hygiene status and respiratory pathogen profile remains unclear. Therefore, this study aimed to compare respiratory pathogen prevalence and Opportunistic Respiratory Pathogenic Index (ORPI) between residents with 'clean' and 'unclean' prostheses. **Methods:** This cross-sectional study recruited 160 removable prostheses wearers from 19 LTCFs in Hong Kong SAR. Participants were categorized into Clean (n=81) and Unclean (n=79) groups using a 50% plaque percentage coverage threshold. Plaque samples were analyzed using the Type IIB Restriction-site Associated DNA Sequencing (2bRAD-M). Respiratory pathogen prevalence and ORPI were compared between groups based on a reference database for opportunistic respiratory pathogens. The ORPI was calculated as the cumulative relative abundance of all identified pathogens. **Results:** Forty-nine opportunistic respiratory pathogens were identified, with all samples harboring at least two species. The overall prevalence levels were similar between groups ($p=0.591$), with *Prevotella denticola* most prevalent (95.0%). Group-specific analysis revealed *Prevotella denticola*, *Arachnia propionica* and *Stenotrophomonas maltophilia* as most prevalent in the Clean group, versus *Prevotella denticola*, *Arachnia propionica* and *Streptococcus agalactiae* in the Unclean group. Notably, the ORPI was significantly higher in the Unclean group than the Clean group ($p=0.013$). **Conclusion:** These findings indicate that respiratory pathogens are prevalent on removable prostheses in LTCF residents, with higher ORPI on unclean prostheses. Therefore, effective prosthesis hygiene to reduce the cumulative opportunistic pathogen burden is essential in this vulnerable population.

Keywords: Respiratory Pathogens, Long-term Care Facilities, Removable Dental Prostheses, Plaque

Presenter ID: OP21

Hyperlipidemia Accelerates Aging of Pulpo-dentinal Complex by Inflammatory Response Stimulation

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Background: Systemic lipid metabolism may stimulate the inflammatory response in local microenvironment, and subsequently influence the cellular senescence and tissue aging. The role of hyperlipidemia in pulpo-dentinal complex was still lack of research evidence. **Methods:** A hyperlipidemia mouse model was established with a high-fat diet for 12 weeks. The thickness of dentine in the incisor was investigated by microCT and HE staining. The microhardness of dentine was tested by a Vickers hardness instrument. SEM was used to observe the microstructure of dentine. The changes of dentine collagen and the degree of pulp fibrosis were analyzed by irius red staining. The expression of CD36 and inflammatory factors were appraised by qRT-PCR and immunofluorescent staining. **Results:** In hyperlipidemia mouse, the incisor dentine was thickened while the dentine microhardness was decreased. SEM showed increased cracks and shrunken dentinal tubule in dentine of hyperlipidemia mice. The type I collagen was mostly replaced by type III collagen in incisor dentin caused by hyperlipidemia. A higher degree of fibrosis in dental pulp were manifested in hyperlipidemia mice. Furthermore, the lipid transporter CD36 was increased in odontoblasts, and subsequently, inflammatory mediators such as TNF- α and IL-1 β were upregulated in hyperlipidemia mice. **Conclusion:** The present study demonstrated that in hyperlipidemia mice, the age-related changes of pulpo-dentinal complex were accelerated by CD36 elevation and inflammatory response stimulation.

Keywords: Hyperlipidemia, Pulpo-dentinal Complex, Aging, Inflammation

Presenter ID: OP22

LPS-Induced Inflammation Upregulates Gene Expression and Alters Primary Cilia Length in Human Periodontal Ligament Fibroblasts

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Background: Primary cilia (PC) are single protruding tubules on mammalian cells that regulate key intracellular signalling pathways for tissue homeostasis. While defective primary cilia cause craniofacial and dental abnormalities, their role in periodontal ligament fibroblasts (PDLFs) during periodontitis remains largely unexplored. **Objectives:** (1) establish and characterise PDLFs from healthy teeth; (2) evaluate the effect of various concentrations of *P. gingivalis* Lipopolysaccharide (PG-LPS) (0 ng/mL, 10 ng/mL, 100 ng/mL, 1 μ g/mL) on gene expression of KIF3A, IFT88 and TNF- α at 48 and 72 hours; (3) determine the incidence and length of PDLF's PC when exposed to PG-LPS at 48 and 72 hours. **Methods:** PDLFs were isolated using the explant method and characterised by immunofluorescence staining for vimentin. PDLFs were seeded at 2.5×10^5 and subjected to serum starvation to induce ciliogenesis, then exposed to PG-LPS. Gene expression was analysed by real-time qPCR, and the incidence and length of PC were quantified by co-immunofluorescence staining for ARL13B (axoneme marker) and γ -tubulin (basal body marker). **Results:** Exposure to 100 ng/mL PG-LPS resulted in the most significant fold change in gene expression for all three targets: KIF3A, IFT88, and TNF- α at both 48 and 72 hours. A significantly higher incidence of ciliated cells ($p \leq 0.05$) and significantly shorter mean PC ($p \leq 0.05$) were observed in PG-LPS-treated PDLFs compared with untreated controls at both 48 and 72 hours. **Conclusion:** PG-LPS infection increases the expression of ciliary and inflammatory genes and alters ciliary incidence and length of PDLF in vitro.

Keywords: Primary Cilia, Periodontal Ligament Fibroblasts, *Porphyromonas gingivalis* LPS, Periodontitis, Primary Cilia Gene Expression, Primary Cilia Frequency, Primary Cilia Length

Presenter ID: OP23

Screening of Bioactive Compounds from *Paris Polyphylla* var. *Yunnanensis* to Prevent *Streptococcus Mutans* Biofilm Formation

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Background: Dental caries represents a major global oral health issue, primarily caused by *Streptococcus mutans* (*S. mutans*). The new antibacterial drugs used for the prevention and treatment of dental caries need further exploration. This study aims to extract and isolate the bioactive compounds of *Paris polyphylla* var. *yunnanensis* against *S. mutans* biofilm formation, and to explore their effect and mechanism. **Methods:** Rhizomes of *Paris polyphylla* var. *yunnanensis* were extracted using water and ethanol gradients (30%-90%) via ultrasonication. Active fractions were screened through minimum inhibitory concentration (MIC) assays and crystal violet (CV) staining. Active fractions were further isolated using liquid-liquid extraction and column chromatography, followed by UPLC/MS-MS identification. Computer aided drug design (CADD) was employed to predict the potential anti-caries compounds based on their binding affinity to the target protein glucosyltransferase C (GtfC), then these compounds were subsequently validated through MIC and CV staining. **Results:** UPLC/MS-MS analysis identified isospirostanol-type steroidal saponins as the primary constituents of *Paris polyphylla* var. *yunnanensis*. Molecular docking revealed several compounds with high binding affinities for GtfC, specifically gracillin (-11.34 kcal/mol), polyphyllin II (-10.918 kcal/mol), polyphyllin I (-9.798 kcal/mol), and diosgenin (-8.877 kcal/mol). Experimental validation confirmed that these compounds significantly inhibited *S. mutans* biofilm formation ($P < 0.05$). **Conclusion:** Gracillin, polyphyllin II/I, and diosgenin are key anti-caries compounds from *Paris polyphylla* var. *yunnanensis*, offering a promising new strategy for the prevention and pharmacological management of dental caries.

Keywords: Dental Caries, *Streptococcus Mutans*, *Paris Polyphylla* var. *Yunnanensis*, Molecular Docking

Presenter ID: OP24

Effect and Mechanism of *Paris Polyphylla* var. *Yunnanensis* Extract on Inhibiting Adhesion of *Streptococcus Mutans*

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Background: Adhesion is a fundamental process by which *Streptococcus mutans* (*S. mutans*) exerts its cariogenicity and causes dental caries. We previously demonstrated that *Paris polyphylla* var. *yunnanensis* effectively inhibits biofilm formation by *S. mutans*. However, the precise mechanisms and inhibitory effects of its on bacterial adhesion remain to be investigated. This study aims to preliminarily explore the inhibitory effect of the extract of *Paris polyphylla* var. *yunnanensis* on *S. mutans* adhesion and the underlying mechanisms. **Methods:** The adhesion inhibition assay was employed to the extract of *Paris polyphylla* var. *yunnanensis* against *S. mutans*. The effects of the extract of *Paris polyphylla* var. *yunnanensis* (0.5-2mg/mL) on cells surface hydrophobicity, biofilm viability, exopolysaccharide (EPS) production, and biofilm architecture were evaluated using hydrophobicity index, confocal laser scanning microscopy (CLSM), and scanning electron microscopy (SEM). Real-time quantitative reverse transcription polymerase chain reaction (RT-qPCR) was performed to assess changes in the expression of adhesion-related virulence genes. An overexpression strain of the key adhesion-related target gene was constructed and analyzed. Data analysis was conducted using Graphpad Prism 9.5.1. **Results:** The extract of *Paris polyphylla* var. *yunnanensis* significantly reduced the hydrophobicity index of *S. mutans* ($P < 0.0001$), disrupted biofilm architecture, decreased viable bacteria and EPS synthesis ($P < 0.0001$). Furthermore, the extract was found to dynamically regulated adhesion-related virulence genes. The UA159/pDL278-*vicR* showed enhanced adhesion, viable bacteria, EPS synthesis, and up-regulated *vicR*, *gtfB*, *gtfC* and *gtfD* expression ($P < 0.05$). Notably, the extract dynamically regulated these genes in the UA159/pDL278-*vicR*. **Conclusion:** The extract of *Paris polyphylla* var. *yunnanensis* could inhibit *S. mutans* adhesion through a multi-target synergistic mechanism, and the *vicR* could serve as a key potential target.

Keywords: *Streptococcus Mutans*, *Paris Polyphylla* var. *Yunnanensis*, Adhesion, Gene Expression

Presenter ID: OP25

Capability and Mechanism of Melanin Synthesis in *Arachnia Rubra* E10012 and *Arachnia Propionica* E1715

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Background: Extrinsic black stain (EBS) is characterized by pigment deposition on primary teeth. *Arachnia rubra* E10012 and *Arachnia propionica* E1715 were the microbial biomarkers for EBS. However, their roles in pathogenesis remain unclear. This study investigated their capability and underlying mechanism of melanin production. **Methods:** Melanin-like substances were extracted from the fermentation supernatants of the two strains using the alkali-soluble acid-precipitation method. Their physicochemical properties were evaluated through solubility testing, FeCl₃ assay, ultraviolet-visible absorption analysis, oxidation and reduction resistance tests, thermal stability testing, light stability testing, and DPPH radical scavenging assays. Fourier transform infrared spectroscopy and pyrolysis gas chromatography–mass spectrometry were employed to analyze their microstructure. In addition, recombinant HppD expression systems derived from the two strains were constructed, and the proteins were purified. An in vitro reaction system was established to monitor changes in absorbance under different substrate concentrations and reaction times. **Results:** Both extracts exhibited the characteristic properties of melanin, including alkali solubility and acid precipitability, insolubility in water, acidic solutions, and most organic solvents. They formed reddish-brown flocculent precipitates in the presence of FeCl₃, and showed stability, oxidation resistance, and a certain degree of free radical scavenging activity. Aromatic pyrolysis products such as benzene and phenol were identified. Recombinant HppD proteins derived from the two strains were successfully expressed and purified. In the Fe²⁺ containing vitro reaction system, changes in absorbance were observed with variations in 4-hydroxyphenylpyruvate concentration and reaction time. **Conclusion:** Both *Arachnia rubra* E10012 and *Arachnia propionica* E1715 were found to produce melanin. The HppD-related pathway may participate in the conversion of melanin precursors.

Keywords: Extrinsic Black Stain, Melanin, *Arachnia*, 4-Hydroxyphenylpyruvate-Dioxygenase

Presenter ID: OP27

A Non-Discoloring Bioactive Hydrogel for Caries Management: An In Vitro Study

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Background: Silver-based anti-caries agents are effective but can cause tooth discolouration. **Methods:** The fCS/nSPS hydrogel was developed by incorporating nanostrontium phosphosilicate (nSPS) into a functionalised chitosan (fCS) matrix, followed by ultraviolet-induced photocrosslinking. The hydrogel's phase formation and composition were identified by X-ray Diffraction spectroscopy (XRD), whereas its functional groups were detected by Fourier Transform Infrared spectroscopy (FTIR). Surface morphology was observed by Scanning Electron Microscopy (SEM). Antibacterial effects of fCS/nSPS were evaluated using minimum inhibitory concentration (MIC) and minimum bactericidal concentration (MBC). Mineralising properties were studied using SEM and XRD on fCS/nSPS-treated dentine blocks after cariogenic challenge. Cytocompatibility of fCS/nSPS-treated dentine was tested towards human gingival fibroblasts (HGF-1). Dentine discolouration after intervention (ΔE) was quantified using colour spectrophotometry. Silver diamine fluoride (SDF) and deionised water were used as controls. **Results:** Nuclear Magnetic Resonance (NMR), XRD, FTIR, and SEM confirmed the synthesis of fCS/nSPS. The MIC/MBC values (mean $\pm$ SD) of fCS/nSPS against *Streptococcus mutans*, *Actinomyces naeslundii*, and *Lactobacillus acidophilus* are 87 $\pm$ 20/200 $\pm$ 35, 129 $\pm$ 38/373 $\pm$ 80, and 110 $\pm$ 36/253 $\pm$ 53 μ g/mL, respectively. SEM revealed a smooth mineralised surface on fCS/nSPS-treated dentine. XRD confirmed deposition of strontium phosphosilicate. HGF-1 viability (mean $\pm$ SD) of the fCS/nSPS, SDF and Water groups were 135.7 $\pm$ 4.9%, 0.5 $\pm$ 0.1% and 131.7 $\pm$ 3.5% ($p < 0.001$, SDF < fCS/nSPS, Water). The ΔE values (mean $\pm$ SD) of fCS/nSPS, SDF, and water groups were 6.4 $\pm$ 2.6, 47.7 $\pm$ 6.6, and 7.5 $\pm$ 2.0, respectively ($p < 0.001$; fCS/nSPS, Water < SDF). **Conclusion:** The fCS/nSPS hydrogel caused no significant tooth discolouration, exhibited antibacterial activity, promoted dentine's surface mineral deposition, and was cytocompatible, supporting further investigation for non-restorative caries management.

Keywords: Caries, Nanoparticles, Mineralisation, Biomaterials, Hydrogel

Presenter ID: OP28

Biomimetic Remineralization with Novel Dual-Function Gels

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Background: Growing concerns about dental fluorosis, systemic toxicity, and cumulative fluoride exposure have sparked interest in biomimetic, fluoride-free alternatives to traditional treatments. Bioactive materials such as nanohydroxyapatite (nHA) and tricalcium phosphate (TCP) may enhance enamel remineralization by mimicking the natural mineral phase of teeth. This study evaluated the remineralization potential of novel dual-function gels (DFGs) compared with fluoride-based varnishes. **Methods:** Sixty demineralized human enamel specimens were randomly allocated into six groups (n=10): DFG-1 containing nHA, DFG-2 containing TCP, DFG-3 containing nHA+TCP, Clinpro™ White Varnish (CWV; fluoride-containing functionalized TCP), Clinpro™ Clear Fluoride (CCF; water-based fluoride varnish), and untreated negative control group (NTG). The specimens underwent a 10-day pH-cycling protocol. The DFGs were applied daily, whereas CWV and CCF were applied once before cycling. Surface microhardness was measured at baseline, post-demineralization, and post-treatment to calculate percentage hardness recovery (%HR) and remineralization potential (%RP). Data were analyzed using ANOVA and the Games–Howell test ($\alpha = 0.05$). **Results:** Significant differences in %HR and %RP were observed among groups ($p < 0.05$). DFG-3 demonstrated the highest %HR and %RP values (7.17 and 18.61), followed by DFG-1 (6.90 and 18.25), CWV (5.48 and 15.46), DFG-2 (6.58 and 12.98), CCF (2.23 and 9.15), and NTG (-7.66 and -7.07). No significant differences were observed among DFG-1, DFG-2, DFG-3, and CWV ($p > 0.05$), whereas all showed significantly greater remineralization than CCF ($p < 0.05$). **Conclusion:** Novel DFG formulations demonstrated remineralization efficacy comparable to CWV and superior to CCF, suggesting their potential as effective biomimetic agents for non-invasive management of early carious lesions.

Keywords: Remineralization, Bioactive Materials, Nanohydroxyapatite, Biomimetic Dentistry, Dual-function Gel

Presenter ID: OP29

Human Deciduous Pulp Stem Cells with CRISPR/Cas9 for Duchenne Muscular Dystrophy Mechanisms

Cui Zhang, Xiaoyong Ren

Background: Duchenne muscular dystrophy (DMD) is caused by mutations in the DMD gene on the X chromosome, and rehabilitation and steroid therapy do not halt disease progression. Although CRISPR/Cas9 gene editing offers potential, viral vectors and stem cell approaches face challenges, including low delivery efficiency, immune rejection, and ethical concerns. Human deciduous pulp stem cells (hDPSCs), derived from exfoliated primary teeth, exhibit proliferative and immunomodulatory capacities than bone marrow-derived mesenchymal stem cells (MSCs), with minimal ethical concerns and low immunogenicity, making them promising carriers for gene editing in children aged 5–12 years. This study evaluated whether hDPSCs could serve as an efficient delivery platform for CRISPR/Cas9 gene editing to restore DMD function and dystrophin expression in a murine model of DMD. **Methods:** We established mdx mice as a classic DMD model and divided animals into four groups: hDPSCs alone, CRISPR/Cas9 alone, combined hDPSCs + CRISPR/Cas9, and controls (PBS/blank). Tail-vein injection was performed, with dynamic evaluation at day 3, 1, 2, 4, and 8 weeks. Molecular repair was assessed by PCR; dystrophin restoration by Western blot and immunofluorescence; and muscle pathology by HE staining of gastrocnemius and quadriceps. **Results:** Combined therapy showed dual benefits. Genetically edited hDPSCs homed to damaged muscle, crossed defective sarcolemma, and reduced fibrosis through paracrine effects, downregulating TNF- α and IL-6 while upregulating IL-10. CRISPR/Cas9 mediated exon skipping/restoration, and the combination group restored dystrophin protein expression most effectively. **Conclusion:** The 'oral stem cells + gene editing' strategy provides a precise, ethical, and preventive approach for DMD. Dentists may evolve from observers to providers of seed cells for systemic diseases, expanding the role of paediatric preventive dentistry in Asia.

Keywords: DMD, hDPSCs, CRISPR/Cas9, Immunogenicity

Presenter ID: OP30

Influence of Adhesive Bonding and Lithium Disilicate Thickness on The Final Color of Ultra-Thin Veneers

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Background: Ultra-thin veneers support minimally invasive restorative treatment by preserving tooth structure; however, their limited thickness may reduce the concealing ability. Although adhesive bonding agents may influence the optical performance of ceramic restorations, their effect on the final color of ultra-thin veneers remains unclear. This study aimed to evaluate the influences of the adhesive bonding agent, lithium disilicate thickness, and their interaction on the final color of ultra thin veneers. **Materials and Methods:** Forty-five lithium disilicate discs (shade A1; diameter, 8 mm; thicknesses, 0.3, 0.6, and 0.9 mm; n = 5/group) were fabricated and cemented onto A1 resin composite substrates using translucent resin cement alone (control) or together with Adper Single Bond 2 (ASB), Single Bond Universal (SBU), or Scotchbond Universal Plus (SUP). Color coordinates (CIE L*a*b*) were measured using a spectrophotometer, and color differences (ΔE) relative to the control were calculated. Data were analyzed using two-way ANOVA followed by Bonferroni post hoc tests ($\alpha = 0.05$). **Results:** Adhesive bonding agent, ceramic thickness, and their interaction significantly affected ΔE ($p < 0.05$). ASB produced the smallest color difference. Increasing ceramic thickness significantly improved color masking, with 0.9-mm veneers exhibiting the lowest ΔE . Within the ultra-thin (0.3 mm) group, ASB yielded the lowest ΔE . **Conclusion:** Both adhesive bonding agent and ceramic thickness significantly influence the final color of lithium disilicate veneers. A 0.9-mm veneer provided the greatest concealing ability, whereas ASB minimized color alteration across thicknesses and produced the most favorable color outcome in ultra-thin (0.3-mm) veneers, supporting its use in minimally invasive esthetic restorations.

Keywords: Lithium Disilicate, Adhesive Bonding Agent, Ceramic Thickness, Tooth Structure Preservation, Minimally Invasive Veneer, Ultra-thin Veneer, Esthetics

Presenter ID: OP31

Comprehensive Analysis of Occupational Burnout, Emotional Intelligence, and Depression Among Dentists in Northwestern China

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Background: High occupational stress threatens dentists' mental health, yet few regional data exist for Northwest Chinese dental practitioners. This study explored correlations between occupational burnout, emotional intelligence (EI), depression, and the moderating effect of EI on work stress-burnout associations to guide mental health interventions. **Methods:** A cross-sectional convenience sampling survey recruited licensed dentists with ≥ 1 year practice from 5 northwest provinces. 1100 questionnaires were distributed; 900 valid responses were collected. Surveys included demographic forms and three validated scales: Maslach Burnout Inventory-General Survey (MBI-GS), Depression Anxiety Stress Scale-21 (DASS-21), Connor-Davidson Resilience Scale (CD-RISC). Ethics approval (2025-XJKQIEC-KY-QT-0047-002) was obtained. SPSS 27.0 and R 4.3.1 ran Pearson correlation and hierarchical regression, with two-tailed $P < .05$ as significance threshold. **Results:** Respondents (81.8% response rate) mostly were female (64.9%), aged 26–35 (39.6%), bachelor's degree or below (77.2%). Mean burnout score = 3.93 ± 0.94 , EI = 1.62 ± 0.54 , depression = 3.55 ± 0.76 . Burnout positively correlated with EI ($\beta = 0.31$, $P < .001$); daily patient volume ($\beta = 0.12$, $P < .01$) and weekly working hours ($\beta = 0.15$, $P < .001$) worsened burnout. EI exerted a significant negative moderating effect ($\beta = -0.10$, $P < .01$), buffering work stress' damage to personal accomplishment. Teaching duties reduced burnout ($\beta = -0.11$, $P < .05$). **Conclusion:** Northwest Chinese dentists face severe burnout and depressive symptoms driven by heavy workloads. Higher EI mitigates stress-related occupational harm. Hospitals should optimize patient scheduling and implement emotional intelligence training to improve dental providers' long-term mental well-being.

Keywords: Occupational Burnout; Emotional Intelligence; Depression; Dentists; Northwestern China

Presenter ID: OP32

Parental KAP, and Caries Experience as Determinants of ECC Among Toddlers in Hulu Langat, Malaysia

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Background: Early childhood caries (ECC) is a prevalent yet preventable condition influenced by parental oral health knowledge, practices, and caries experience. This study aimed to assess parental knowledge, practices and caries experience and to determine the relationship with caries experience among their children aged 1–4 years. **Methods:** A cross-sectional study was conducted among 409 parent–child pairs attending government antenatal healthcare facilities in Hulu Langat. Parents completed validated questionnaires assessing oral health knowledge, attitudes, and practices. Meanwhile, clinical examination recorded both parental and children decayed, missing, and filled teeth (DMFT /dft) scores. Descriptive statistics, chi-square tests, and correlation analyses were used to evaluate associations between the outcomes of interest. **Results:** The majority of participants were Malay (90.7%), aged 25–34 years (56.5%), and from the B40 income group (69.7%). Parents generally demonstrated good oral health knowledge despite inconsistent preventive practices. Although 81.2% of parents recognised the importance of a first dental visit by 12 months of age, only 46.7% had brought their child within period. Delayed first dental visits and delayed cessation of bottle feeding were significantly associated with a higher prevalence of dental caries among children ($p < 0.001$). A weak but statistically significant positive correlation was observed between parental DMFT scores and child dft scores ($r = 0.142$, $p = 0.004$). **Conclusion:** Parental knowledge regarding early childhood oral health was generally satisfactory, however, preventive practices remain suboptimal. Poor parental oral health status and delayed preventive dental visits were associated with a higher prevalence of ECC, highlighting the need for targeted early oral health promotion.

Keywords: Parental KAP, Early Childhood Caries, Toddler, Oral Health

Presenter ID: OP33

An Epidemiological Investigation of Hypomineralized Second Primary Molars in Shaanxi, China

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Background: This cross-sectional study aimed to determine the prevalence of hypomineralized second primary molar (HSPM) in 3–5 year-old children in Shaanxi Province, northwest of China, and identify associated factors, with particular attention to the potential role of maternal vomiting during pregnancy. **Methods:** Children were randomly sampled from Xi'an, Yan'an, and Weinan. HSPM was assessed through oral examinations, and health-related information was collected using questionnaires. Chi-square tests and binary logistic regression were performed to identify independent risk factors. **Results:** A total of 1,022 questionnaires were collected. The sample included 48.97% males and 51.03% females, with regional distributions of 45.71% (Xi'an), 27.78% (Weinan), and 26.51% (Yan'an) of the sample. Teeth with HSPM had a higher caries rate (55.6%) than those without (45.3%), but not significantly. Univariate analysis linked HSPM to gestational age ($p = 0.022$), asthma in the first year of life ($p = 0.03$), measles ($p = 0.003$) and maternal nausea ($p = 0.018$). In the adjusted multivariable model, maternal education (bachelor's degree, OR = 0.297, 95%CI: 0.122, 0.72) and maternal vomiting (OR = 1.694, 95%CI: 1.027, 2.795) associated with HSPM. **Conclusion:** The prevalence of HSPM in Shaanxi falls within the reported global range. Lower maternal education and maternal vomiting during pregnancy were significant risk factors for HSPM, providing new evidence for targeted prenatal prevention strategies.

Keywords: Hypomineralized Second Primary Molar, Prevalence, Pregnancy Vomiting, Maternal Education, Pediatric Dentistry

Presenter ID: OP34

Association between Chewing Difficulty, Probable Depression, and Suicidal Ideation in Korean Adults

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Background: While oral health is linked to psychological well-being, the independent contribution of chewing difficulty to suicidal ideation beyond depressive symptoms remains unclear. This study investigated the age-specific dose-response relationship between chewing difficulty and mental health outcomes in the Korean population. **Methods:** This cross-sectional study utilized data from the Community Health Survey (N=750,962). Probable depression was defined as a Patient Health Questionnaire-9 (PHQ-9) score ≥ 10 , and suicidal ideation via PHQ-9 item 9. Survey-weighted logistic regression, incorporating stratification and clustering, adjusted for socioeconomic status, lifestyle, and clinical comorbidities. Dose-response (5 levels) and age-stratified (19–64 vs ≥ 65 years) analyses were performed. **Results:** Chewing difficulty was significantly associated with depression (aOR: 2.46; 95% CI: 2.36–2.56), with a higher risk in the elderly (aOR: 2.58; 95% CI: 2.43–2.74) than younger adults (aOR: 2.16; 95% CI: 2.06–2.28). After adjusting for non-suicidal depressive symptoms (PHQ-9 items 1–8), chewing difficulty remained a significant independent predictor of suicidal ideation (aOR: 1.44; 95% CI: 1.43–1.44). A significant dose-response relationship showed that depression risk increased as chewing function worsened (P-trend < 0.001). The "Very difficult" group had up to 5.89 times higher odds of depression, with a steeper gradient in older adults (P-interaction < 0.001). **Conclusion:** Chewing difficulty is a potent, independent risk factor for both depression and suicidal ideation. Integrating masticatory rehabilitation with mental health support systems is critical, particularly for older populations vulnerable to oral functional decline.

Keywords: Chewing Difficulty, Depression, Suicidal Ideation, Community Health Survey

Presenter ID: OP35

Frailty and Oral Health-Related Quality of Life in Community-Dwelling Older Adults of Shanghai, China

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Background: Previous studies indicated the associations between oral health and frailty. We aimed to investigate their associations among Chinese community-dwelling older adults in Shanghai. **Methods:** In the cross-sectional study, we enrolled 1122 Chinese community-dwelling older adults for oral examinations and questionnaires in Huangpu District, Shanghai. Oral examinations were conducted according to the World Health Organization's Basic Methods for Oral Health Surveys (5th edition). Frailty was assessed using FRAIL scale and oral-health related quality of life using the Chinese version of OHIP-14 (Oral Health Impact Profile-14). **Results:** Among the 1122 participants, 225 were diagnosed with frailty, accounting for 20.05% of the participants. Age, sex, education, BMI, and number of missing teeth were significantly different between participants with and without frailty (P<0.01). After adjusting for covariates, logistic regression models showed that OHIP-14 was significantly associated with frailty as the dependent variable (OR=1.08; 95% CI 1.07-1.10). An exploratory mediation analysis was conducted with tooth loss as a potential mediator of the association between OHIP-14 and frailty, and the model yielded statistically significant results. The exploratory analysis indicated that tooth loss may partially explain the association between OHIP-14 and frailty, accounting for approximately 11.70% of this association. **Conclusion:** There might be a potential association between oral health assessed by OHIP-14 and frailty, and tooth loss may play a possible partial mediating role in this relationship. Oral health should be a key focus for both the elderly and healthcare providers, especially tooth loss.

Keywords: Older Adults, Missing Teeth, Depression, Cross-Sectional Study

Presenter ID: OP36

Early Childhood Caries Negatively Correlates with Intelligence Quotient in Preschool-Aged Children: A Birth Cohort Study

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Background: Accumulating evidence suggests that early childhood oral health, particularly dental caries, may adversely affect children's cognitive development. However, findings remain inconsistent, especially among preschool-aged populations. This study investigated the associations between early childhood caries and intelligence quotient (IQ) in four-year-old children. **Methods:** This cross-sectional study analyzed data from 1,099 four-year-old children enrolled in the Shanghai Birth Cohort. Oral health was assessed using the decayed, missing, filled teeth (dmft) index and caries status. Child IQ was measured with the Wechsler Preschool and Primary Scale of Intelligence, Fourth Edition (WPPSI-IV). Multivariable linear regression models were employed to estimate the associations between oral health indicators and child IQ, while restricted cubic spline models were used to evaluate non-linear relationships. Subgroup analyses were conducted to examine potential effect modifications by maternal education, breastfeeding status, and child sex. **Results:** After adjusting for potential confounders, higher dmft scores were significantly associated with lower full-scale IQ (FSIQ) ($\beta = -0.23$; 95% CI: $-0.46, -0.01$) and verbal comprehension index (VCI) scores ($\beta = -0.29$; 95% CI: $-0.54, -0.03$). Consistently, children with caries exhibited significantly lower FSIQ and VCI scores than their caries-free counterparts. The associations appeared more pronounced in girls, children with less educated mothers, and those with shorter breastfeeding duration, although no statistically significant interaction effects were observed. **Conclusion:** Dental caries, as quantified by the dmft index, are associated with lower IQ scores in preschool children.

Keywords: Early Childhood Caries, Intelligence Quotient, Preschool Children, dmft Index, Birth Cohort

Presenter ID: OP37

Tobacco Use and Awareness Among High School and Preuniversity Students of Belagavi District, Karnataka, India

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Background: Creating awareness of tobacco among adolescents, who are an easy prey for this habit, is crucial, and hence the current study is planned with the aim of assessing awareness of tobacco among this group. **Methods:** This cross-sectional study was conducted on 1772 participants aged 13- to 18-year-old students of Belagavi district. Information was digitally filled using the KOBO collect app. Information pertaining to tobacco, its contents and harmful effects and habit history were collected, followed by oral examination. At the end of the clinical examination, a standardized awareness talk on 'Ill-effects of tobacco' which emphasized harmful effects of tobacco, oral cancer's signs, symptoms, diagnosis, treatment, and preventive measures was made. Data was transferred to Microsoft Excel and descriptive statistics and chi square test were applied. **Results:** Out of 1,772 participants, 55% were female and 45% were male. Of these, 912 were college students and 860 were school students. Overall, 93% had heard of tobacco, while 87.75% were aware that tobacco is harmful to health. Only 15.12% knew that nicotine causes addiction. None of the participants reported ever using tobacco in any form. Awareness that nicotine causes addiction was significantly higher among college students (64%) than school students (6%) ($p < 0.05$). **Conclusion:** College students had better awareness in comparison to school students. However, none of the participants gave a positive history of tobacco consumption. Further qualitative studies can be undertaken to find out the real picture of the community.

Keywords: Tobacco Awareness, School Students, Preuniversity Students, India

Presenter ID: OP38

The Silver Tsunami: Root Canal Treatment Uptake and its Implications for Malaysia's 60:20 Tooth-Retention Goal

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Background: As Malaysia approaches ageing nation status by 2030, achieving the national “60:20” goal of retaining at least 20 natural teeth at age 60 is a public health priority. Root Canal Treatment (RCT) is a key strategy to prevent tooth loss, yet its uptake remains suboptimal. This study aimed to determine the prevalence of RCT uptake and explore factors influencing patients’ RCT decisions in Malaysian public dental clinics. **Methods:** A two-phase mixed-methods study was conducted across public dental clinics in three districts. Phase 1 assessed RCT uptake among patients with RCT-indicated teeth using universal sampling (n=326). Phase 2 involved in-depth interviews (n=14) via purposive sampling, guided by the Health Belief Model to explore facilitators and barriers influencing treatment decisions. Quantitative data were analysed using multivariable logistic regression, while qualitative data underwent thematic analysis. **Results:** RCT uptake was low (38.0%). Older adults were less likely to accept RCT (aOR=0.10; 95%CI:0.03–0.36). Higher uptake was associated with Chinese ethnicity (aOR=2.65; 95%CI:1.21–5.77), tertiary education (aOR=3.87; 95%CI:2.10–7.12), maxillary anterior teeth (aOR=16.13; 95%CI:6.04–43.10), and greater clinical complexity (aOR=7.64; 95%CI:2.77–21.10). Facilitators included desire to preserve natural teeth, restoration of function and aesthetics, and perceived cost affordability of RCT. Barriers comprised appointment difficulties, preference for immediate pain relief via extraction, and inadequate treatment information. **Conclusion:** Low RCT uptake may hinder progress towards Malaysia’s 60:20 tooth retention goal. Strategies addressing barriers related to treatment burden and patient perceptions are needed to improve acceptance of tooth-preserving treatment and support healthy ageing amidst the silver tsunami.

Keywords: Ageing/Older Adults, Natural Tooth Retention, Malaysia, Public Dental Services, Root Canal Treatment

Presenter ID: OP40

Association between Work-Related Stress and Oral Diseases among Workers in Japan: Inverse Probability Weighting Analysis

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Background: The high prevalence of oral diseases imposes a significant burden on healthcare costs and workforce productivity. Stress is considered to be one of the mechanisms linking social determinants to health. However, few studies have investigated the impact of occupational stress on oral diseases. **Methods:** An online survey was conducted in January 2026. The data of 8,957 workers aged 20–69 years. Oral health-related quality of life (OHRQoL), temporomandibular disorder (TMD), periodontal disease, toothache, and dental attendance were assessed. Occupational stress was measured using the Brief Job Stress Questionnaire (57-item version). The associations between occupational stress and oral health outcomes were estimated using inverse probability weighting, adjusting for age, sex, socioeconomic status, employment status, and health status. **Results:** The results showed that occupational stress was significantly associated with reduced OHRQoL, TMD, periodontal disease, and toothache, with clear dose-response relationships. Compared with the lowest tertile, the highest stress group had a higher prevalence of reduced OHRQoL (prevalence ratio [PR] = 3.07; 95% confidence interval [CI]: 2.54–3.60), TMD (PR = 2.81; 95% CI: 2.23–3.39), periodontal disease (PR = 1.53; 95% CI: 1.32–1.75), and toothache (PR = 1.89; 95% CI: 1.65–2.12). Regarding dental attendance, the prevalence of no dental checkup within the past year was significantly higher in the high-stress group (PR = 1.12; 95% CI: 1.05–1.19), whereas no significant association was observed with treatment visits. **Conclusion:** In conclusion, occupational stress was associated with multiple oral health outcomes.

Keywords: Work-related Stress, Epidemiologic Study, Quality of Life

Presenter ID: OP41

Agreement and Diagnostic Accuracy of Panorama-Based Periodontitis Staging in Korean Adults: A Cross-Sectional Study

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Background: Population-level periodontitis surveillance requires reliable case ascertainment. The 2018 classification of periodontitis assigns a patient-level stage using clinical and radiographic criteria, with a 33% radiographic bone loss (RBL) threshold central to the severity assessment. Panoramic radiography is widely used in oral-health monitoring, but its diagnostic performance for 2018 staging in East Asian adults and the cross-population validity of the 33% RBL threshold remains insufficiently tested. **Methods:** In a single-center cross-sectional study of 144 Korean adults, full-mouth periodontal examination provided the reference standard (probing-based Tonetti stage). Panoramic radiographs, acquired within 30 days of examination, were measured at least three months later. Agreement was assessed by Cohen κ . Nested logistic regression added demographic/socioeconomic (Model 1), systemic (Model 2), and dental behavioral variables (Model 3) to panorama-based Stage III/IV; discrimination was quantified by AUROC. The Youden-optimal RBL cut-off was 33%. **Results:** Among 144 participants (Stage II, 51; Stage III/IV, 93), absolute agreement was 83.3% (Cohen $\kappa=0.645$). Panorama-based Stage III/IV alone yielded an AUROC of 0.836 (95% CI 0.761-0.910). Sequential models reached AUROCs of 0.863, 0.875, and 0.902, but every 95% CI overlapped that of the primary model, indicating no clear incremental discrimination. For continuous maximum RBL (AUROC 0.904), the 33% threshold gave the best balance (sensitivity 0.84, specificity 0.86), and the Youden-optimal cut-off (33.4%) was essentially identical. **Conclusion:** Panorama-based Tonetti staging showed substantial agreement with probing-based staging in Korean adults, and the 33% RBL threshold was empirically supported. Panoramic radiography is a feasible adjunct for periodontitis surveillance in preventive-dentistry settings.

Keywords: Periodontitis, 2018 Classification, Panoramic Radiography, Diagnostic Accuracy, Korean Adults

Presenter ID: OP42

Oral Health Status, Betel Quid Chewing and Food Insecurity among Indigenous Adults in Pahang

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Background: Oral health (OH) inequalities among the indigenous populations are shaped by a complex interplay of cultural, environmental and socioeconomic factors. In Malaysia, data on the OH status of indigenous communities and its associated determinants remain limited. This study aimed to assess OH status, betel quid chewing habits, food insecurity, and their associated factors among indigenous adults in Pahang. **Methods:** A cross-sectional study was conducted among 180 indigenous adults in Pahang. OH status was assessed through clinical examination based on WHO criteria. Betel quid chewing habits were assessed using questionnaires, while food insecurity was evaluated using the validated Household Food Insecurity Access Scale. Data were analysed using descriptive and multivariate statistical analyses. **Results:** Dental caries affected 76.7% of participants, with a mean DMFT score of 5.17 (SD=6.23). Prevalence of unhealthy periodontium and tooth loss was 66.7% and 38.9%, respectively. More than half of indigenous adults (60.0%) practiced betel quid chewing, with most initiating the habit between 10-19 years of age (62.1%). Food insecurity affected 40.6% of participants, including 25.0% experiencing moderate-to-severe levels. Caries prevalence was highest among participants with severe food insecurity (88.9%), although the association was not statistically significant. Multiple logistic regression showed that age was significantly associated with dental caries, unhealthy periodontium, the presence of calculus and tooth loss ($p<0.05$). **Conclusion:** Poor OH status was prevalent among indigenous adults, with age identified as a significant determinant. The high prevalence of betel quid chewing habit and food insecurity reflects the broader cultural and socioeconomic challenges faced by this community.

Keywords: Betel Quid Chewing, Food Insecurity, Indigenous, Oral Health Status

Presenter ID: OP43

Designing Equity Through a School-Based Prevention: The SPARKLE Model

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Background: Oral health inequities persist among primary schoolchildren from rural and low socioeconomic communities in Malaysia, with preventable oral diseases disproportionately affecting disadvantaged populations. Existing school oral health initiatives have not sufficiently reduced disparities, highlighting the need for proportionate, school-based prevention strategies that are evidence-based and feasible. **Objectives:** To identify effective promotive and preventive interventions for disadvantaged children; to develop a context-sensitive, evidence-informed intervention package; and to evaluate and refine its implementation feasibility from the perspective of oral healthcare providers as key implementers. **Methods:** A three-phase translational design was used. A systematic review assessed interventions targeting dental caries, periodontal disease, oral hygiene practices, sugar consumption, and tobacco use. Eleven higher-quality studies informed the development of SPARKLE (School-based Prevention for Kids from Rural and Low Socioeconomic Backgrounds), a multi-component school-based model. Focus group discussions and in-depth interviews with oral healthcare providers explored the acceptability, appropriateness, and practicality of the intervention. Thematic analysis guided iterative refinement. **Results:** SPARKLE integrates oral health education, supervised toothbrushing, fluoride varnish, fissure sealants, motivational interviewing, school food policy enhancement, and a healthy food initiative. Feasibility findings indicated that the model was largely acceptable, appropriate, and practical, with refinements to address workforce capacity, training needs, resource allocation, and intersectoral collaboration. **Conclusion:** SPARKLE operationalises equity by translating evidence into a school-based prevention model capable of improving oral health among rural and low socioeconomic primary schoolchildren through sustainable, proportionate strategies.

Keywords: Oral Health, Dental Caries, Preventive Dentistry, Low Socioeconomic, Schoolchildren

Presenter ID: OP44

Facilitators, Barriers and Strategies in Toddler Oral Health Anticipatory Guidance Practices Among Dental Therapists

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Background: Early childhood caries remains highly prevalent among Malaysian children, especially in Terengganu. Anticipatory Guidance (AG) is a proven caries preventive strategy that provides developmentally tailored oral health information, delivered primarily by Dental Therapists (DTs). Despite widespread implementation, a critical gap remained in understanding the dynamics of effective AG delivery. Hence, this study aimed to explore facilitators, barriers, and strategies for improvement in AG practices on oral health for toddlers from DTs' perspectives in Terengganu. **Methods:** A phenomenological qualitative study involved DTs recruited via purposive sampling from government dental clinics in Kuala Terengganu and Besut districts. FGDs were conducted using a validated semi-structured guide until data saturation was achieved. Audio recordings were transcribed verbatim, translated into English, and thematically analyzed using ATLAS.ti version 23. Validity was ensured through investigator triangulation, data triangulation, and member checking. **Results:** A total of 40 female DTs participated in eight FGDs. Facilitators included experience and organizational support, strong parent-therapist partnership, availability of AG educational tools, and collaboration with other dental and medical personnel. Barriers encompassed five themes: misconceptions about AG, environmental barriers, workforce challenges, limited knowledge and training in AG, and poor parental commitment and socioeconomic barriers. Four strategies emerged to improve AG practices: provide training and professional development, use digital tools and social media, empower parental involvement, and strengthen and increase human resources for AG. **Conclusion:** Optimizing AG practices requires leveraging existing facilitators, addressing multidimensional barriers, and implementing these strategies to improve oral health outcomes among toddlers.

Keywords: Anticipatory Guidance, Toddler, Oral Health, Dental Therapists

Presenter ID: OP45

Province-Wide Oral Health Promotion Model in Zhejiang, China: A Public Health Practice

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Background: Oral diseases remain a major public health concern in China, particularly among children and older adults. Zhejiang Province established a province-wide oral health promotion program integrating prevention, health education, community services, and digital health management to improve oral health equity and accessibility. **Methods:** From 2014 to 2024, Zhejiang established a four-level oral public health network involving provincial, municipal, county, and community institutions. Major interventions included school pit-and-fissure sealant programs, fluoride varnish application for preschool children, oral health surveillance, community oral health hospitals, and digital oral health record systems. Program implementation was supported through standardized training, hierarchical supervision, annual evaluation, and multidisciplinary collaboration. **Results:** By the end of 2024, Zhejiang achieved province-wide coverage of the school pit-and-fissure sealant program, involving 6,926 schools, 4.99 million screened students, and over 11.29 million sealed teeth, with a retention rate exceeding 90%. Preschool fluoride programs reached nearly 200,000 children and were associated with reduced new caries occurrence. Oral health surveillance included 5,559 participants across 11 monitoring sites and identified disparities in caries status, oral hygiene, and treatment awareness across age groups. Community oral health hospitals and digital oral health records further improved access to preventive care and long-term oral health management, particularly in rural and underserved areas. **Conclusion:** The Zhejiang oral health promotion model demonstrates the feasibility of integrating prevention-oriented strategies, community-based care, and digital health management into regional oral public health systems. This model may provide a practical framework for improving oral health equity and population-based prevention in other regions.

Keywords: Oral Health Promotion, Public Health, Community Oral Care, Preventive Dentistry

Presenter ID: OP46

Effect of Interactive Shared Book Reading on Toddler's Oral Health

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Background: Effective teeth cleaning is a key strategy for preventing dental caries, but toddlers often struggle to cooperate. Interactive shared book reading (ISBR) serve as important learning method. Nevertheless, evidence-based toothbrushing resources and the effects of ISBR remain limited. This study aimed to screen existing toothbrushing picture books and evaluate the effect of ISBR. **Methods:** Chinese online platforms were searched using keywords including 'oral picture books,' 'toothbrushing picture books,' and 'teeth picture books.' The selected books were evaluated according to established criteria. Based on these findings and current caries prevention guidelines, a tailored picture book was developed. A 3-month randomized controlled trial enrolled children aged 2-3 years with a baseline Simplified Debris Index (DI-S) ≥ 1 . The intervention group received oral health education, parental training, and ISBR; the control group received education and training only. After 3 months, children's oral hygiene status, oral hygiene-related behaviors and parents' oral health-related attitudes were evaluated. **Results:** Among 145 identified books, 9 were included, but none provided detailed brushing methods. We developed a Let's Brush Our Teeth picture book and an ISBR curriculum, which were used for intervention involving 186 participants. At 3-month follow-up (dropout rate 10.2%; 83 intervention, 84 control), the intervention group had significantly lower mean DI-S scores ($P=0.024$), with greater improvement in girls ($P=0.033$). Brushing behaviors improved but not significantly versus controls. **Conclusion:** Existing toothbrushing picture books lack detailed guidance. ISBR using a tailored book improves oral hygiene in toddlers, particularly girls, but has limited impact on behavior change.

Keywords: Toddler, Interactive Shared Book Reading, Picture Books, Toothbrushing Behavior, Oral Hygiene

Presenter ID: OP47

Oral Health Education Among Older Adults with and without Depressive Symptoms: A Randomized Controlled Trial

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Background: Depression is associated with poor oral hygiene in older adults. However, whether it modifies responses to oral health education (OHE) remains unclear. This study evaluated the 18-month effectiveness of OHE and examined effect modification by depressive symptoms. **Methods:** A randomized controlled trial was conducted among community-dwelling older adults aged ≥ 60 years in Yangon. Participants were stratified by Geriatric Depression Scale-15 (Dep+: GDS ≥ 5 ; Dep-: GDS < 5) and randomly allocated into Dep+/OHE, Dep+/control, Dep-/OHE, or Dep-/control. OHE included group sessions at 3-month intervals, a photo-illustrated booklet, and weekly SMS reminders. Outcomes were Marginal Plaque Index (MPI), Bleeding on Probing (BOP), oral health knowledge (OHK), and Oral Health Impact Profile-14 (OHIP-14). ANCOVA estimated overall OHE effects and OHE $\times$ depressive symptoms interactions. Within-group changes were assessed using Wilcoxon signed-rank tests, with Hedges' g effect sizes. **Results:** Of 244 enrolled, 225 (92.2%) completed follow-up. OHE showed significant overall improvements in MPI ($\beta = -0.124$), BOP ($\beta = -0.083$), OHK ($\beta = +1.77$) and OHIP-14 ($\beta = -2.14$) (all $p < 0.001$). OHE $\times$ depressive symptoms interactions were not significant. MPI significantly decreased in both OHE groups, with a larger within-group effect in Dep+/OHE ($\Delta = -0.15$, $g = -0.94$) than Dep-/OHE ($\Delta = -0.06$, $g = -0.32$). BOP also improved in both OHE groups. OHK improved only in Dep-/OHE ($\Delta = +1.97$, $g = 0.78$), while OHIP-14 did not improve significantly in either OHE group. Controls showed no favorable changes. **Conclusion:** OHE improved oral health outcomes regardless of depressive symptoms status. However, depressed participants showed greater behavioral improvements in plaque control while non-depressed participants demonstrated greater knowledge gains, suggesting OHE may work through different pathways by psychological status.

Keywords: Oral Health Education, Older Adults, Depressive Symptoms, Oral Hygiene, Oral Health Literacy, Myanmar

Presenter ID: OP48

The Effect of Personalized Oral Health Education on Oral Hygiene Behavior and Periodontal Health

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Background: The benefits of personalized oral health education on periodontal health remains controversial. This study aims to systematically evaluate the impact of personalized oral health education on periodontal health in the general population through meta-analysis, and to provide evidence-based guidance for governments, medical institutions, and healthcare workers on implementing primary prevention of periodontal problems. **Methods:** We conducted a search of Chinese and English databases to identify relevant literature published from the inception of each database to December 31, 2023, regarding the impact of personalized oral health education on periodontal health in the general population. Randomized controlled trials that met our inclusion and exclusion criteria were collected, and a quality assessment of the literature was performed. The risk of bias was assessed using the Cochrane risk of bias tool Version 2.0. The meta-analysis was conducted using RevMan 5.4 and Stata 18.0 after the literature quality evaluation. **Results:** After rigorous screening, a total of eight out of 2693 papers were included, involving 3384 participants. The meta-analysis showed that personalized oral health education, compared with conventional methods, significantly decreased plaque index (PI) (mean difference [MD] = -0.42 , 95% confidence interval [CI]: -0.63 , -0.21), $p < 0.0001$) and gingival index (GI) (MD = -0.47 , 95% CI $[-0.67, -0.27]$, $p < 0.0001$). It also enhanced self-efficacy and oral health-related quality of life. Moreover, it increased the frequency of toothbrushing and dental flossing, and improved oral health knowledge, attitude, and behavior among participants. **Conclusion:** This study suggests that personalized oral health education can significantly improve periodontal health.

Keywords: Meta-analysis, Periodontal Health, Personalized Oral Health Education, Systematic Review

Presenter ID: OP49

Knowing, Believing, Doing: Insights from a Validated Parental Oral Health Knowledge, Attitude and Practices Questionnaire

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Background: Instruments assessing family oral health knowledge, attitudes, and practices (KAP) among parents of primary schoolchildren remain scarce, leaving key behavioral gaps unaddressed in family-centered interventions. This cross-sectional study evaluated parental oral health KAP using a validated questionnaire among parents of children aged 7–12 years. **Methods:** Data were obtained from 106 Malaysian parents recruited from dental clinics, schools and community settings. Domain scores and item-level response patterns were derived from the final retained items: 11 knowledge, 9 attitude, 4 family dietary practices, 2 family oral hygiene practices and 2 fluoride-use items. **Results:** Overall knowledge was high (median score: 9/11). Most parents (98.1%) identified frequent sugar intake as increasing tooth decay risk; however, more than one-third reported that their children consumed sugary food and drinks several times a week or more. Oral cancer-related knowledge was weaker: 53.8% identified betel quid chewing as a cause, 57.5% recognized that avoiding alcohol could reduce their risk, and 43.4% recognized a non-healing ulcer as a warning sign. Attitudes were mostly positive (median score: 41/45), although agreement was lower for betel quid- and alcohol-related oral health beliefs. Parent-reported oral hygiene practices were favorable, with 87.7% of children and 88.7% of parents brushing at least twice daily. Most parents (92.5%), and children (94.3%), used fluoridated toothpaste. **Conclusion:** Parents of primary schoolchildren had good oral health knowledge and positive attitudes, but item-level analysis identified specific gaps in oral cancer awareness and sugar consumption. This KAP questionnaire may help clinicians and public health teams conduct more targeted family-centered counselling.

Keywords: Knowledge, Attitudes and Practices, Family-centred Counselling, Sugar Consumption, Oral Cancer Awareness, Preventive Dentistry

Presenter ID: OP50

Oral Health Literacy among Special Needs Primary School Teachers and Assistants in Putrajaya, Malaysia

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Background: Oral health literacy (OHL) is an important determinant of oral health behaviours and outcomes. Teachers and assistants working with children with special needs can influence the adoption of healthy oral health practices among their students. However, evidence on OHL among this population in Malaysia remains limited. This study assessed baseline OHL levels and examined their associations with selected sociodemographic characteristics. **Methods:** A cross-sectional study was conducted among 96 teachers and assistants from six public primary schools with special education programmes in Putrajaya, Malaysia, using purposive sampling. OHL was assessed using the validated Malay version of the Oral Health Literacy Instrument (OHLI-M), which measures reading comprehension and numeracy. Total scores range from 0 to 100 and are classified as inadequate (0 to 59), marginal (60 to 74), or adequate (75 to 100). Descriptive statistics, independent samples t-tests, one-way ANOVA, and Pearson's correlation were used for data analysis. **Results:** Most participants were female (91.7%), had tertiary education (82.3%), and had a mean teaching experience of 11.8 (SD 6.3) years. Nearly half (49.0%) had visited a dentist within the past year, while 67.7% had attended a health course. The mean OHLI-M score was 80.91 ± 9.43, indicating adequate OHL. OHL scores were not significantly associated with educational level, teaching experience, attendance at health courses, or time since the last dental visit (all $p > 0.05$). **Conclusion:** Most special needs primary school teachers and assistants demonstrated adequate OHL. Further studies involving larger and more diverse samples are warranted to identify factors associated with oral health literacy among special needs educators

Keywords: Oral Health Literacy, Special Needs Teachers, Teachers' Assistants, Primary School

Presenter ID: OP51

Breaking Free from Vaping: Cessation Barriers and Support Needs among Malaysian University Students

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Background: E-cigarette use has surged among young adults globally, with the highest prevalence reported among those aged 20–24 in the GATS Malaysia 2023 survey. Understanding the barriers and enablers to cessation in this population is essential for designing tailored and effective public health interventions. **Methods:** Twelve focus group discussions (FGDs) were conducted with 57 young adults aged 18–25 years from private universities in Kuala Lumpur and Selangor. Participants were divided into two groups; health science and non-health science academic background. The FGDs were further stratified by gender and user type (exclusive and dual users). Participants were recruited through purposive and snowball sampling until data saturation was achieved. Data were analysed using inductive thematic analysis. **Results:** Regarding barriers to cessation, both groups identified social influence, accessibility, withdrawal symptoms, and dependency as main challenges. Non-health science students additionally cited weak regulation, limited cessation support, and minimal risk perception as barriers to quitting. Clinical and pharmacological support emerged as the main enablers to cessation across both groups, alongside social support, oral substitute, and physical activities. Health science students additionally highlighted online community support, whereas non-health science students identified restricted access to vaping products and religious influence as important enablers. **Conclusion:** Targeted prevention strategies that address peer influence, mental health, awareness of cessation support and misconceptions surrounding e-cigarette use are essential. Empowering young adults through education and effective communication is key to breaking the cycle of e-cigarette use.

Keywords: Young Adults, E-Cigarettes Initiation, Barriers, Enablers To Cessation

Presenter ID: OP52

Impact of Supervised Toothbrushing on Brushing Behaviours, Dental Caries and Oral Cleanliness among Preschool Children

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Background: Supervised toothbrushing programme (STP) promotes effective toothbrushing skills among preschool children. However, evidence regarding their impact among Malaysian preschool children remains limited. Thus, this study evaluated caries experience, brushing behaviours and oral cleanliness among preschool children with and without school-based STP. **Methods:** A cross-sectional study was conducted among children aged 5–6 years from private preschools in Terengganu, with and without school-based STP, using stratified random cluster sampling. Data were collected through parental questionnaires, plaque charting, caries assessment and video-recorded toothbrushing observation. Clinical examiners and raters were trained and calibrated. Data were analysed using descriptive and multivariate statistics. **Results:** A total of 145 preschool children participated in this study. Mean dmft was higher in the non-supervised group (3.96 ± 4.04) than in the supervised group (3.46 ± 4.04). Most children used fluoridated toothpaste (82.8%), brushed for less than 1 minute (66.2%), used horizontal stroke (89.0%) and showed a non-systematic brushing pattern (100%). Plaque score reduction after toothbrushing activities was slightly higher in the supervised group (18.75%) than the non-supervised group (17.50%). Children using half or full-length amounts of toothpaste had 86.6% lower odds of poor oral cleanliness than those using a smear or peas-sized amount (AOR=0.134; 95% CI: 0.023–0.795; $p=0.027$). Higher caries scores were observed among children who brushed less than twice daily, used non-fluoridated toothpaste and stopped milk bottle use after 2 years of age. **Conclusion:** School-based STP may improve plaque control among preschool children, although its impact on caries experience remains inconclusive. Fluoride exposure and dietary control remain important preventive factors.

Keywords: Preschool Children, Supervised Toothbrushing, Toothbrushing Behaviour, Video Observation, Oral Cleanliness

Presenter ID: OP54

Empowering Caregivers to Improve Oral Health in Vulnerable Populations: A Multicentre Study

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Background: Individuals with vulnerabilities experience a higher burden of oral diseases due to physical, sensory, behavioural, and medication-related challenges. Those managing these groups serve as frontline caregivers, promoting oral hygiene, supporting healthy dietary practices, and identifying oral health problems. However, oral health education is often inadequately integrated into caregiver training, resulting in gaps in knowledge and practices. Assessing caregivers' Knowledge, Attitudes, and Practices (KAP) is therefore essential to guide targeted oral health promotion strategies. This study aimed to evaluate oral health KAP among individuals managing vulnerable groups. **Methods:** A quasi-experimental pre-post study was conducted among individuals managing vulnerable groups from six centres providing special education programmes. Participants completed a validated, structured self-administered questionnaire adapted from WHO oral health surveys and established dental KAP literature. An oral health awareness intervention was subsequently delivered by a dentist, followed by administration of the same questionnaire three months post-intervention to assess changes in oral health-related KAP. **Results:** Pre-intervention findings revealed knowledge gaps: 17% believed dental caries cannot occur in primary teeth, 11% believed vulnerable groups are not more susceptible to oral health problems, and 65% perceived that their oral healthcare needs do not differ from those of the general population. Regarding attitudes, 16% believed a balanced diet was not essential for oral health. In practice, 20% did not use any additional oral healthcare products beyond toothbrushing. **Conclusion:** These findings highlight important gaps in caregivers' understanding of vulnerable groups' oral health needs and support the importance of targeted educational interventions to improve oral health awareness and caregiving practice

Keywords: Caregiver Empowerment, Oral Health Promotion, Vulnerable Populations, Preventive Dentistry, Community-Based Intervention

Presenter ID: OP56

Exploring the Features for AI-Powered App in Delivering Oral Health Messages to Hearing Impaired Patients

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Background: Hearing-impaired individuals often face barriers to oral health education because it relies heavily on verbal communication. Caregivers therefore play an important role in supporting oral hygiene. Although Artificial Intelligence (AI) has potential to improve accessible oral health education, evidence among hearing-impaired populations in Malaysia remains limited. This study explored oral hygiene practices, communication challenges, caregiver roles, and expectations for AI-powered oral health education applications. **Methods:** Semi-structured interviews were conducted with four hearing-impaired individuals (aged 12–17 years) and four caregivers recruited through hearing-impaired support organisations in Malaysia. Data were analysed using thematic analysis. **Results:** Participants reported communication barriers, limited awareness of proper oral hygiene techniques, and reliance on caregivers. Visual learning was preferred, and both hearing-impaired individuals and caregivers expressed positive attitudes towards AI-powered applications. Recommended features included visual content, sign language support, interactive learning, affordability, and communication-bridging functions. **Conclusion:** AI-powered oral health education applications may improve access to oral health information, enhance independence among hearing-impaired individuals, and support caregivers.

Keywords: AI-powered Applications, Communication Barriers, Hearing Impairment, Oral Health Education, Visual Learning

Presenter ID: OP58

CBCT-Based Automated 3D Maxillofacial Assessment System for Landmark Detection and Malocclusion Diagnosis

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Background: Malocclusion is a highly prevalent dentofacial condition, affecting an estimated 56% globally and 48% in Asia, with potential impacts on facial aesthetics, oral function, and quality of life. Accurate 3D maxillofacial assessment is essential for orthodontic care, yet CBCT-based cephalometric analysis remains time-consuming and operator-dependent, limiting its clinical value. To address this gap, we developed and internally validated an automated CBCT-based system for 3D landmark detection, cephalometric analysis, and malocclusion diagnosis. **Methods:** This single-center study included 756 CBCT scans from patients with malocclusion. Fifty-nine dental, skeletal, and soft-tissue landmarks were annotated by clinicians. A two-stage coarse-to-fine AI model localized landmarks by integrating image features and anatomical relationships. The system generated 82 3D cephalometric measurements and classified eight malocclusion patterns. Performance was evaluated by landmark accuracy, measurement reliability, and diagnostic accuracy in an internal test set of 70 cases. **Results:** The system achieved an overall mean radial error of 1.07 ± 0.56 mm and a 2-mm success detection rate of 90.71%. Among 82 measurements, 68 (82.92%) achieved success detection rates above 90%, including 31 (37.80%) with 100% success detection. For sagittal skeletal classification, accuracies were 0.890, 0.926, and 0.912 for skeletal Class I, II, and III, respectively. For all remaining diagnostic categories, accuracy exceeded 0.78. **Conclusion:** By automating landmark-based 3D maxillofacial assessment, this system has the potential to reduce the workload and operator-dependent variability of CBCT-based cephalometric analysis and support standardized orthodontic diagnosis and treatment planning. Multicenter external validation is warranted before clinical implementation.

Keywords: Maxillofacial Deformity; Cone-Beam Computed Tomography; Three-dimensional Cephalometry

Presenter ID: OP60

Comparative of Caries Risk Assessments and MinDTain for Predicting Children's Permanent Caries Risk in Indonesia

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Background: Dental caries remains one of the most prevalent oral health problems among school-aged children, highlighting the importance of accurate Caries Risk Assessment (CRA) for early prevention. Various conventional CRA tools have been widely used; however, digital multilevel approaches such as the MinDTain Mobile Application offer broader assessment dimensions. This study aimed to compare the results of permanent dental caries risk assessment between several conventional CRA methods—AAPD Care Pathways, Cariogram, CAMBRA, CARDEC-04, and Irene's Donut—and the MinDTain Mobile Application among children aged 6–12 years. **Methods:** A quantitative analytical observational study with a cross-sectional design was conducted involving 79 children from MI Ma'arif 1 and 2 Sukun, Malang City. Caries risk assessment was performed using each conventional CRA tool and the MinDTain Mobile Application. Data were collected through clinical oral examinations, structured interviews, and completion of assessment instruments. Statistical analyses were performed using Chi-Square and Kendall Tau tests with a significance level of $p < 0.05$. **Results:** Significant differences were identified between all conventional CRA tools and the MinDTain Mobile Application in assessing permanent dental caries risk ($p < 0.05$). Most conventional methods classified children as moderate-to-high risk, whereas MinDTain predominantly categorized children as low risk. The differences were associated with variations in assessment approaches and evaluated risk factors. **Conclusion:** The MinDTain Mobile Application demonstrated significantly different caries risk assessment outcomes compared with conventional CRA tools, indicating the importance of selecting assessment methods based on population characteristics and preventive program objectives.

Keywords: Dental Caries, Caries Risk Assessment, MinDTain, School-Aged Children, Preventive Dentistry

Presenter ID: OP61

Application of an AI-Assisted 2D-to-3D Lesion Reconstruction Model in the Postoperative Follow-Up of Micro-Apical Surgery

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Background: After micro-apical surgery, close follow-up is essential for managing refractory apical periodontitis. Conventional 2D radiographs are insensitive to subtle bone changes, whereas CBCT involves higher radiation. We developed an AI model that reconstructs 3D lesions from routine 2D radiographs without additional radiation to enable sensitive detection of progression.

Methods: A total of 450 paired preoperative CBCT scans and periapical/paralleling radiographs from micro-apical surgery cases were collected. Lesions and relevant tooth/bone structures were manually annotated. A tooth-position-specific alignment strategy matched 2D annotations with 3D projections. A 3D U-Net-ASPP-based 2D-to-3D reconstruction model with boundary refinement was trained and evaluated using the Dice coefficient, maximum diameter relative error, relative volume error, and average surface distance. An observational study was designed to compare clinicians' assessment of treatment outcomes with and without 3D reconstructions. **Results:** On the test set, the model achieved a mean Dice score of 0.706 ± 0.177 , a maximum diameter relative error of $13.52\% \pm 9.85\%$, a relative volume error of $27.17\% \pm 22.05\%$, and an average surface distance of 0.434 ± 0.290 mm, comparable to CT-based benchmarks. **Conclusions:** This AI workflow reconstructs 3D lesion geometry from standard radiographs without extra radiation, potentially enabling more precise monitoring and earlier intervention. The planned observational study will further validate whether access to 3D reconstructions can elevate general clinicians' evaluation accuracy to an expert level.

Keywords: AI 2D-to-3D Reconstruction, Micro-apical Surgery Follow-up, Radiation Reduction

Presenter ID: OP62

Silent Smile Program: Inclusive Oral Health Education with 21-Day Calendar for Kolok Community in Bali

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Background: The Kolok community (congenitally deaf-mute) in Bengkala Village, Bali possesses a distinct cultural identity, relying exclusively on the local sign language 'Kata Kolok' for communication. Consequently, conventional oral health education is often ineffective. Lack of inclusive information access contributes to poor oral health knowledge and suboptimal oral care behaviors. This study assessed the effects of the 'Silent Smile' program in addressing this disparity through culturally and linguistically tailored education, reinforced by a 21-day tooth-brushing calendar to strengthen daily habits. **Methods:** A one-group pretest-posttest design was conducted among 34 Kolok community members selected through random sampling. The intervention included: local sign-language interpreter during oral health education, demonstration of correct tooth-brushing technique, and distribution of a 21-day calendar as behavioural reinforcement. Data collection involved pre- and post-intervention knowledge tests, daily tooth-brushing adherence logs, and oral hygiene examinations (OHIS). Statistical analysis used paired t-test and Wilcoxon signed-rank test. **Results:** Significant improvements were observed: oral health knowledge scores increased substantially, twice-daily tooth brushing adherence (morning and night) improved markedly, and oral hygiene status improved with reduced OHIS scores. The 21-day calendar effectively strengthened brushing habits, while the local sign-language interpreter ensured comprehension of oral health messages. **Conclusion:** The Silent Smile Program is an effective, inclusive oral health education model for the Kolok community. Selecting oral health cadres from within the Kolok community ensures program sustainability. This intervention is suitable for adaptation in similar communities, as it is grounded in local language, culture, and behavioral practices.

Keywords: Deaf Persons, Inclusive Oral Health Education, Kolok Community, Oral Hygiene, Toothbrushing Behaviour

Presenter ID: OP64

Aligning Teledentistry with Clinical Practice: Dental Healthcare Professionals' Requirements for Paediatric Oral Health Systems

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Background: Teledentistry has emerged as a promising approach to improve access to paediatric oral healthcare, particularly in settings where barriers such as limited access, workforce constraints, and missed appointments persist. However, many digital health systems fail due to poor alignment with clinical workflows and user requirements. Evidence on dental healthcare professionals' (DHCPs) perspectives in designing teledentistry systems for children remains limited. **Methods:** This qualitative study explored DHCPs' requirements for a teledentistry system for paediatric oral health screening and monitoring. Fourteen DHCPs, including paediatric dental specialists, dental public health specialists, dental officers and dental nurses, participated in focus group discussions guided by integrated Human-Centred Design (HCD) and User-Centred Design (UCD) principles. Low-fidelity web-based prototypes were used to facilitate discussion. Data were analysed using thematic analysis. **Results:** Four main themes were identified: (1) needs and preferences, (2) requirements and implementation in the Malaysian context, (3) perceived benefits, and (4) perceived challenges. DHCPs emphasised the importance of structured screening and monitoring workflows, integration with existing clinical pathways, and features such as guided image capture, teleconsultation, and patient monitoring dashboards. Key concerns included data security, infrastructure limitations, and potential increases in workload. Both functional and non-functional requirements were identified to ensure usability, feasibility, and clinical relevance. **Conclusion:** Findings highlight the importance of incorporating DHCP perspectives in teledentistry system design. Integrating HCD and UCD principles supports alignment with both clinical workflows and usability needs. Addressing contextual and operational challenges is essential to enhance adoption and sustainability of teledentistry in paediatric oral healthcare.

Keywords: Teledentistry; Paediatric Dentistry; Qualitative Research; User-Centred Design; Human-Centred Design; Digital Health

Presenter ID: OP65

PlaqueSLAM: A Deep Learning Model for Plaque Detection in Preschoolers Using Unstained Smartphone Intraoral images

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Background: Limited access to routine paediatric dental care highlights a critical need for accessible oral health management tools. This study aimed to develop and evaluate a deep learning model for dental plaque screening using unstained smartphone-acquired intraoral images acquired in community outreach settings. **Methods:** A total of 3888 unstained intraoral images were collected from 648 preschoolers (352 males and 296 females; mean age 4.80 ± 0.97 years) using a smartphone (iPhone 7, Apple Inc.) equipped with standard flash illumination. According to the standard photography protocol, six images were captured per child, encompassing the buccal surfaces of the maxillary and mandibular incisors, canines, and molars. 567 children (3,402 images) and 81 children (486 images) were assigned to the training and test dataset respectively. PlaqueSLAM, a novel deep learning algorithm, was developed employing a three-stage training strategy based on a multi-task learning paradigm. Diagnostic performance was evaluated using accuracy, sensitivity, and specificity. **Results:** Despite the absence of disclosing agent, PlaqueSLAM still demonstrated an accuracy of 79.6% (95% CI: 77.43%–80.42%), a sensitivity of 76.58% (95% CI: 74.36%–78.68%), and a specificity of 82.47% (95% CI: 80.44%–84.49%) in plaque detection among preschool children. **Conclusion:** PlaqueSLAM offers an acceptable dental plaque detection method using standard smartphone images for young children, eliminating the requirement for disclosing agents. This non-invasive, scalable teledentistry solution is highly practical for kindergarten-based outreach programs. By facilitating remote, dynamic oral hygiene monitoring, PlaqueSLAM offers a cost-effective strategy to promote paediatric dental health and decrease the burden of early childhood oral diseases.

Keywords: Deep learning, Dental plaque, Intraoral images, Preschooler

Presenter ID: OP87

Deep-Learning Radiographic Bone Loss Quantification for Periodontitis Staging: Tooth- and Patient-Level Validation

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Background: Conventional radiographic bone loss (RBL) measurement for periodontitis classification is time-consuming and subject to inter-examiner variability, motivating automated artificial-intelligence approaches. This study aimed to validate a deep-learning model that automatically predicts alveolar bone and root landmarks against expert manual measurement at the tooth and patient levels. **Methods:** A two-stage pipeline was used. First, a YOLOv8-seg model performed tooth detection on panoramic radiographs. Second, after CLAHE filtering of each tooth patch, a ResNet-34 backbone with a heatmap-regression head localized six landmarks per tooth (alveolar bone crest, cemento-enamel junction, and root apex on the mesial and distal aspects), from which RBL% was computed. Radiographs from 146 adults were split by patient; on the held-out test set (n=30), performance under the 2018 classification (Tonetti et al., 2018) was evaluated against manual RBL% using Pearson correlation, intraclass correlation (ICC), AUC, Cohen's κ , sensitivity, and specificity. **Results:** At the tooth level, predicted and manual RBL% agreed well ($r=0.71$; ICC=0.69; MAE=4.87 percentage points). The AUC for Stage III sites was 0.91; however, at the 33% threshold, sensitivity was 0.41, specificity 0.99, and $\kappa=0.51$, indicating underestimation of severely affected sites. At the patient level, agreement was moderate ($r=0.71$; ICC=0.72; AUC=0.79; $\kappa=0.46$; sensitivity 0.72, specificity 0.75, accuracy 0.73). **Conclusions:** The model quantified RBL and discriminated Stage III lesions well (AUC 0.91), but patient-level agreement was moderate, with underestimation of severe sites—attributable to a regression-to-mean bias and the sensitivity of maximum-RBL staging to single-site error. Further validation with larger, balanced datasets is warranted.

Keywords: Alveolar Bone Loss, Periodontitis, Deep Learning, Diagnostic Accuracy, Panoramic Radiography

Presenter ID: OP66

Translating Evidence into Practice: Strengthening the Adopted Village Oral Health Programme in Malaysia

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Background: Community-based oral health programmes are central to preventive dentistry; however, translating programme objectives into practice remains challenging. Malaysia's Adopted Village Oral Health Programme (Program Kampung Angkat Pergigian), introduced nationally in 2020, aimed to improve oral health knowledge, expand care access, and strengthen community networks through outreach and home visits. Early implementation revealed limited uptake and community engagement challenges. This study generated implementation evidence to strengthen programme delivery and inform guideline refinement. **Methods:** A qualitative study was conducted from June 2023 to June 2024 in Negeri Sembilan. 56 participants, including policymakers, programme managers, oral healthcare providers, community leaders, oral healthcare volunteers, and community members, were purposively recruited. 39 in depth interviews and three focus group discussions were conducted. All data were recorded, transcribed verbatim and analysed thematically. An integrated knowledge translation approach facilitated co-interpretation of findings and translation of evidence into policy and practice. **Results:** Community acceptance of this programme was facilitated by perceived benefits, improved access, and trust mediated through local leadership, particularly for vulnerable groups. Barriers included fear of treatment, preference for curative care, privacy concerns, and limited continuity of care. Implementation challenges highlighted the need for community prioritisation criteria, governance, strengthened monitoring and evaluation, and provider–community partnerships. These findings informed refinement of the 2026 national guideline, incorporating targeted home visits, governance mechanisms, continuity of care, and outcome indicators measuring orally fit status. **Conclusion:** Implementation findings informed policy changes through integrated knowledge translation, showing evidence can strengthen programme responsiveness, sustainability, and equity in oral health systems.

Keywords: Community-based Oral Health Programme, Health Systems Strengthening, Oral Health Policy, Implementation Research, Knowledge Translation

Presenter ID: OP67

Spatial Distribution of Community Oral Health Resources and Oral Health Status in Shanghai, China

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Background: This research evaluated the distribution and equity of community oral health resources in Shanghai, and analyzed their correlations with population oral health metrics including caries prevalence and periodontal health level. **Methods:** Data covering staffing, equipment and public oral health services were gathered from 249 community health centers across Shanghai's 16 districts, paired with residents' sampled clinical oral health records. Gini coefficients, Lorenz curves and Theil indices measured resource equity, and Pearson correlation tests clarified links between resource inputs and oral health status. **Results:** Oral health practitioners were mostly young females (78.2% female; 63.9% aged ≤ 44). Though 76.6% attained bachelor's or higher degrees, merely 4.2% held senior professional titles. Suburban areas possessed 66.0% of dental chairs and 57.7% of clinical premises, with more basic and advanced devices such as implant equipment, operating microscopes and laser therapeutic instruments. Geography, population- and economy-based Gini coefficients of independent dental departments were 0.151, 0.357 and 0.362; those of dental chairs were 0.356, 0.420 and 0.400. The overall Theil index for dental chair geographic distribution reached 0.7128, where suburban and urban sub-indices were 0.3809 and 0.0359. Suburban regions accounted for 92.66% of total geographic inequity versus 7.34% from urban zones. Strong positive correlations were found between manpower index and periodontal health rate ($r=0.881$, $P=0.020$), as well as dentist-population ratio and periodontal health ($r=0.870$, $P=0.024$). **Conclusions:** Shanghai's community oral health resources display obvious spatial imbalance, adversely affecting public oral health. Targeted policies should expand premium resources in disadvantaged regions and boost primary oral care capacity to realize equitable, efficient oral healthcare.

Keywords: Community oral healthcare resources, Health resource allocation, Gini coefficient, Lorenz curve, Theil index, Caries prevalence, Oral health status

Presenter ID: OP68

Maternal Oral Health Management Project: A Prevention-Centered Model from Shenzhen

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Background: Oral diseases during pregnancy remain a public health concern due to hormonal and dietary changes that increase risks of caries, periodontal disease, pericoronitis, etc. Research has confirmed that oral diseases during pregnancy not only harm maternal health but also raise the risk of adverse pregnancy outcomes. This study aimed to integrate oral health evaluation into prenatal care that is essential for prevention. **Methods:** A tiered training approach focusing on risk assessment, preventive and therapeutic procedures, and safe referral pathways was implemented. Non-dental specialists (obstetricians, gynaecologists, prenatal care providers) received training using photos and descriptions of common oral issues to improve their questioning and referral skills (non-diagnostic), enabling them to identify oral health risks during routine exams. Dentists were trained in diagnosis and treatment. **Results:** Four training sessions trained nearly 1,000 professionals. A three-level service network was established: municipal maternity and child healthcare hospitals as the technical core, district-level hospitals as hubs, and community health centres as the grassroots base. Screening services were built up at preconception, prenatal, and postpartum stages with a unified and simplified oral examination form. **Conclusions:** Incorporating oral health into routine prenatal care, equipping non-dental healthcare workers and oral professionals with the necessary skills, and establishing a three-level service system is conducive to the prevention and treatment of oral diseases during pregnancy and provides a replicable "Shenzhen model" globally.

Keywords: Oral diseases during pregnancy, Prenatal care, Tiered training, Non-dental specialists, Referral pathways, Three-level service network, Screening services, Shenzhen model

Presenter ID: OP69

Oral Health Policy and Priorities for Practice Guideline Development in Asia

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Background: Oral diseases are among the most prevalent noncommunicable diseases (NCDs) and disproportionately affect disadvantaged populations. They share major modifiable risk factors with other NCDs, including high free sugar intake, tobacco and alcohol use. Despite growing global attention, many Asian countries face persistent challenges in translating evidence and high-level commitments into oral health strategies, essential service packages, and practical, context-specific guidelines. This review synthesizes evidence and proposes priorities for oral health policy and practice guideline development in Asia. **Methods:** A narrative review of peer-reviewed literature and institutional documents on oral disease burden, policy frameworks, prevention strategies, service delivery, and implementation in Asian settings was conducted. World Health Organization global and regional frameworks guided interpretation. Thailand, Indonesia, Singapore, and Japan were selected as country exemplars reflecting diverse health system contexts and stages of policy maturity. **Results:** Findings were organized into five areas: (1) burden and inequities; (2) policy architecture; (3) prevention and service delivery; (4) financing and workforce; and (5) guideline development and implementation. Consistent priorities included population-level and targeted fluoride use, school-based prevention, minimally invasive caries management, counselling on shared risk factors, and early detection with clear referral pathways. Implementation commonly leveraged schools, primary care, community services, task-sharing, reliable supply of preventive commodities, and pragmatic monitoring systems. Country exemplars showed progress, but gaps remained in sustainable financing, workforce distribution, and equitable access. **Conclusions:** Region-specific, context-adapted guidelines are urgently needed to support effective implementation of oral health policies and programmes across Asia.

Keywords: Health System, Policy, Oral Health, Development

Presenter ID: OP72

Feasibility, Acceptability and Effectiveness of a Pilot Betel Quid Cessation Programme in Sabah, Malaysia

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Background: Sabah bears the highest prevalence of betel quid (BQ) use in Malaysia, placing many communities at increased risk of oral cancer. Despite this significant public health burden, Malaysia lacks a structured BQ cessation programme, limiting efforts to support cessation among high-risk populations. This study evaluated the feasibility, acceptability and preliminary effectiveness of a culturally-adapted BQ cessation programme among an indigenous community in Sabah. **Methods:** A quasi-experimental pilot study was conducted among 46 BQ users from the Bajau community in Kota Belud, Sabah, with 23 participants in each group. The intervention group received a culturally-adapted cessation programme comprising five support sessions and an educational booklet, while the controls received the booklet only. Feasibility, acceptability and preliminary effectiveness were evaluated using interviewer-assisted questionnaires, combined cessation outcomes and researcher observations. Differences were analysed using Cochran's Q, McNemar and Chi-square tests. **Results:** Data collection procedures and programme delivery were feasible. However, implementation was initially challenged by limited support from village leaders, difficulties recruiting eligible participants, and a shortage of dental personnel conversant in the indigenous language to serve as facilitators. Acceptability was high, with all intervention components rated as very helpful, 100% programme satisfaction, and all participants willing to recommend the programme. As most participants were dual users of BQ and chewable tobacco, the intervention achieved a higher combined cessation rate at Day-22 (65.2% vs. 34.8%) and a significantly higher prolonged abstinence rate at Month-3 (52.2% vs. 26.1%). **Conclusions:** The programme offers a promising model for community-based BQ cessation in Sabah, Malaysia.

Keywords: Betel Quid, Cessation Programme, Feasibility, Acceptability, Cessation Rate

Presenter ID: OP73

Oral Health Burden and Service Delivery among School Personnel and Families in Sabah's PPKPS Programme

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Background: School dental clinics in Malaysia primarily serve students, which may limit convenient access to dental care for teachers, supporting staff and their families. The School Dental Clinic Community Service Expansion (PPKPS) initiative was introduced to extend selected services to this wider school community. In Sabah, where geography may affect service delivery, programme performance has not been formally evaluated. This study examined the utilisation pattern and treatment profile of PPKPS across nine districts in Sabah. **Methods:** A descriptive analysis of secondary data from PPKPS 1/2023 records was conducted from January to December 2025 across nine areas in Sabah. Variables included new and repeat attendance, age group, dental caries status using DMFT criteria, Basic Periodontal Examination (BPE) status, treatments provided, case completion and oral cancer screening. Frequencies, percentages and means were used to describe utilisation, oral health status and treatment patterns. **Results:** A total of 1,962 patient visits were recorded, including 1,323 new cases. Most new patients were aged 30 - 49 years (51.4%). The mean DMFT score was 5.4 with the "Missing" component increasingly dominant from age 30 onwards, accounting for 64% of total DMFT among those aged ≥60 years. Only 6.4% had healthy periodontal status. Treatment was predominantly curative while preventive care accounted for less than 5% of new patients. Case completion was achieved in 53.9% and oral cancer screening reached 80.7% of new cases. Urban districts accounted for 59.9% of new cases and had the highest screening coverage (87.6%). **Conclusion:** PPKPS met its intended aim of extending access to school teachers, supporting staff and their families, reaching a population with substantial oral disease burden. However, service delivery remained largely curative.

Keywords: School Dental Clinic, Community Service Expansion Programme, Oral Health Burden, Dental Service Utilisation, DMFT, PPKPS

Presenter ID: OP74

Voices from the Longhouse: Perceived Dental Accessibility in Rural Sarawak

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Background: Improving dental access for rural and Indigenous communities is vital in reducing oral health disparities. In Sarawak, infrastructure improvements and outreach programmes have expanded service delivery, but community perspectives remain underexplored. This study explored perceived accessibility of dental services among rural Iban longhouse communities in Sarawak, Malaysia. **Methods:** A qualitative study using focus group discussions (FGD) was conducted among a purposively selected Iban longhouse community in rural Sarawak. Three FGDs (n = 19) were conducted in the Iban language, audio-recorded, and transcribed verbatim. Data were analysed using hybrid thematic analysis guided by Levesque et al.'s healthcare accessibility framework. Coding utilised original Iban transcripts to preserve linguistic nuance and contextual meaning. **Results:** Participants perceived government dental services as increasingly accessible due to improved road connectivity, expanded clinic hours, and outreach programmes. Travel times of 30–45 minutes via motorcycles were generally acceptable. *Program Kampung Angkat Pergigian* was highly valued for reducing travel burden and improving service reach. Government services were trusted and perceived as safe, affordable, and community-oriented, particularly due to routine blood pressure screening and improved treatment experiences. However, dental attendance remained largely symptom-driven and seeking care only when pain became severe. Utilisation was further hindered by mobility and physical access barriers among elderly, individuals with disabilities, pregnant mothers, and caregivers with young children. Participants preferred outreach and community-based services over static clinics to improve access in rural settings. **Conclusion:** Although accessibility has improved, preventive utilisation remains influenced by mobility-related, psychological, and behavioural barriers. Addressing these highlights the need for more inclusive outreach and community-based service delivery models in rural and Indigenous communities.

Keywords: Healthcare Accessibility, Iban Longhouse, Oral Health Equity, Qualitative Research, Sarawak

Presenter ID: OP75

The Preventive Dentistry Research Landscape in Malaysia: An Analysis of the National Oral Health Research Compendium (2020-2025)

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Background: Preventive dentistry plays a pivotal role in improving population oral health and reducing the burden of oral diseases. Understanding the current landscape of preventive dentistry research is important for identifying research priorities and guiding future preventive oral healthcare initiatives. This study aimed to analyse the characteristics of preventive dentistry research represented in the National Oral Health Research Compendium between 2020 and 2025. **Methods:** A descriptive document review was conducted using abstracts compiled in the National Oral Health Research Compendium (2020–2025). Preventive dentistry abstracts were identified using predefined eligibility criteria developed for this review. Eligible abstracts were categorised according to the type of research and preventive strategy. Descriptive statistics were used to summarise the findings. **Results:** Of 766 abstracts screened, 69 met the eligibility criteria for preventive dentistry research. Epidemiological and observational studies constituted the largest proportion (30.4%), followed by implementation and health services research (24.6%) and intervention research (20.3%). Health promotion and behavioural interventions were the most common preventive strategy (27.5%), followed by programme implementation and health systems (23.2%), preventive clinical interventions (21.7%), preventive products and technologies (20.3%), and screening, early detection and risk assessment (7.2%). **Conclusions:** Preventive dentistry research represented in the National Oral Health Research Compendium demonstrates a diverse range of preventive approaches, with particular emphasis on health promotion and implementation research. This analysis provides an overview of current research directions and may inform future research prioritisation and strengthen preventive oral health research in Malaysia.

Keywords: Preventive Dentistry, Oral Health Research, Research Priorities, Evidence Mapping, Malaysia

Presenter ID: OP76

A Comparative Analysis of National Teledentistry Regulatory Guidelines in Japan, Thailand, China, Malaysia, and Korea

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Background: Teledentistry expanded after COVID-19 beyond remote screening to consultation, triage, monitoring, oral-health education, and referral coordination. However, national approaches to teledentistry governance remain poorly understood. **Methods:** We comparatively analyzed national-level regulatory policy documents from Japan's Ministry of Health, Labour and Welfare; Thailand's Dental Council and Ministry of Public Health; China's National Health Commission; Malaysia's Ministry of Health; and South Korea's Ministry of Health and Welfare. The analytical framework was derived from the 11 implementation steps in the WHO Consolidated Telemedicine Implementation Guide. The steps were consolidated into eight dimensions for cross-country comparison by merging overlapping concepts and incorporating concepts from three WHO–ITU and FDI documents and academic literature: legal authority; accessibility and equity; service delivery and clinical scope; safety and quality; workforce roles; data and technology governance; financing and sustainability; and monitoring and evaluation. **Results:** All countries addressed legal authority, service delivery, safety, and workforce responsibilities, although scope and dental specificity varied. Japan provided the most comprehensive dentistry-specific framework, including hybrid care planning, assisted participation, referral pathways, and quality evaluation. Thailand established dentistry-specific operational standards. China emphasized institutional supervision, data governance, and monitoring without dentistry-specific provisions. Malaysia prioritized platform governance, standards, and data protection. South Korea recognized dentists within remote-care provisions but lacked dentistry-specific pathways and evaluation frameworks. Financing, accessibility, digital-health literacy, and equity were least consistently addressed. **Conclusion:** The five countries are at different stages of developing dentistry-specific guidance. Future policies should strengthen hybrid referral pathways, digital health literacy, equitable access, and systematic evaluation.

Keywords: Teledentistry, Digital Health Governance, Health Policy, Patient Safety, Health Equity

Presenter ID: OP78

Improving Oral Health Status of Orphanage Children Through Oral Probiotics and Motivational Interviewing-Based Teacher Training

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Background: Dental caries is influenced by behavioral and environmental factors, poor oral hygiene, and low parental education and socioeconomic status. This study aimed to improve the oral health status of schoolchildren through the combined use of oral probiotics, motivational interviewing (MI), and teacher training using a training on trainer (TOT) approach. **Methods:** A cross-sectional study was conducted at Sekolah Rakyat SMP Negeri 8 Cimahi using total sampling. Twenty-five students with a salivary pH below 7 were included. Oral hygiene status was assessed using the Oral Hygiene Index-Simplified (OHI-S) on Day 1 and Day 21. Oral probiotics were administered on Days 7 and 14, while oral health education using a motivational interviewing approach was delivered by trained teachers. Teachers underwent a training program, and inter-rater reliability was evaluated before implementation. **Results:** The mean DMFT index was 4.88, indicating a high level of dental caries according to the World Health Organization (WHO) classification. The mean OHI-S score improved from 2.55 before the intervention to 1.17 after the intervention. Paired *t*-test analysis demonstrated a statistically significant improvement in oral hygiene status following the intervention ($p < 0.05$). **Conclusion:** The combination of oral probiotic administration, motivational interviewing, and teacher-based training was effective in improving children's oral health status by enhancing oral health education, awareness, toothbrushing practices, and overall oral hygiene.

Keywords: Dental Caries, Oral Probiotics, Motivational Interviewing, Oral Hygiene, OHI-S, Teacher Training, Schoolchildren

Presenter ID: OP80

A Systematic Review of Caries Status of Older Adults in Asian Countries

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Background: Dental caries among older adults presents a significant public health challenge. As Asia is expected to become the global epicenter of population aging, up-to-date data on caries status in this region are essential for developing targeted public health interventions. **Methods:** Two independent researchers systematically searched three electronic databases (PubMed, Scopus, and EMBASE) and official government websites for national/regional oral health surveys conducted in Asia from 2016 onwards. Additional surveys were obtained from delegates of the Southeast Asia Association for Dental Education. Oral health surveys reporting caries prevalence and/or caries experience, using decayed, missing, filled teeth (DMFT) or carious/filled root, in older adults aged 60 or above were included. The risk of bias was evaluated using the Joanna Briggs Institute Critical Appraisal Checklist for Studies Reporting Prevalence Data. A random-effects model was employed to pool the prevalence of root caries and analyze effect sizes. **Results:** The review included 13 oral health surveys from 12 Asian countries. The proportion of older adults with caries experience ranged from 66.4% (Vietnam) to 100% (Mongolia). The pooled prevalence of untreated dental caries and root caries among Asian older adults was 51% (95% CI: 39%–63%) and 24% (95% CI: 8%–55%), respectively. All included studies demonstrated a low risk of bias. **Conclusion:** The findings indicate that the prevalence of both coronal and root caries among older adults in Asia remains high and is unevenly distributed across different regions.

Keywords: Dental Caries, Older Adults, Elderly, Oral Health, Prevention

Presenter ID: OP81

Early Interceptive Management of Single Tooth Anterior Crossbite in Primary Care: A Case Series

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Background: Anterior crossbite is an occlusal anomaly characterized by the palatal positioning of one or more maxillary anterior teeth relative to the mandibular incisors. Among Malaysian orthodontic patients, its prevalence is reported at 30.4%. This condition rarely resolves spontaneously and, if untreated, may lead to occlusal trauma, periodontal damage, and unfavourable craniofacial growth. Early interceptive management is therefore essential. This case series compares the effectiveness of removable appliances with Z-springs and bonded resin-composite slopes in correcting single-tooth anterior crossbite in a primary care setting. **Case Description:** Four pediatric patients aged 8–11 years with single-tooth anterior dental crossbite and Class I skeletal relationships were treated. Two cases with deep reverse overbite and mobile mandibular incisor received upper removable appliances with Z-springs, while two cases with minimal reverse overbite and firm mandibular incisors were managed using bonded resin-composite slopes. Patients were reviewed for up to one year. Outcomes assessed included treatment duration, correction of overjet and overbite, periodontal health, tooth mobility, and patient cooperation. All cases achieved successful correction with stable positive overjet and overbite. Treatment duration ranged from 1–3 months for removable appliances and approximately 2 weeks for bonded resin-composite slopes. No adverse periodontal or occlusal effects were observed. **Conclusion:** Both interceptive treatment modalities were minimally invasive, well tolerated, and effective in correcting single-tooth anterior crossbite in children in primary care. Treatment selection should be guided by clinical presentation, patient factors, and clinician experience. Early interceptive management in primary care may reduce the need for specialist referral.

Keywords: Anterior Crossbite, Removable Appliances, Bonded-resin Composite Slopes

Presenter ID: OP82

Preventing the Preventable: A Case Series of Preschool Children with Severe Early Childhood Caries

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Background: Severe Early Childhood Caries (S-ECC) is an aggressive form of dental caries that poses a major international health burden and challenges. Despite being largely preventable, the Global Burden of Disease Study 2017 estimated that over 530 million children worldwide suffered from primary tooth decay. Locally, the National Oral Health Survey of Preschool Children (NOHPS) Malaysia 2015 reported a caries prevalence of 71.3% among preschool children. This case series presents three preschool children, aged 3 to 5 years, who attended primary pediatric dental clinics with varying severities and complications of S-ECC. Clinical presentations ranged from arrested caries and chronic dental abscesses to severe, acute orofacial infection requiring urgent intervention. Across all three cases, shared modifiable risk factors included prolonged bottle-feeding, frequent consumption of sugary snacks and drinks, poor oral hygiene, limited fluoride exposure, and poor dental attendance. Despite differences in disease progression, these cases collectively highlight missed opportunities for timely preventive care during crucial early childhood developmental stages. In summary, S-ECC is a preventable disease with a substantial impact on children and their families. This case series underscores the importance of preventing the preventable via various strategies including collaboration among policymakers, healthcare professionals, teachers, and parents to promote oral health education, healthy behaviours, and regular dental visits in shifting paediatric dentistry from a restorative to a preventive approach. Recognizing and addressing modifiable risk factors early are essential to prevent disease progression and improving long-term oral health outcomes, ultimately reducing the burden of S-ECC nationally and globally.

Keywords: Severe Early Childhood Caries, Preventive, Risk Factors, Preventive Measures

Presenter ID: OP83

Caries Arrest and Acceptability of Community-Based Silver Diamine Fluoride for Early Childhood Caries in Sabah

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Background: Silver diamine fluoride (SDF) offers a simple, minimally invasive strategy for caries control among children. However, evidence on different formulations and acceptability in community settings remains limited. This study compared short-term caries arrest rates and treatment acceptability following Riva Star and Riva Star Aqua application among preschool children in Sabah. **Methods:** This non-randomised quasi-experimental study included preschool children with ICDAS 5 or 6 lesions receiving single-step Riva Star (SDF group) or Riva Star Aqua (AgF group). Child cooperation and parental acceptance were assessed using the Venham Behaviour Rating Scale and a validated questionnaire. Caries arrest was assessed by a trained and calibrated examiner at three months based on lesion consistency, colour, and signs and symptoms. Data were analysed using descriptive statistics and generalised estimating equations. **Results:** Among 275 children with 957 treated lesions, 85.1% demonstrated total cooperation, with no significant difference between groups (84.0% SDF vs 86.3% AgF; $p=0.582$). At one week, 272 parents completed the questionnaire. Favourable parental acceptance was higher in the AgF group than SDF group; however the difference was not statistically significant (52.4% SDF vs 59.7% AgF; $p=0.230$). At three months, 866 lesions were assessed, and 71.7% were arrested. Caries arrest was significantly higher in the SDF group than the AgF group (77.0% vs 66.5%; $p=0.045$). **Conclusion:** Most lesions were arrested at three months, with SDF demonstrating significantly higher arrest rate than AgF. Both materials demonstrated comparable acceptance in community settings. These findings support community-based SDF therapy for caries control in Sabah and similar access-limited settings.

Keywords: Early Childhood Caries, Silver Diamine Fluoride, Silver Fluoride, Preschool Children

Presenter ID: OP84

Prevention and Prenatal Screening of Rare Congenital Neonatal Oral Tumors: A Multi-Case Series from Asia

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Background: Congenital granular cell epulis (CGCE) and melanotic neuroectodermal tumor of infancy (MNTI) are among the rarest neonatal oral tumors, with fewer than 250 and approximately 200–300 cumulative global reports, respectively. While CGCE is histologically benign, large lesions compromise feeding and, rarely, respiration; MNTI exhibits local aggressiveness with recurrence rates near 15%. Despite high obstetric ultrasound coverage in Asia, prenatal detection of fetal oral lesions remains inconsistent — a preventable gap directly relevant to the preventive-dentistry mission of ICAAPD. **Case Description:** We synthesized 23 cases across three Asian tertiary centers (2008–2022): Zhejiang Children's Hospital ($n=9$ CGCE, all female, 6/9 detected at 28–39 weeks), Qingdao Women & Children's Hospital ($n=1$ three-foci CGCE of lip, tongue, and maxillary gingiva, 4.8 cm, showing the "bubble-blowing" prenatal ultrasound sign with radial CDFI flow), and Peking University ($n=13$ MNTI, 2 recurred within 1 month). We additionally report a 5-foci CGCE in a 3-day-old female, S-100 scattered-positive, managed by MDT-staged excision with no recurrence at 2-year follow-up — a case missed prenatally despite normal maternal ultrasound, underscoring blind spots in routine fetal oral sweep. **Conclusion:** Rare congenital oral tumors expose critical gaps in Asian prenatal prevention systems. Integrating a 28–32-week fetal oral gingival sweep into routine obstetric ultrasound, AI-assisted triage, and delivery-suite multidisciplinary team (MDT) consultation forms a low-cost preventive chain benefiting CGCE, MNTI, cleft palate, and high-burden paediatric oral disease alike. We propose establishment of a China–Japan–Korea–Southeast Asia congenital oral lesion registry to standardize early detection and long-term follow-up.

Keywords: Congenital Oral Tumor; Congenital Granular Cell Epulis; Melanotic Neuroectodermal Tumor of Infancy; Prenatal Screening; Multidisciplinary Team; Preventive Dentistry

Presenter ID: OP85

Time-Stratified Network Meta-Analysis of School-Based Oral Health Interventions for the Prevention of Dental Caries

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Background: This study aimed to combine and synthesize the available evidence to compare and rank school-based oral health interventions for preventing dental caries among children and adolescents using direct and indirect comparisons. **Methods:** A systematic search of PubMed, Embase, Web of Science and the Cochrane Library was conducted to identify randomized controlled trials. The review was prospectively registered in PROSPERO and data were synthesized using a frequentist network meta-analysis. Risk of bias was evaluated using the Cochrane RoB 2 tool, and the certainty of evidence was assessed using the CINeMA framework. Heterogeneity, inconsistency, sensitivity analyses, and SUCRA-based treatment rankings were evaluated. **Results:** A total of 44 randomized controlled trials out of 28,375 records screened, evaluating 20 school-based oral health interventions were included. Across ≥ 3 years follow-up periods, active interventions showed small and largely overlapping treatment effects. SUCRA rankings identified xylitol candy and fluoridated toothbrushing as the highest-ranked interventions at ≥ 3 years, while erythritol ranked highest at 2–2.5 years, followed by fluoridated toothpaste, xylitol candy, and fluoride varnish. Estimates at 1–1.5 years were less precise due to the limited number of studies. Overall, the certainty of evidence ranged from low to very low due to high risk of bias and indirectness. **Conclusions:** Despite low-to-moderate-certainty evidence, sealants, topical fluorides, and sugar alcohol-based interventions consistently demonstrated greater potential for reducing dental caries over extended follow-up periods.

Keywords: Adolescents, Children, Dental caries, Fluoride, Network Meta-analysis, Oral Health Programs, Preventive Dentistry, School Based Interventions, Sealants, Sugar Alcohols

Presenter ID: OP86

Early Childhood Caries as a Biofilm-Driven Disease

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Background: Early childhood caries (ECC) is among the most common chronic diseases of children. The condition is understood as a biofilm-mediated process driven by microbial interactions and facilitated by sugar exposure. This review summarizes the microbial and ecological basis of ECC and translates it into practical prevention strategies to guide clinical decision-making and risk assessment. **Method:** A comprehensive literature search was conducted across the PubMed and Scopus databases for English-language articles published between January 2007 and June 2026. Literature searches focused on ECC, oral biofilms, and cariogenic microorganisms to synthesize the current evidence. **Result:** Upon tooth eruption, primary enamel is rapidly colonized by oral microorganisms, establishing a structured dental biofilm. Repeated exposure to fermentable carbohydrates accelerates acid production, leading to prolonged low-pH microenvironments that drive enamel demineralization when host buffering and remineralization capacity are exceeded. *Streptococcus mutans* serve as a primary pathogen due to its acidogenicity, aciduricity, and capacity to synthesize extracellular polysaccharides from sucrose. *Streptococcus sobrinus* further enhances cariogenic potential, whereas *Lactobacillus* species are predominantly associated with the progression of established lesions. Furthermore, interkingdom interactions, such as the pathogenic synergy between *Candida albicans* and *S. mutans* within mixed bacterial-fungal biofilms, significantly increase biofilm biomass, matrix stability, and localized acid retention. **Conclusion:** ECC reflects dynamic interactions among polymicrobial biofilms, host diet, salivary factors, fluoride exposure, tooth susceptibility, oral hygiene, and caregiver behaviors. Effective prevention requires early risk assessment, caregiver-targeted counseling, and tailored interventions to maintain a healthy oral microenvironment.

Keywords: Biofilm, Caries, Childhood, Children

Presenter ID: OP10

Polymer Fluoride in Dentistry - A Scoping Review

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Background: Polymer fluoride has recently attracted increasing attention in dentistry because they can be incorporated into dental materials and enhance their clinical performance. This scoping review summarizes the dental applications of polymer fluoride. **Methods:** Two researchers independently searched Web of Science, Medline, and Scopus for English-language articles. **Results:** This review identified 3,166 records and included 43 publications. Among the 43 publications, 41 are laboratory studies. The laboratory studies showed that fluorine has high electronegativity, and fluorinated monomers confer lower surface energy and higher hydrophobicity compared with non-fluorinated analogues. These characteristics are associated with reduced polymerization shrinkage, decreased water sorption, and diminished microbial adhesion, thereby improving the mechanical and physical performance of dental materials. The studies also showed fluoridated cationic polymers have potential use as immobilized components in resin matrices or as active additives in oral care formulations. These materials also show promising antimicrobial properties, remineralizing potential, favorable biocompatibility, and good biostability, supporting their suitability for future oral healthcare applications. Fluorinated/fluoridated polymers have been synthesized and incorporated into various dental materials, including resin composites, denture base resins, adhesives, endodontic paper points, oral care products, hemostatic agents, sutures, and guided tissue regeneration membranes. **Conclusion:** The literature found polymer fluoride shows considerable promise for broad dental application, offering advantages in physical properties, antimicrobial activity, remineralizing effects, and biocompatibility. However, most studies are in vitro with limited clinical evidence.

Keywords: Polymer, Fluoride, Dentistry, Remineralization, Electronegativity

Presenter ID: OP39

Smoking and Periodontal Bleeding in Chinese Adolescents: The Mediating Role of Dental Calculus

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Background: This study aimed to investigate the association between smoking habits and gingival bleeding among Chinese adolescents aged 12–15 years. **Methods:** Drawing on data from the Fourth National Oral Health Epidemiological Survey, a multistage cluster sampling design was employed to recruit middle school students aged 12–15 years. The study comprised two components: clinical examinations and a structured questionnaire survey. Clinical examinations assessed gingival bleeding and calculus status, while the questionnaire collected information on sociodemographic characteristics, oral health behaviors, particularly smoking habits. Mediation analysis was conducted to elucidate the relationship between smoking and gingival bleeding. **Results:** A total of 118,514 students completed both the oral examination and the questionnaire. Compared with non-smokers, the smoking group exhibited significantly higher prevalence of gingival bleeding and calculus. Mediation analysis revealed that the indirect effect through calculus was significantly amplified, whereas the direct effect was significantly attenuated. The total effect of smoking on gingival bleeding was not statistically significant. **Conclusion:** Adolescent smokers presented with more severe periodontal conditions compared to their non-smoking peers. These findings underscore the necessity of strengthening smoking cessation interventions targeted at adolescents.

Keywords: Adolescents, Smoking, Gingival Bleeding, Dental Calculus, Mediation Analysis

Presenter ID: OP57

Application of Digital Management System for Early Childhood Caries Using Caries Risk Assessment in Shanghai

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Background: Early childhood caries (ECC) is a highly prevalent oral disease among preschool children in China, severely impairing children's masticatory function, speech development, and quality of life. This study aimed to implement effective, scalable and accessible prevention strategies which were urgently needed. **Methods:** In this study, we implemented a digital management system for ECC, which delivered stratified interventions based on the results of caries risk assessment (CRA). Children with moderate-to-low caries risk received 2 sessions of topical fluoride application per year, while those with high caries risk received 4 sessions annually. The system integrates multiple functions including oral examination documentation, delivery of personalized oral health education content, and regular examination reminders, to provide children with tailored, risk-based caries management services. **Results:** We rolled out this digital system in Shanghai, enrolling a total of 5060 preschool children (2580 males and 2480 females) as study participants, with a one-year follow-up conducted to evaluate the intervention effect. CRA classified 496 (9.80%) as high-risk, 1015 (20.06%) as moderate-risk, and 3549 (70.14%) as low-risk. The follow-up new caries incidence was 20.44%, significantly lower than the 40%~50% incidence of peer controls in the same region, demonstrating remarkable prevention efficacy. **Conclusion:** This digital system enables effective risk-stratified ECC management, enhancing intervention efficacy via personalized supports and stratified fluoride intervention. It provides a low-cost, scalable model with wide population coverage that supports “early detection and prevention” and reduces the treatment burden of pediatric oral diseases.

Keywords: Preschool Children; Early Childhood Caries; Caries Risk Management; Digital Health; Fluoride Intervention

POSTER PRESENTATION

Presenter ID: PP1

Bridging the Gap: Dental Specialist Workforce Projections in Malaysia (2021–2030)

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Background: Effective health human resource planning is vital for ensuring "the right people with the right skills" are available to meet national oral health goals. This study projected the supply and requirements for dental specialists in Malaysia from 2021 to 2030 across nine key specialties, including Periodontics, Orthodontics, Restorative Dentistry, Dental Public Health, Paediatric Dentistry, Forensic Odontology, Special Care Dentistry, Oral Pathology & Oral Medicine and Oral Maxillofacial Surgery. **Methods:** Using a system dynamics modeling approach within a needs-based framework, a generic projection model was developed in Vensim simulation software. The model integrated three main components: a supply model (encompassing training, population, and Full-Time Equivalent sub-models), a requirement model (addressing demand based on population health status), and a gap model to identify workforce discrepancies. **Results:** Simulation results indicated significant workforce shortages by 2030 across several specialties. For instance, Paediatric Dentistry showed a projected supply of 133 against a requirement of 220, while Dental Public Health faced a gap of 114 specialists. Oral and Maxillofacial Surgery (OMFS) exhibited the highest demand, with a projected requirement of 353 specialists compared to a supply of 180. Accuracy was validated using root mean square error (RMSE), with values ranging from 0.1 to 0.3. **Conclusion:** A critical gap exists between the projected supply and the required number of dental specialists in Malaysia. To bridge this gap, the study recommends increasing postgraduate intake annually, expanding postgraduate training through international collaborations, and implementing strategic deployment programmes for underserved regions such as Sabah and Sarawak.

Keywords: Dental Specialist Workforce; Workforce Projection; System Dynamics Modelling; Human Resource Planning; Oral Health Workforce; Malaysia

Presenter ID: PP2

Global Research Trends on Sugar Reduction Policies for Dental Caries Prevention: A Bibliometric Analysis

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Background: Excessive sugar intake and sugar-sweetened beverages are established risk factors for dental caries, and policy-based prevention has become an important public health strategy. However, the global research landscape on sugar reduction policies in relation to dental caries prevention remains insufficiently mapped. **Methods:** A bibliometric analysis was conducted using Scopus and Web of Science. Searches were performed until May 2026 using terms related to dental caries/oral health, sugar reduction, sugar sweetened beverages, sugar taxation, food labelling, school food policies, prevention, intervention, and public health. Only English language articles and reviews within dentistry related records were included. After screening and duplicate removal, 203 publications from 1990 to May 2026 were analysed. **Results:** The final dataset comprised 147 articles and 38 reviews, with an annual growth rate of 8.82%. Publication output increased markedly in recent years, peaking in 2025 with 34 publications. The United States produced the highest number of documents, followed by the United Kingdom and Australia. Saudi Arabia recorded the highest total citations, while Sweden showed high citation impact. Community Dentistry and Oral Epidemiology, BMC Oral Health, and the British Dental Journal were among the most productive journals. The most cited publication was Moynihan's 2014 systematic review on sugar restriction and caries prevention. **Conclusion:** Research on sugar reduction policies for dental caries prevention has expanded substantially, reflecting growing global interest in policy-driven oral health promotion. Findings highlight influential countries, journals, and publications, while supporting the need for stronger multidisciplinary research on sugar-related caries prevention strategies.

Keywords: Bibliometric Analysis, Dental Caries, Sugar Reduction, Sugar-Sweetened Beverages, Oral Health Policy

Presenter ID: PP3

Trust Before Treatment: Building Community Trust in Oral Healthcare Outreach Among Orang Asli in Selangor

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Background: Establishing trust is essential in healthcare delivery among marginalised Indigenous populations. In *Orang Asli* (OA) communities, oral healthcare outreach programmes frequently encounter challenges related to fear, communication barriers, and limited community engagement. Understanding how frontline oral health personnel build trust within these communities may improve programme effectiveness and long-term participation. This study explored trust-related challenges and trust-building strategies experienced by oral health personnel involved in OA outreach programmes in Selangor. **Methods:** A qualitative study was conducted through seven focus group discussions involving 39 purposively selected oral health personnel from seven districts in Selangor. Participants included dental officers, dental therapists, dental surgery assistants, healthcare assistants, and drivers directly engaged in OA outreach programmes. Data were analysed thematically, with data collection continuing until thematic saturation was reached. **Results:** Participants identified trust as a key factor influencing community participation in oral healthcare programmes. Fear of dental treatment, avoidance of dental personnel, language barriers, and limited oral health awareness reduced engagement. Trust-building approaches included early engagement with Tok Batin and community leaders, learning basic OA language phrases, culturally sensitive communication, and ongoing interactions. Informal engagement activities, incentives, and collaboration with local agencies enhanced rapport and encouraged participation. Culturally appropriate communication platforms, including OA radio channels, improved information dissemination and increased community acceptance. **Conclusion:** Trust was perceived as a critical determinant of successful oral healthcare outreach among OA communities. Relationship-centred and culturally responsive approaches may strengthen community engagement, reduce barriers to care, and support equitable oral healthcare delivery for Indigenous populations.

Keywords: Orang Asli; Trust-building; Indigenous Health; Oral Healthcare Outreach; Qualitative Study; Community Engagement

Presenter ID: PP5

Malay Version of the Patient Denture Satisfaction Questionnaire (M-PDSQ): Cross-Cultural Adaptation and Psychometric Evaluation

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Background: Generic oral health-related quality of life tools and single-item scales may overlook specific prosthodontic patient-reported concerns. Consequently, developing a culturally adapted, denture-specific instrument can help more accurately assess patient satisfaction and improve service delivery. This study aimed to develop and psychometrically evaluate the Malay version of the Patient Satisfaction Questionnaire for Denture Wearers (M-PDSQ) among Malaysian adults. **Methods:** The M-PDSQ was developed and validated in three phases: (1) item adoption/adaptation from literatures; (2) cross-cultural adaptation involving forward/backward translation finalised by a linguist; and (3) psychometric evaluation. The psychometric evaluation phase included content validation by five dental experts and face validation with six patients. To establish construct validity and internal consistency reliability, a pilot study with a minimum sample of 147 participants was executed. Exploratory Factor Analysis (EFA) and internal consistency (IC) using Cronbach's alpha were analysed with SPSS Version 30. **Results:** Cross-cultural adaptation successfully achieved semantic equivalence. The resulting M-PDSQ comprises four domains (Aesthetics, Physical Function/Stability, Comfort, and Psychosocial) demonstrating strong content validity (mean I-CVI = 0.91). Exploratory Factor Analysis confirmed a robust unidimensional structure for each domain across upper and lower dentures, yielding high Kaiser-Meyer-Olkin values (0.782–0.874, $p < 0.001$) and explaining >71% of the total variance. The instrument demonstrated high internal consistency reliability (Cronbach's $\alpha = 0.908$ –0.953). **Conclusion:** The M-PDSQ is a valid, reliable, and culturally tailored psychometric tool for evaluating denture satisfaction. Standardising this patient-reported outcome measure in primary dental care will facilitate evidence-based improvements in denture service delivery across Malaysia.

Keywords: Denture Satisfaction, Cross-cultural Adaptation, Psychometrics, Questionnaires, Adult

Presenter ID: PP6

Out-of-Hours Paediatric Dental Referrals at Tertiary Referral Centre in Central Malaysia: A Retrospective Cross-Sectional Study

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Background: Out-of-hours (OOH) paediatric dental referrals contribute to the workload of hospital emergency services. However, there is limited local data on the utilisation of these services. This study aimed to describe the characteristics of OOH paediatric dental referrals and evaluate response times and clinical management at a tertiary referral centre. **Methods:** A cross-sectional study was conducted at the Department of Paediatric Dentistry, Hospital Tunku Azizah, involving patients aged below 16 years. Data were collected from on-call records. A total of 219 cases were included using convenience sampling. Data collected included patient demographics, referral sources, clinical need categories (emergency, urgent, or routine) based on the Commissioning Standard for Urgent Dental Care, response times, and treatment outcomes. Data were analysed using descriptive statistics and the Chi-square test. **Results:** Majority of patients were male (62.1%), with most referrals originating from the Emergency Department Green Zone (81.7%). Emergency cases accounted for 50.7% of referrals, while 49.3% were classified as urgent (13.2%) or routine (36.1%), suggesting that many emergency department referrals did not require immediate emergency dental intervention. Overall, 93.2% of cases met the recommended response time standards. A significant association was found between clinical need category and treatment provided ($p < 0.05$), with emergency cases more likely to receive immediate active treatment and routine cases managed with follow-up care. **Conclusion:** The OOH paediatric dental service demonstrated good compliance with response time standards, with treatment decisions appropriately aligned with patients' clinical needs. However, almost half of the referrals involved non-emergency conditions. Optimising referral pathways and improving access to primary dental services may help reduce inappropriate OOH utilisation and ensure that resources remain available for genuine dental emergencies.

Keywords: Out-of-hours (OOH), Paediatric Dental, Referrals

Presenter ID: PP7

Cost-Effectiveness of a Program to Empower Dental Health Cadres in Caries Prevention

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Background: Oral diseases remain a major public health challenge in Indonesia. The 2023 Indonesian Health Survey reported that 56.9% of the population experienced oral health problems, while only 6.2% practiced toothbrushing at the recommended times. Given the high cost of restorative dental care and limited access to oral health services, community-based preventive interventions are needed. This study evaluated the effectiveness and cost-effectiveness of a Dental Health Cadre Empowerment Program for caries prevention. **Methods:** A quasi-experimental one-group pre-test-post-test design was conducted involving 43 of 62 registered dental health cadres. The intervention consisted of structured training in oral health promotion and caries prevention, after which cadres provided oral health education to 108 community members. Changes in cadre knowledge were assessed using pre- and post-test questionnaires and analyzed with the Wilcoxon signed-rank test. **Results:** The mean cadre knowledge score increased significantly from 8.6 before training to 9.8 after training ($p = 0.001$), representing a 13.9% improvement. Following the intervention, 108 community members received oral health education, with a mean knowledge score of 8.6. The total program cost was IDR 593,000, equivalent to IDR 13,791 per trained cadre and IDR 5,491 per community beneficiary. The cost required to achieve a one-point increase in cadre knowledge was IDR 494,167, while the cost per 1% improvement in knowledge was IDR 42,662. **Conclusion:** The Dental Health Cadre Empowerment Program improved cadre knowledge while requiring relatively low implementation costs. These findings indicate that cadre empowerment is an effective and cost-efficient community-based strategy to strengthen oral health promotion and support caries prevention, particularly in resource-limited settings.

Keywords: Dental Health Cadres, Oral Health Promotion, Caries Prevention, Cost-effectiveness Analysis, Community Empowerment

Presenter ID: PP8

Effectiveness and Safety of Alternative Restorative Materials versus Dental Amalgam: Systematic Review and Meta-Analysis

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Background: The global commitment to phase down dental amalgam has accelerated the adoption of mercury-free restorative materials. Evaluating their clinical effectiveness and safety compared with dental amalgam is essential to support evidence-based implementation of mercury-free dentistry. **Methods:** A systematic review and meta-analysis were conducted to evaluate alternative restorative materials and dental amalgam for the treatment of dental caries. Studies published between 2006 and May 2025 were identified through searches of Cochrane CENTRAL, PubMed, and OVID databases. Randomized controlled trials comparing alternative restorative materials with dental amalgam were included. Outcomes assessed included clinical effectiveness and safety. Risk of bias was assessed using the Cochrane Risk of Bias 2.0 tool and certainty of evidence using GRADE. Meta-analysis was conducted for eligible outcomes. **Results:** Fourteen studies were included, with four studies eligible for meta-analysis. Most studies involved children aged 6–12 years, and the overall certainty of evidence was low. Composite and compomer restorations were associated with a higher risk of secondary caries than amalgam restorations (RR 3.42). No significant difference in restoration survival was observed between glass ionomer cement and amalgam restorations (RR 1.00; 95% CI 0.96–1.04). Although amalgam restorations were associated with higher urinary mercury concentrations and increased microalbuminuria, no clinically significant adverse effects on renal, neuropsychological, immunological, or physical developmental outcomes were identified. **Conclusion:** Mercury-free restorative materials showed variable clinical performance compared with dental amalgam. No clinically significant adverse health effects were identified; however, the low certainty of the evidence highlights the need for further high-quality studies.

Keywords: Effectiveness, Alternative Restorative Materials, Safety, Dental Amalgam

Presenter ID: PP9

Integration of Preventive Oral Health into Antenatal Care in Indonesia: A Scoping Review

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Background: Preventive oral healthcare is an important component of antenatal care because maternal oral health affects well-being and may be associated with adverse pregnancy outcomes. However, its integration into antenatal services across Indonesia remains unclear. This scoping review examined integration approaches, implementation factors, and maternal oral-health outcomes. **Methods:** PubMed, Web of Science, and GARUDA were searched for evidence regarding pregnancy, antenatal care, oral health, and Indonesia. Eligible sources involved pregnant women, antenatal services, or healthcare providers and reported integration, implementation, or maternal oral-health outcomes. Records were deduplicated and screened by title, abstract, and full text. Findings were synthesized. **Results:** Searches yielded 56 records, of which 27 reports were sought and 26 full texts were assessed and included. Integration was reported through guideline-recommended dental examinations, community health centres, multidisciplinary training, the Maternal and Child Health handbook, and digital education and self-screening. However, one study documented no dental examination in 84.6% of antenatal records, while another reported dental service utilisation during pregnancy of only 25.9%. Barriers included limited patient and provider knowledge, low perceived need, financial and geographic constraints, workforce limitations, unclear responsibilities, and gaps in policy implementation. Reported outcomes included limited dental attendance, suboptimal oral health practices, dental caries, erosion, gingival inflammation, periodontal treatment needs, and poorer oral-health-related quality of life. Digital interventions improved oral health knowledge, with one study also reporting improvements in self-reported behaviour. **Conclusion:** Preventive oral health is incorporated into some Indonesian antenatal services, but implementation remains inconsistent. Stronger provider training, referral pathways, consistent policy implementation, and accessible education are needed.

Keywords: Oral Health, Prenatal Care, Preventive Dentistry, Integrated Health Care, Indonesia

Presenter ID: PP10

Managing Unmet Dental Treatment Needs in Kota Belud and Ranau: Mobile Dental Operators' Perspectives

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Background: The mobile dental team provides essential oral healthcare for schoolchildren in rural areas of Kota Belud and Ranau. This study explored operators' perspectives on strategies and barriers in managing unmet dental treatment needs among schoolchildren in both districts. **Methods:** Semi-structured in-depth interviews were conducted with 6 operators (3 dental officers and 3 dental therapists) involved in the school mobile dental programme. The interviews were transcribed, anonymised, and thematically analysed using an inductive method to identify emerging themes. **Results:** Operators frequently reported severe untreated caries and complex dental anomalies. Within restricted school schedules and limited chairside time, operators employed strict triage prioritising active permanent tooth caries, completing restorations, delayed extractions to the final day of treatment and emphasising on preventive care such as fissure sealants. However, operations were heavily compromised by poor road conditions and unstable school electricity supply that disrupted the service delivery and the use of essential equipment. Additional hurdles include limited dental equipment, unexpected restorative material shortages due to higher than anticipated treatment demand, difficult moisture control and challenging work environments. While complex cases were referred to the dental clinics, referral attendance was observed extremely low due to transport and financial barriers. **Conclusion:** Mobile dental team operators address unmet treatment needs through strategic clinical prioritization and preventive care but delivery is heavily constrained by logistical, environmental, and resource challenges. Low referral adherence remains a critical concern. These findings suggest that improving mobile dental care requires not only better operational support but mechanisms to help schoolchildren complete referred treatment after the mobile visits.

Keywords: Unmet dental needs, mobile dental services, dental care accessibility, qualitative research, rural health, Sabah

Presenter ID: PP11

Who You Live With Matters: Living Arrangements Shape Access to Oral Healthcare Among Older Adults

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Background: Living arrangement is an important social determinant of health that may influence older people's access to oral healthcare. While its association with general health has been widely explored, evidence describing how living arrangements affect access to oral healthcare remains fragmented. This scoping review aimed to map the existing evidence on types of living arrangement among community-dwelling older adults and examine how these arrangements influence access to oral healthcare. **Methods:** A scoping review was conducted following the Joanna Briggs Institute methodology and reported according to the PRISMA Extension for Scoping Reviews (PRISMA-ScR). Three electronic databases; PubMed, Scopus and Web of Science were systematically searched for studies published between 1996 and 2025. Studies were selected based on the Population–Concept–Context framework. Data were charted and synthesised to identify recurring types of living arrangements and their barriers to oral healthcare. **Results:** Of the 399 records screened, 12 studies met the eligibility criteria. Four categories of living arrangements were identified; living alone, living with a spouse or partner, living with family members or adult children, and living with non-family members. Across the included studies, living arrangements influenced oral healthcare access through availability in social support, financial resources, transportation availability and, household and caregiver context. **Conclusion:** Living arrangements influence access to oral healthcare through multiple interconnected social and environmental pathways. Integrating this social determinant into oral health policies and service planning may improve equitable access to dental care and support healthy ageing among older populations.

Keywords: Living Arrangement, Older Adults, Oral Healthcare

Presenter ID: PP12

A Decade of Oral Health Change: Dental Caries Trends among School Children, Sibü, Sarawak 2015-2024

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Background: Dental caries is a widespread oral disease sharing common risk factors, social and commercial determinants with many non-communicable conditions. The 2017 national survey highlighted that school children in Sarawak suffered from poorer dental caries status compared to their peers in other states. Because localized data is essential for evidence-based healthcare planning in a diverse state like Sarawak, this study aimed to evaluate the dental caries status and treatment needs of 12-year-old students enrolled in the Ministry of Health Malaysia (MOH) incremental oral health care programme in Sibü Division between 2015 and 2024. **Methods:** This cross-sectional study examined eight cohorts of 12-year-olds from Ministry of Education Malaysia (MOE) primary schools in Sibü. Data spanning 2015 to 2024 (excluding the 2020 - 2021 pandemic school closures) was extracted from the MOH Health Information Management System. Diagnoses followed World Health Organization (WHO) guidelines, utilizing the Decayed, Missing, and Filled Teeth (DMFT) index. **Results:** Over the 10-year period, caries prevalence among the cohorts steadily declined from 35.9% in 2015 to 23.0% in 2024. Similarly, caries experience (mean DMFT) dropped from 0.79 to 0.48, and the corresponding treatment needs (mean decayed teeth, mean D) decreased from 0.22 to 0.17. **Conclusion:** While the dental caries status of 12-year-olds in Sibü Division demonstrated gradual improvement from 2015 to 2019, further proactive measures are required to ensure these children maintain sustainable, long-term oral health after their initial treatments are completed.

Keywords: Dental Caries, Caries Prevalence, Caries Experience, School Children, Sibü, Sarawak

Presenter ID: PP13

Prevalence of Impacted Maxillary Canines and Associated Dental Anomalies: A Single-Centre Orthodontic Study

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Background: Impacted maxillary canines (IMC) frequently cause eruption and occlusal complications. This study aimed to determine the prevalence of IMC and its association with other dental anomalies among orthodontic patients in Sabah, Malaysia. **Methods:** A retrospective cross-sectional study analyzed records of 200 new orthodontic patients attending Klinik Pergigian Ortodontik Kota Kinabalu (KPOKK) from January to December 2019. Data collected included gender, ethnicity, and specific dental anomalies (tooth number, shape, size, and eruption/exfoliation patterns). Associations were evaluated using Chi-square tests, Fisher's Exact tests, and odds ratios (OR), with significance set at $p < 0.05$. **Results:** The overall prevalence of IMC was 8.5%. Statistical analysis revealed no significant association between IMC and demographics, including gender ($p = 0.300$) and ethnicity ($p = 0.395$). However, IMC was highly associated with concurrent dental anomalies ($p < 0.05$). Patients exhibited significantly higher odds of canine impaction when presenting with number (OR = 13.51), shape (OR = 9.11), eruption/exfoliation (OR = 8.28), and size anomalies (OR = 6.58). **Conclusion:** The findings demonstrate a relatively high prevalence of IMC strongly associated with specific dental anomalies. Early identification of these anomalies serves as a vital clinical indicator for potential canine impaction, emphasizing the need for comprehensive diagnostic assessments to support early preventive interventions.

Keywords: Impacted Canine, Impacted Maxillary Canine, Dental Anomalies

Presenter ID: PP14

Oral Health Status of Japanese and Foreign Children Residing in Japan

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Background: The purpose of this study is to examine the oral health status of children of Japanese nationals and foreign residents.

Methods: In Japan, infant and toddler health checkups are conducted in accordance with the law. The subjects of this study were 25,170 (24,694 Japanese and 476 foreign nationality) 18-month-olds and 37,794 (37,081 Japanese and 713 foreign nationality) 3-year-olds who underwent health checkups in Kobe City. This study was approved by the Ethics Committee of Osaka Dental University (Approval #111353) and the Research Ethics Review Committee for Kobe City Health Projects (Approval #6-2).

Results: In the 18-month health checkups, the mean number of present teeth was 15.3 ± 2.0 in Japanese children and 15.1 ± 2.4 in foreign children, with no significant difference. However, df prevalence (1.0% vs. 6.3%), df index (0.0 ± 0.3 vs. 0.1 ± 0.8), and df teeth rate (0.2% vs. 1.2%) were significantly higher in foreign children ($P < 0.05$). No filled teeth were observed in foreign children. In the 3-year-old, the mean number of present teeth was 19.9 ± 0.4 in Japanese and 19.9 ± 0.3 in foreign children, with a significant difference favoring foreign children ($P < 0.05$). df prevalence (8.7% vs. 29.7%), df index (0.3 ± 1.2 vs. 1.2 ± 2.5), and df teeth rate (1.4% vs. 6.1%) were significantly higher in foreign children ($P < 0.05$). **Conclusion:** Based on these findings, foreign children residing in Japan were suggested to be a high-risk group for dental caries compared with Japanese children.

Keywords: dental caries, Maternal and Child Health, Community Public Health

Presenter ID: PP15

The Interaction Between The Effects of Toothbrushing Behavior Patterns, Halitosis Symptoms, and Gum Health

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Background: Halitosis affects social confidence and quality of life, while gingival health is central to oral health. About 30% of Chinese adults experience halitosis. As toothbrushing is essential for daily oral hygiene, examining its impact on halitosis and the role of gingival health has important clinical and practical significance. This study explored the association between toothbrushing behavior patterns, halitosis symptoms, and gum health. **Methods:** A cross-sectional survey and retrospective analysis were conducted among outpatients, accompanying family members, and community volunteers. Data were collected through questionnaires, oral examinations, and breath analysis using a Breathalyzer Model 8170. **Results:** Among 281 participants who completed questionnaires and oral examinations, 17.8% used the Bass brushing technique and 44.5% brushed casually. The Bass group had a lower mean Gingival Index than the casual brushing group (0.185 vs. 0.378), and a lower reported gingival bleeding rate (26.4% vs. 40.0%). Among 111 participants who underwent breath analysis, mean volatile sulfur compound levels were lower in the Bass group than in the casual brushing group (193.80 vs. 260.99 ppb). Breath scores 2 were also less frequent in the Bass group (31.8%) than in the casual brushing group (53.7%). Trends in gingival index and breath values were consistent across groups. **Conclusion:** Toothbrushing method is associated with halitosis severity and gingival inflammation. The Bass brushing technique may help improve gingival health and reduce halitosis-related indicators.

Keywords: Toothbrushing Behavior, Bass Brushing Technique, Halitosis, Gingival Index, Oral Health

Presenter ID: PP16

Epidemiological Study of Socioeconomic and Oral Health Behavioural Determinants of Caries in Children in Brunei

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Background: There was limited evidence on the associations between dental caries and its determinants among children in Brunei. This study assessed the associations of the prevalence of caries (dmft>0) among children in Brunei with socioeconomic characteristics and oral health behaviours. **Methods:** A cross-sectional study was conducted in 2023 involving 345 5-year-old children from 18 government and private schools. Data collection included parental questionnaires and dental examinations conducted using the British Community Dentistry index. Chi-squared tests examined associations between the prevalence of caries and socioeconomic characteristics. Meanwhile, logistic regression examined associations between the prevalence of caries and oral health behaviours, adjusting for sex, school type, parental education and household income. **Results:** A higher prevalence of caries among children was associated with attending government schools (72.3%), having parents with the lowest education level (76.0%) and coming from the lowest-income households (71.4%) (all $p<0.001$). Starting toothbrushing after 6 months of age increased the odds of caries experience prevalence (OR: 2.67; 95% CI: 1.31–5.45; $p=0.007$) compared with starting at 6 months or earlier. Parental supervision during toothbrushing (OR: 0.24; 95% CI: 0.09–0.63; $p=0.004$) compared with no supervision, brushing twice or more daily (OR: 0.24; 95% CI: 0.09–0.65; $p=0.005$) compared with once daily, and stopping breastfeeding between 12–24 months (OR: 0.49; 95% CI: 0.24–0.99; $p=0.046$) compared with stopping after 24 months were associated with lower odds of caries experience prevalence. **Conclusion:** Several socioeconomic characteristics and oral health behaviours were associated with the prevalence of caries experience among children in Brunei.

Keywords: Caries, Children, Socioeconomic, Behaviours, Inequalities

Presenter ID: PP17

The Prevalence of Malocclusion and Orthodontic Treatment Needs among 14-Year-Old Schoolchildren in The Federal Territory of Labuan, Malaysia

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Background: Malocclusion is a common oral health condition that may affect function, aesthetics, and psychosocial well-being. However, low referral rates to the orthodontic unit and limited epidemiological data raise concerns. Therefore, this study aimed to determine the prevalence of malocclusion and orthodontic treatment need among schoolchildren in Labuan. **Methods:** A cross-sectional study was conducted among 379 14-year-old students from ten public secondary schools in Labuan, selected using simple random sampling. Malocclusion and orthodontic treatment need were assessed using the Index of Orthodontic Treatment Need (IOTN), comprising the Dental Health Component (DHC) and Aesthetic Component (AC), alongside sociodemographic data collected using a data collection sheet. Clinical examinations were performed by two calibrated examiners following specialist calibration, achieving kappa values of 0.792–1.000. **Results:** Malocclusion traits were common, with crowding as the most prevalent finding, affecting 56.7% of the upper and 64.9% of the lower dentition. Upper arch crowding was more common among females, whereas lower arch crowding was more common among males. Based on the DHC, 48.3% of students demonstrated a definite treatment need (Grades 4–5), whereas only 6.9% perceived a high need based on the AC. **Conclusion:** There is a high prevalence of malocclusion and orthodontic treatment need among schoolchildren in Labuan. The discrepancy between normative and self-perceived treatment need suggests limited awareness of malocclusion, highlighting the importance of professional screening, early detection and oral health promotion activities. Low referral rates further highlight the need for enhanced training of primary care dental officers to support appropriate orthodontic referral.

Keywords: Index of Orthodontic Treatment Need (IOTN), Orthodontic, Malocclusion, Schoolchildren

Presenter ID: PP18

Pattern and Distribution of Cleft Lip and Palate Patient Seeking Orthodontic Management

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Background: This study aim to investigate the new cases and distribution of various oral clefts of patients with cleft lip and/or palate (CLP) receiving Orthodontic treatment in the Orthodontic Department in Queen Elizabeth II Hospital, Kota Kinabalu. **Methodology:** A retrospective study was performed on 100 records of patients with cleft lip and/or palate that received Orthodontic treatment in the Orthodontic Department in Queen Elizabeth II from Hospital, Kota Kinabalu from Jan 2017 to Jan 2020. Descriptive analysis and chi-square test were used in this study to evaluate the association between oral clefts with gender, races and locality. **Results:** In 100 patient's records, the majority of 75% of male outweigh 25% of females who have cleft lip and/or palate. The highest number of types of oral cleft recorded was cleft lip and palate (89%). But, there is no significant association between the type of oral cleft and gender (P-value 0.160). Among all the races in Sabah, Other Bumiputera has 25% oral cleft followed by Dusun 22%, chinese 16%, Kadazan 13%, bajau 10%, Malay 7%, Rungus 4% and Murut 3% whereas there is none oral cleft recorded for Indian. Oral cleft has the most distribution in Area 1 (52%) as compared to Area 2 (10%) and Area 3 (38%). **Conclusion:** This study will provide a baseline insight for future research regarding oral cleft in Sabah, Malaysia and to come up with comprehensive treatment for the benefit of oral cleft patients in this study area.

Keywords: Sociodemographic, Cleft Lip, Cleft Palate, Cleft Lip and Palate, Cleft Alveolar, Queen Elizabeth Hospital II

Presenter ID: PP20

Dental Treatment Needs Among Patients Receiving Methadone Replacement Therapy: Evidence from A Scoping Review

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Background: Patients receiving Methadone Replacement Therapy (MRT) often experience substantial oral health problems and dental treatment needs, yet oral healthcare remains inadequately integrated into addiction treatment services. Although these issues among this population are increasingly recognised, the characteristics of the existing evidence and the spectrum of the reported dental treatment needs have not been comprehensively mapped. This study aimed to examine the available oral health evidence and summarise reported dental treatment needs among MRT patients. **Methods:** A scoping review was conducted following the Joanna Briggs Institute (JBI) framework. Four databases (PubMed, Ovid MEDLINE, Scopus, and Web of Science) were systematically searched for English-language studies related to oral health among MRT patients. Eligible studies underwent screening, data extraction, and descriptive synthesis, focusing on study characteristics, populations, settings, geographical distribution, and reported dental treatment needs. **Results:** Twenty-eight studies published between 1984 and 2025 were included. The available evidence was predominantly derived from cross-sectional studies and originated mainly from Europe (39.3%) and Asia (32.1%). Most studies directly involved patients receiving MRT, while others included comparison groups (i.e., prison populations and inpatient rehabilitation settings). Dental treatment needs were broad but inconsistently reported, with preventive and prosthodontic care being the most frequently identified treatment needs (21.4% of studies each), followed by restorative treatment (14.3%) and extractions (10.7%). **Conclusion:** Current evidence on oral health among MRT patients remains limited and largely descriptive but consistently indicates unmet dental treatment needs. Strengthening the integration of preventive, restorative, and rehabilitative oral healthcare within MRT services may improve access to comprehensive dental care for this population.

Keywords: Methadone Replacement Therapy, Oral Health, Dental Treatment Needs, Preventive Dentistry, Scoping Review

Presenter ID: PP21

Handgrip Strength Mediates the Association between Edentulism and Depressive Symptoms among Older Indonesian Adults

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Background: Edentulism, defined as complete tooth loss, is a prevalent oral condition among older adults linked to depressive symptoms, though underlying mechanisms remain unclear. Physical function, particularly handgrip strength, may represent a critical pathway linking oral to mental health. This study examined whether handgrip strength mediates the association between edentulism and depressive symptoms among older adults. **Methods:** This cross-sectional study used data from the Indonesian Family Life Survey-5 (IFLS-5, N=3,663). Handgrip strength was measured via hand dynamometer and analyzed as a continuous z-score, depressive symptoms were assessed using the 10-item Center for Epidemiologic Studies Depression Scale as a binary outcome. Missing data were addressed via multiple imputation. Generalized structural equation modeling estimated the mediation model, specifying a Gaussian family with an identity link for handgrip strength and a Bernoulli family with a logit link for depressive symptoms. Direct and indirect effects were estimated via the product-of-coefficients method, while adjusting for sociodemographic and health-related factors. **Results:** Edentulism was significantly associated with lower handgrip strength ($\beta=-0.115$, 95% CI: $-0.204, -0.026$, $p=0.011$), and lower handgrip strength was associated with greater depressive symptoms ($\beta=-0.270$, 95% CI: $-0.455, -0.086$, $p=0.004$). The indirect effect approached significance ($\beta=0.031$, 95% CI: $-0.002, 0.064$, $p=0.060$). Both the direct effect ($\beta=-0.277$, 95% CI: $-0.710, 0.156$, $p=0.209$) and total effect ($\beta=-0.246$, 95% CI: $-0.675, 0.183$, $p=0.261$) were non-significant. **Conclusion:** These findings suggest handgrip strength may partly explain the edentulism–depression link, though the evidence was not statistically conclusive. Physical function remains a promising, modifiable target for future longitudinal and intervention studies.

Keywords: Edentulism, Handgrip Strength, Depressive Symptoms, Older Adults, Mediation Analysis

Presenter ID: PP22

Decade of Change: Trends in Dental Caries among 12-Year-Old SchoolChildren in Kapit Division, Sarawak, Malaysia

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Background: Dental caries is the most prevalent oral health condition globally. The previous three series of National Oral Health Survey of School Children (NOHSS) in 1997, 2007 and 2017 only reported dental caries status at state level but not at divisional level in Sarawak, such as Kapit Division. The study aimed to determine the changes in dental caries prevalence (%), dental caries experience (mean DMFT) and dental caries treatment need (mean D) of nine cohorts of 12-year-olds from over an 11-year period (2015 - 2025). **Methods:** This study involved secondary data analysis using oral health data extracted from the Ministry of Health Malaysia Health Information Management System (MOH-HIMS) and MOH Dental Information System from 2015 to 2024 (no data for 2020 and 2021 due to school closure during COVID-19 pandemic). Dental caries was diagnosed by trained Dental Therapists in accordance with the World Health Organization (WHO) criteria using the Decayed, Missing, and Filled Teeth (DMFT) index. **Results:** The results demonstrate that caries prevalence among 9 cohorts of 12-year-olds displaying a decreasing trend from 2015 (54.6%) to 2025 (43.7%). The mean DMFT also showed a reduction from 2015 (1.55) to 1.22 (2025). The dental treatment needs (mean D) decreased from 0.44 in 2015 to 0.32 in 2025 with the exception of 2022 (0.68). **Conclusion:** The dental caries status of 12-year-olds in Kapit Division is showing a reducing trend, and this trend should be sustained with more pragmatic and proactive preventive strategies towards lifetime of health smiles among Kapit population.

Keywords: Dental Caries, Caries Prevalence, Caries Experience, School Children, Kapit, Sarawak

Presenter ID: PP23

Oral Health Status of 12-Year-Old Schoolchildren in Tawau From 2017 To 2024: A Six-Year Review

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Background: Dental caries is a progressive oral disease that can negatively affect children's quality of life. The Incremental Dental Care (IDC) Program was introduced in 1985 to improve the oral health status of schoolchildren. This study aimed to compare the oral health status of 12-year-old schoolchildren in urban, rural, and outskirt areas of Tawau. **Methods:** Annual reports from the Health Information Management System (HIMS) from 2017 to 2024 were reviewed. The mean DMFX score, total decayed teeth, dental caries prevalence, and caries-free prevalence among 12-year-old schoolchildren were collected. Primary schools that had implemented the IDC Program before 2017 were included, while private schools were excluded. Data were analysed using SPSS Version 29 with a significance level of $p < 0.05$. **Results:** Mean DMFX scores initially increased from 2017, then declined in later years in urban regions from 1.9 to 1.67, whereas a worsening trend was observed in both the rural (1.92 to 2.12) and outskirt regions (2.5 to 2.55). Caries prevalence in urban areas declined from 60.58% to 54.75% in 2024, while caries-free prevalence in outskirt areas decreased from 38.02% to 28.55% in 2024. Significant differences in mean DMFX and mean decay were observed across different regions, with a positive correlation. **Conclusion:** Despite continuous implementation of the IDC Program, disparities in oral health status persisted among schools in different geographical regions. Enhanced preventive measures such as implementation of water fluoridation and oral health outreach initiatives are essential, particularly in rural and outskirt regions of Tawau.

Keywords: Oral Health Status, Tawau, 12-year-old Schoolchildren

Presenter ID: PP24

Parity and Tooth Loss in Women: A Scoping Review of Biological and Socio-Behavioral Mechanisms

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Background: The saying 'a tooth per child' encapsulates the paradox of women having fewer teeth despite being periodontally healthier than men. Parity (number of births) has been shown to be associated with missing teeth, and understanding the mechanisms behind this relationship is paramount to advancing SDGs 3.4, 3.8, and 5.6. This scoping review aims to evaluate the literature on the socio-behavioral and biological mechanisms that explain the association between parity and missing teeth. **Methods:** A PubMed search was conducted using MeSH terms for parity, gravidity, and tooth loss. An initial 14 matches were screened using the Population-Concept-Context (PCC) framework, and 9 studies met inclusion criteria. **Results:** Synthesis of the literature reveals two overarching pathways. First, the association persisted after adjusting for socioeconomic status, smoking, dental visit frequency, age, and diabetes, suggesting a biological mechanism involving hormonal fluctuations, gingival inflammation, and increased bone turnover with reduced mineral density. Second, socio-behavioral mechanisms—including age at first pregnancy, socioeconomic status, education, misconceptions, and clinician/patient attitudes toward dental care during pregnancy—were also significant contributors. Notably, this association was consistently stronger in women than in men and strengthened with age and menopausal status, suggesting hormonal and reproductive-stage-specific vulnerability in women. **Conclusion:** Findings suggest tooth loss is likely driven by a combination of biological and socio-behavioral pathways, compounded by the timing of the reproductive stage. While existing studies strengthen evidence for this relationship, a gap remains in the literature on causal pathways.

Keywords: Parity, Tooth Loss, Women's Oral Health, Scoping Review

Presenter ID: PP25

Brief Chairside Preventive Counseling in Paediatric Dentistry: Effects on Parental Oral Health Knowledge in Bangladesh

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Background: Dental caries remains highly prevalent among children in Bangladesh, largely due to inadequate parental awareness and poor preventive practices. Brief chairside preventive counselling during routine dental visits may offer a practical and low-cost strategy to improve oral health literacy. This study evaluated the immediate effect of structured preventive counselling on oral health knowledge among parents of paediatric dental patients. **Methods:** A hospital-based interventional study was conducted among 120 parents of children aged 5-9 years attending the dental outpatient department of Sir Salimullah Medical College Mitford Hospital, Dhaka, Bangladesh, from July to December 2025, using consecutive sampling. Oral health knowledge was assessed using a structured questionnaire developed from WHO oral health guidelines and relevant literature. Content validity was ensured through expert review. Participants received a 5–7-minute chairside counselling session using illustrated educational posters. Knowledge was reassessed immediately after the intervention. Data were analysed using a paired t-test. **Results:** Mean knowledge score increased significantly from 4.40 ± 0.86 to 7.38 ± 1.14 out of 10 following counselling ($p < 0.001$). Improvements were observed across all assessed domains, particularly in awareness regarding regular dental visits (16% to 64%), correct toothbrushing frequency (28% to 76%), fluoridated toothpaste use (12% to 44%), and recognition of sugary foods as harmful to teeth (60% to 86%). **Conclusion:** A brief chairside counselling session significantly improved oral health knowledge among parents of paediatric dental patients. Integrating structured preventive education into routine paediatric dental visits may represent a practical, cost-effective strategy in resource-limited settings such as Bangladesh.

Keywords: Chairside Counselling, Preventive Dentistry, Oral Health Knowledge, Paediatric Dentistry, Bangladesh

Presenter ID: PP27

Oral Health Behaviors and Hyperglycemia Among Community-dwelling Older Adults in Yogyakarta, Indonesia: A Cross-sectional Study

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Background: Hyperglycemia is a highly prevalent component of metabolic syndrome among older adults, contributing to increased cardiovascular and diabetes risk. Oral health behaviors are hypothesized to influence glycemic regulation through systemic inflammatory pathways, yet evidence in community-dwelling elderly in Indonesia remains limited. This study examined the association between oral health behaviors and hyperglycemia among community-dwelling older adults in Sleman Regency, Yogyakarta, Indonesia. **Method:** An observational cross-sectional study was conducted in 2022 during the COVID-19 pandemic transition period, nested within the Health Demographic Surveillance System (HDSS) Sleman, Universitas Gadjah Mada. A total of 139 older adults aged ≥ 60 years were included. Independent variables comprised tooth brushing frequency, oral hygiene adjunct use, and dental visit regularity, assessed through structured interviews. Hyperglycemia, defined as fasting blood glucose ≥ 100 mg/dL per International Diabetes Federation (IDF) criteria, served as the dependent variable. Hierarchical binary logistic regression was performed adjusting for Body Mass Index (BMI), income, and smoking status. **Results:** Hyperglycemia was identified in 58 participants (41.7%). In bivariate analysis, tooth brushing frequency was significantly associated with hyperglycemia ($p=0.007$). Following multivariate adjustment, no oral health behavior was independently associated with hyperglycemia (all $p>0.05$), although brushing ≥ 3 times/day demonstrated a protective tendency (OR=0.264; 95% CI: 0.060–1.157; $p=0.077$). The model demonstrated good fit (Nagelkerke $R^2=0.206$; Hosmer-Lemeshow $p=0.497$; accuracy 70.5%). **Conclusion:** Oral health behaviors were not independently associated with hyperglycemia after multivariate adjustment. Longitudinal studies with larger samples and clinical oral health assessments are needed to confirm these findings.

Keywords: Older Adults, Oral Health Behavior, Hyperglycemia

Presenter ID: PP29

Unveiling the Oral Health Footprint of Modern Lifestyles: A Systematic Review

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Background: Sedentary lifestyles and unhealthy dietary patterns have become increasingly prevalent worldwide and are recognized as shared risk factors for both non-communicable and oral diseases. The evidence regarding their combined impact on oral health outcomes, however, is still fragmented. This systematic review aimed to synthesize current evidence on the association between sedentary behavior, unhealthy dietary patterns, and oral health outcomes. **Methods:** A systematic review was conducted in accordance with the PRISMA 2020 guidelines. Relevant literature published between 2016 and 2025 was retrieved from PubMed and Scopus databases. Studies examining sedentary behavior, dietary patterns, and oral health outcomes were screened according to predetermined eligibility criteria and analyzed narratively. **Results:** Studies involving diverse populations were included. Prolonged sedentary behaviour was consistently linked to unhealthy dietary habits, particularly an increased consumption of sugar-rich foods and beverages, as evidenced by the majority of studies. These behavioral patterns were linked to a higher prevalence of dental caries, poorer periodontal health, and reduced oral health-related quality of life. The interrelationship between oral and systemic health was also underscored by numerous studies, which emphasised the clustering of lifestyle-related risk factors. **Conclusion:** Adverse oral health outcomes are consistently linked to sedentary behaviour and unhealthy dietary habits. These results corroborate the Common Risk Factor Approach and underscore the importance of integrating oral health promotion with interventions that promote healthy eating and regular physical activity to alleviate the burden of oral diseases and contribute to the realisation of Sustainable Development Goal 3 (SDG 3) for good health and well-being.

Keywords: Physical Activity, Nutrition, Dental Public Health, Dental Caries, SDG 3.

Presenter ID: PP30

Knowledge, Attitudes, and Practices on Oral Health of Public High School Students in Ternate, Cavite

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Background: Ternate, Cavite is a fourth-class municipality in the Philippines where adolescents comprise 20.6% of the population as of 2022. Adolescents exhibit risk behaviors for dental caries, periodontal disease, and tooth wear, including frequent sugary food and drink consumption, irregular toothbrushing, smoking, alcohol use, and infrequent dental visits. This study determined the oral health knowledge, attitudes, and practices of public high school students in Ternate, Cavite, and examined correlations among these domains. Findings may serve as baseline data to guide future oral health programs. **Methods:** This was a non-experimental, descriptive cross-sectional study using multi-stage stratified systematic sampling; 307 of 350 questionnaires distributed among public high school students 12-19 years old were returned valid (87.7% usable response rate). A paper-based, self-administered questionnaire was used, and data were analyzed using frequency distribution and Spearman's rank correlation. **Results:** The mean age of the participants was 14.8 years with a median age of 15 years. Among the 307 participants, 98.05% had high knowledge, 94.46% had positive attitudes, and 71.66% had adequate self-reported oral health practices. Spearman's rank correlation revealed a moderate positive correlation between knowledge and attitude ($p=0.373$, $p<0.001$), with weaker correlations between knowledge and practice ($p=0.2536$, $p<0.001$) and between attitude and practice ($p=0.192$, $p<0.001$). **Conclusion:** Public high school students in Ternate, Cavite demonstrated strong foundational oral health knowledge and generally positive attitudes. Most reported adequate oral health practices; however, a notable minority may benefit from further improvement in this regard.

Keywords: Oral Health, Adolescent, Health Knowledge, Attitudes, Practice

Presenter ID: PP31

Evaluating The Customized MyVISION Module In Improving Oral Health KAP Among Malaysian Blind Adults

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Background: Visual impairment affects approximately 1.3 billion people worldwide and often limits an individual's ability to maintain oral health (OH). This study evaluated the effectiveness of the customized oral health education module, "MyVISION" in improving oral health knowledge, attitudes, and practices (OHKAP) among blind adults in Malaysia. **Methods:** A community-based quasi-experimental intervention study was conducted among blind adults aged 18 years and above registered with associations under the National Council of the Blind Malaysia (NCBM). Participants were recruited through convenience sampling and assigned to intervention (n=131) and control (n=122) groups. The intervention group received oral health education using the "MyVISION" module. OHKAP was assessed at baseline, post-intervention-1, and post-intervention-2 using a validated questionnaire specifically developed for blind individuals. Data were analysed using IBM-SPSS version 24.0. Descriptive statistics and repeated-measures ANOVA were applied, with statistical significance set at $P < 0.05$. **Results:** A total of 253 blind adults participated in the study. The mean (SD) age was 30.9 (11.66) years in the intervention-group and 38.5 (13.11) years in the control-group. Most participants were male, Malay, and had at least secondary-level education. Oral health knowledge scores increased significantly in the intervention-group following the intervention ($P = 0.001$), whereas no significant change was observed in the control-group ($P = 0.625$). Overall OHKAP scores also improved significantly in the intervention-group between baseline and post-intervention assessments ($P < 0.05$). **Conclusion:** The "MyVISION" module effectively improved OHKAP among blind adults and may serve as a useful educational resource for blind individuals, healthcare providers, caregivers, educators, institutions, and organizations supporting the visually impaired community.

Keywords: Blind adults, Knowledge, Attitude, Practice, Oral Health

Presenter ID: PP32

Integrated Intraoperative Behavioural Model Preventing Musculoskeletal Disorders and Eye Strain among Dental Practitioners

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Background: Occupational musculoskeletal disorders (MSDs) and eye strain are common among dental practitioners, contributing to reduced productivity and shortened clinical careers. Existing ergonomic recommendations often address isolated risk factors without providing an integrated behavioural framework for intraoperative practice. This study aimed to develop an evidence-informed intraoperative behavioural model (BM: Intra-op) to prevent MSDs and eye strain among dental practitioners. **Methods:** A qualitative study employing purposive sampling recruited a multidisciplinary expert panel comprising seven orthopaedic surgeons, ten optometrists, and eight dental specialists ($n = 25$) (IREC No.: 2021-078). Six focus group discussions explored occupational risk factors and preventive strategies. Discussions continued until thematic saturation was achieved. Audio-recorded transcripts were analysed using a five-stage framework analysis, and the findings were synthesised into an integrated behavioural model. **Results:** Framework analysis identified three interdependent behavioural domains underpinning occupational risk reduction: Postural Integrity, Temporal Regulation and Visual Hygiene. Collectively, these domains constitute the BM: Intra-op, integrating ergonomic positioning, four-handed dentistry, scheduled microbreaks, stretching exercises, appropriate magnification, balanced illumination, and the 20-20-20 visual recovery strategy into a unified intraoperative prevention framework. Unlike conventional ergonomic recommendations, the model integrates multiple behavioural domains into a coherent strategy that supports safer clinical practice throughout dental procedures. **Conclusions:** The BM: Intra-op provides a novel, evidence-informed behavioural framework for preventing MSDs and eye strain among dental practitioners. It offers a practical and scalable preventive model to support occupational health programmes, guide future intervention studies, and promote workforce sustainability, clinical longevity, and quality of patient care.

Keywords: Behavioural Intervention, Dental Workforce, Framework Analysis, Occupational Ergonomics, Prevention Model

Presenter ID: PP33

Knowledge And Attitudes of Health Sciences Students Regarding Chronic Kidney Disease

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Background: Chronic kidney disease (CKD) constitutes a substantial global health issue. Students presently enrolled in health science will become future medical professionals and will be crucial in preventing chronic kidney disease at primary, secondary, and tertiary levels within healthcare institutions. Future healthcare workers engaging directly with patients afflicted by chronic renal disease must acquire the information to do their duties efficiently and have a positive demeanor that corresponds with their competencies in managing CKD patients. This study aimed to assess the knowledge and attitudes of health science students concerning chronic renal disease. **Methods:** This research utilized a cross-sectional descriptive design and was conducted using a self-administered online questionnaire. Three hundred sixty-nine students completed the questionnaire. **Results:** The mean overall knowledge score was 8.68 ± 2.706 , derived from 15 questions (p-value of 0.001). The total attitude score of the students was 33.56 (± 2.958). A statistically significant difference was found between the knowledge score and the year of enrollment, as evidenced by a p-value of 0.001 ($p < 0.05$). A statistically significant difference ($p < 0.05$) was identified between the knowledge scores derived from books, papers, and journals vs. those from conferences and seminars. The lowest attitude score documented was 22, while the highest was 41 (p-value 0.001). **Conclusion:** A notable statistical difference was identified between the attitude score and the gender variable, with a p-value of 0.009.

Keywords: Chronic Kidney Disease, Knowledge, Attitude, Students

Presenter ID: PP34

Industry Collaboration Catalyzes Future-Ready Dental Graduates through a SULAM Oral Health Promotion Project

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Background: The Service-Learning Malaysia–University for Society (SULAM) initiative, introduced by the Ministry of Higher Education Malaysia, requires universities to integrate academic learning with community service through collaboration with industry and community stakeholders. Evidence suggests that industry engagement enhances graduate employability by developing leadership, communication, teamwork, negotiation, innovation, entrepreneurship, and problem-solving skills. This paper describes how multidisciplinary collaboration was embedded within Projek SUSU, a preschool oral health promotion (OHP) project, to enhance student competency while addressing community oral health needs. **Methods:** Guided by the Ottawa Charter for Health Promotion (WHO, 1986), dental students collaborated with engineering, industry, business, religious and private healthcare partners, together with preschool teachers, parents and caregivers, throughout project planning and implementation. Each collaboration was designed to cultivate specific competencies. Students co-designed an innovative toothbrushing supervision device with engineering experts, securing sponsorships, explored commercialisation strategies with business academics, validated Islamic oral health educational materials with an accredited religious scholar, and partnered with private dental clinics to establish a preventive birthday dental check-up programme for preschool children. **Results:** These collaborations enabled students to apply classroom knowledge in real-world settings while developing teamwork, stakeholder engagement, negotiation, leadership, design thinking, entrepreneurship, project management, and social marketing skills, alignment with programme learning outcomes. The project demonstrated community outcomes, with acceptability among stakeholders and oral health improvements among preschool children. **Conclusion:** Projek SUSU demonstrates that structured university-industry-community collaboration transforms SULAM-OHP from a community service activity into an authentic learning ecosystem that prepares future-ready oral healthcare professionals while delivering sustainable community impact.

Keywords: Oral Health Promotion, Preventive Dentistry, Service-Learning (SULAM), Preschool children, Community Engagement

Presenter ID: PP35

Operationalising the Ottawa Charter: Building a Sustainable School Oral Health Ecosystem for Preschool Children

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Background: The Ottawa Charter for Health Promotion, introduced four decades ago, remains a cornerstone of health promotion practice. However, translating its principles into coordinated, sustainable interventions engaging children, caregivers, schools and healthcare providers remains challenging, particularly in preschool oral health promotion. This paper describes how PROJEK SUSU, a school-based oral health promotion initiative, operationalised the Ottawa Charter to establish a sustainable school oral health ecosystem promoting lifelong oral health behaviours among preschool children. **Methods:** Following a situational analysis, the project was planned using the Planning Cycle (Ewles & Simnett, 2023) and implemented in an Islamic preschool. The Ottawa Charter was operationalised by implementing a supervised toothbrushing programme, delivering oral health education and toothbrushing training, conducting workshops parents and teachers, establishing a toothbrushing corner with toothbrush holders, a brushing scoreboard and Islamic oral health educational materials, and partnering with dental clinics to provide free birthday dental check-ups. These interventions embedded oral health promotion into children's school and home routines through collaboration among teachers, parents, healthcare providers, and community partners. **Results:** Process evaluation demonstrated acceptability among preschool children, parents, teachers, and collaborating stakeholders. Outcome evaluation showed 86.8% of children improved plaque scores, oral health knowledge, attitude and behaviour exceeded the 80% target, parental unsupervised toothbrushing declined from 33.3% to 2.7%, and school-based toothbrushing compliance reached 100%. **Conclusion:** Four decades after its introduction, Projek SUSU demonstrates that the Ottawa Charter remains a framework for creating school oral health ecosystems that foster healthy behaviours, strengthen community ownership and improve preschool oral health.

Keywords: Oral Health Promotion, Ottawa Charter, Preschool Children, School Based Oral Health, Health Promotion, Preventive Dentistry

Presenter ID: PP36

Early Childhood Caries Burden and Oral Health-Related Behaviours among Preschool Children in Jakarta: Cross-Sectional Study

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Background: Early childhood caries (ECC) remains a public health problem in Jakarta and may affect children's growth, nutrition, learning, and quality of life. Previous evidence reported ECC in 60.1% of children under five. This study described caries experience, oral hygiene, dental service utilization, oral health behaviours, and oral health-related quality of life among preschool children in Jakarta. **Methods:** A cross-sectional study was conducted in three Jakarta districts from February to March 2026 using multistage cluster sampling. Clinical examinations were performed among 302 preschool children to assess caries experience using dmft, bleeding on probing, and oral hygiene status. A validated parental questionnaire adapted from the WHO Oral Health Questionnaire for Children was completed for 148 children to assess dental attendance, toothbrushing, fluoride exposure, sugar-related behaviours, and oral health-related quality of life. Data were analysed descriptively. **Results:** Caries prevalence was 78.1%, with a mean dmft score of 5.89. Although 91.1% of children had good oral hygiene, 43.9% had more than one decayed or missing tooth due to caries, and only 27.7% routinely visited a dentist. Most children brushed at least twice daily (84.5%); however, 14.2% consumed milk until falling asleep. Reported impacts included dissatisfaction with dental appearance (29.7%), difficulty biting hard foods (14.2%), and school absence due to dental pain or discomfort (9.5%). **Conclusion:** Preschool children in Jakarta experienced a high ECC burden despite favourable reported toothbrushing practices. Findings support integrated prevention combining early screening, fluoride-based care, dietary counselling, routine dental attendance, and coordinated referral pathways through schools, communities, and primary healthcare services.

Keywords: Early Childhood Caries, Oral Health, Oral Hygiene, Children, Indonesia

Presenter ID: PP39

Determinants of Dentists' Intention to Use Teledentistry in Aceh, Indonesia: An UTAUT-Based Cross-Sectional Study

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Background: Teledentistry can expand access to oral health services, support remote consultation, and improve continuity of care, particularly in geographically dispersed settings. However, its successful implementation depends on dentists' readiness and intention to adopt digital technologies. This study examined dentists' behavioral intention to use teledentistry in Aceh, Indonesia, using the Unified Theory of Acceptance and Use of Technology (UTAUT). **Methods:** A cross-sectional study was conducted in November 2022 among 491 registered and active dentists in Aceh using total sampling. Data were collected using a structured questionnaire covering sociodemographic characteristics, teledentistry use, and UTAUT determinants, including performance expectancy, effort expectancy, social influence, and facilitating conditions. Age, sex, education, geographical area, and experience were assessed as moderating factors. Data were analyzed using Mann–Whitney, Kruskal–Wallis, and structural equation modeling–partial least squares (SEM-PLS). **Results:** The UTAUT model demonstrated acceptable validity, reliability, and model fit. The model explained 54.6% of the variance in dentists' behavioral intention to use teledentistry. Social influence ($\beta = 0.265$, $p < 0.05$), facilitating conditions ($\beta = 0.262$, $p < 0.05$), and performance expectancy ($\beta = 0.225$, $p < 0.05$) significantly associated with intention to use teledentistry, although effect sizes were small. Moderating factors did not significantly influence the relationships between UTAUT determinants and behavioral intention ($p > 0.05$). **Conclusions:** Dentists' intention to adopt teledentistry in Aceh was mainly shaped by professional/social encouragement, supportive infrastructure, and perceived clinical usefulness. Strengthening digital training, peer endorsement, and practice-level support may accelerate teledentistry adoption and improve access to oral health care.

Keywords: Teledentistry, Utaut, Dentist, Indonesia

Presenter ID: PP40

Interaction of Cerium Nitrate and Amine Fluoride with Pellicle-Covered Dentin

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Background: With increasing life expectancy leading to higher natural tooth retention, root caries is becoming a significant challenge for the elderly. Non-invasive treatments like silver diamine fluoride (SDF) are essential due to age-related complications, yet the need for alternatives like cerium nitrate is growing due to SDF's aesthetic drawbacks and limited availability. **Methods:** In this preliminary study, dentin specimens ($n = 8$) were prepared from bovine incisors and exposed to the oral cavity of two subjects for 3 min to allow initial pellicle formation. The specimens were then treated for 1 min with 20 μ L of either water (control) or 25% cerium nitrate and 5% amine fluoride, applied either sequentially or simultaneously. Interactions with the pellicle-covered dentin were analysed using transmission electron microscopy and energy-dispersive X-ray analysis. **Results:** Following intraoral exposure, the dentin surface was covered by a thin pellicle. Treatment with cerium nitrate and amine fluoride, applied either sequentially or simultaneously, resulted in the formation of needle-shaped, cerium-containing precipitates within and on the pellicle. **Conclusion:** For the first time, a cerium-rich precipitate layer was visualized on pellicle-covered dentin, which may contribute to the protective effect of cerium against demineralization reported in the literature.

Keywords: Dental Pellicle, Dentin, Root Caries, Cerium, Fluoride

Presenter ID: PP42

Stress-Related Biomarker Levels associated with Maternal Periodontal Disease

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Background: Stress is recognized as an exacerbating factor in periodontal disease. However, few studies have investigated the effect of gestational stress on maternal periodontitis. This study aimed to evaluate the potential associations between stress biomarkers (Corticotropin-releasing hormone, CRH ; Dehydroepiandrosterone, DHEA ; Cortisol, Cor; Aldosterone, ALD) and maternal periodontal disease during the third trimester of pregnancy. **Methods:** A total of 170 pregnant women participated in this study. demographic characteristics, systemic conditions, and serum stress-related biomarkers were assessed alongside periodontal clinical measurements. Participants were grouped according to their periodontal status. Both univariate and multivariate regression analyses were performed. **Results:** CRH levels were positively correlated with the risk of periodontitis ($p < 0.05$). Additionally, DHEA levels were positively correlated with the number of remaining teeth ($p < 0.05$). **Conclusion:** Maternal psychological stress is associated with periodontal status during pregnancy.

Keywords: Stress, Pregnancy, Periodontitis, CRH DHEA Biomarker

Presenter ID: PP43

Research Gaps in Early Childhood Caries: A Comprehensive Review

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Background: Early childhood caries (ECC) remains a common and preventable childhood disease, with a higher burden among disadvantaged populations. Although ECC is known to be multifactorial, evidence remains fragmented across biological, behavioural, social, diagnostic, and health-system domains. This limits effective prevention, early detection, and equitable care. To identify key research gaps in ECC and propose priority areas for integrated prevention-oriented research. **Methods:** This narrative review synthesised peer-reviewed literature on ECC epidemiology, pathogenesis, risk factors, diagnosis, prevention, treatment, and service delivery. Evidence was assessed to identify areas that remain limited, inconsistent, or insufficient to guide clinical practice and public health policy. **Results:** Several major gaps were identified. Longitudinal evidence linking microbiome changes, host susceptibility, diet, fluoride exposure, and lesion progression remains limited. Genetic, epigenetic, and early-life determinants are still poorly defined, especially in diverse and underserved populations. Although social inequalities in ECC are well established, pathways linking structural disadvantage with biological risk and access to care remain unclear. Emerging diagnostic tools, including salivary biomarkers, digital imaging, and artificial intelligence, need stronger validation and evidence of clinical utility. Implementation research on scalable, cost-effective, culturally appropriate, and equity-focused prevention models remains inadequate, particularly in low-resource settings. **Conclusion:** Future ECC research should move beyond isolated risk factors and adopt integrated, longitudinal, and equity-oriented frameworks. Stronger evidence across biological mechanisms, social determinants, diagnostic innovation, and implementation science is essential to improve prevention and reduce disparities in child oral health.

Keywords: Caries, Oral Health, Prevention, Children, Childhood, Preschool, Health Equity, Research Gap

Presenter ID: PP44

The Association between Obesity and Periodontitis: An Insight from A Case control Study

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Background: Obesity is a major global health challenge, with over half of Malaysian adults classified as overweight or obese (NHMS 2023). Although international evidence suggests obesity may increase periodontitis risk through shared inflammatory and microbial pathways, local data are limited. Thus, this study investigated this association within the Malaysian population. **Methods:** This multicenter case-control study, conducted in Selangor, included 120 participants aged 18–56 years ($M = 34.37$, $SD = 9.78$). Participants completed structured questionnaires regarding medical health, dental health, and habits and underwent full periodontal examinations. Data were analyzed using correlation, chi-square, odds ratios, and logistic regression. **Results:** Periodontitis prevalence was 65%, higher among obese participants (75.9%) compared to non-obese (61.5%). Obese individuals had 42% higher odds of periodontitis ($OR = 1.42$, 95% CI [0.87, 2.31]), though it was not statistically significant ($p = 0.159$). BMI and body weight correlated positively with plaque index but not with the gingival index. Regression analyses identified age as the most consistent predictor of disease severity ($p < .001$). **Conclusion:** Age is the most consistent predictor of periodontitis severity. Obesity was modestly associated with plaque accumulation but no significant effect on disease severity. Despite non-significant findings, the elevated odds ratio suggests a possible association. Larger studies with refined obesity measures and confounder adjustment are recommended to clarify this relationship.

Keywords: Obesity, Periodontitis, Case-control Study, Multicenter, Plaque Index, Malaysia

Presenter ID: PP45

Modern Approaches to Manage Dental Caries: A Scoping Review

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Background: Caries management has shifted from a surgically focused model to a biologically based approach centred on prevention, early intervention, and preservation of tooth structure. This scoping review summarises current strategies for caries management, with emphasis on minimally invasive care, biomimetic remineralisation, and targeted antimicrobial therapies. **Methods:** A literature search was conducted in PubMed and Scopus to identify English-language studies and reviews addressing contemporary concepts, materials, and technologies for caries prevention and management. Articles were screened for relevance to non-invasive or micro-invasive interventions, remineralising agents, and antimicrobial modalities targeting cariogenic biofilms. **Results:** Three interrelated strategies were identified. First, minimally invasive approaches, including selective caries removal, atraumatic restorative treatment and biophotonic technologies, prioritise preservation of sound tooth structure and may improve long-term tooth survival. Second, biomimetic remineralisation involves bioactive materials such as fluoride-based agents, casein phosphopeptide-amorphous calcium phosphate, and other ion-releasing systems that promote remineralisation and repair of non-cavitated lesions. Third, novel antimicrobial approaches, including antimicrobial peptides and smart delivery systems, aim to selectively suppress cariogenic bacteria, maintain a balanced oral microbiome, and limit lesion progression. **Conclusion:** Contemporary caries management increasingly integrates minimally invasive techniques, remineralising therapies, and selective antimicrobial strategies. These approaches may arrest or reverse early lesions, reduce the need for extensive restorative treatment, and support long-term tooth preservation and improved oral health outcomes.

Keywords: Antimicrobial, Remineralisation, Caries, Peptides, Prevention, Bioactive Materials, LASER

Presenter ID: PP46

Integrating Injection Moulding Technique, Fixed and Removable Prosthesis for An Aesthetic and Functional Rehabilitation

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Background: Managing full-mouth rehabilitation in patients with tooth surface loss (TSL) and prolonged posterior edentulism presents significant clinical challenges. This case report describes a multidisciplinary approach combining surgical crown lengthening, minimally invasive restorative techniques, and fixed and removable prostheses to restore aesthetics and function. **Case Description:** A 33-year-old Malay female presented with aesthetic concerns due to anterior TSL and masticatory discomfort from long-standing, unreplaced missing posterior teeth. Clinical examination revealed reduced clinical crown height and insufficient interocclusal space for posterior tooth replacement. To address these issues, the vertical dimension of occlusion was re-established to create adequate restorative space. The injection moulding technique—utilizing a light-cured injectable resin composite polymerized through a transparent silicone index—was employed to accurately reproduce tooth morphology, maximize aesthetics, and preserve tooth structure. Posterior support and function were subsequently restored using a mandibular cobalt-chromium partial denture and a maxillary porcelain-fused-to-metal (PFM) resin-bonded fixed dental prosthesis (RBFDP). **Conclusion/Clinical Lesson:** Successful functional and aesthetic outcomes were achieved through meticulous, comprehensive treatment planning. Managing complex cases involving TSL, multiple missing teeth, and lost vertical dimension requires a synergistic integration of surgical and restorative protocols. The injection moulding technique offers a predictable, minimally invasive, and highly durable solution for anterior aesthetic rehabilitation. When combined with targeted surgical interventions and mixed prosthodontic options, it ensures a comprehensive, stable rehabilitation that highlights the value of multidisciplinary dental care.

Keywords: Injection Moulding Technique, Surgical Crown Lengthening, Tooth Surface Loss, Full Mouth Rehabilitation, Injectable Resin Composite, Fixed and Removable Prosthesis

Presenter ID: PP47

Water Fluoride Concentration and Periodontal Health in Community-Dwelling Elderly in Yogyakarta

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Background: Fluoride in drinking water contributes to oral health by inhibiting demineralization and supporting remineralization of dental tissues. Evidence regarding its association with periodontal health in the elderly, particularly in Indonesia, remains limited. This study aimed to analyze the association between drinking water fluoride concentration and periodontal health status in elderly residents of Sleman Regency and Yogyakarta City. **Methods:** This observational analytic study involved 388 community-dwelling elderly participants. Water fluoride levels were measured from each participant's primary drinking water source and periodontal health was assessed using the Modified Gingival Index (MGI). Bivariate analysis used Pearson and Spearman correlation tests, followed by multiple linear regression. **Results:** Mean water fluoride level was 0.33 ± 0.10 mg/L and mean MGI score was 1.642 ± 0.85 . Bivariate analysis showed significant negative correlations between MGI and both water fluoride ($r = -0.144$; $p = 0.004$) and educational attainment ($r = -0.168$; $p = 0.001$). Multivariate regression confirmed both remained independently significant after controlling for economic status, diabetes, and smoking (fluoride: $B = -1.102$; $p = 0.009$; education: $B = -0.265$; $p = 0.004$; $R^2 = 0.057$; $p < 0.001$). **Conclusion:** Water fluoride concentration and educational attainment are independently associated with periodontal health status in community-dwelling elderly. These findings suggest that optimization of drinking water fluoride levels and oral health education should be integrated into elderly health programs.

Keywords: Fluoride, Drinking Water, Periodontal Health, Modified Gingival Index, Elderly, Educational Attainment

Presenter ID: PP48

Cerium Oxide-Based Nanomaterials for Periodontitis Management: A Systematic Review

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Background: Periodontitis is a chronic inflammatory disease associated with dysbiotic biofilms, oxidative stress, and dysregulated host responses, leading to periodontal tissue destruction and alveolar bone loss. Cerium oxide (CeO₂)-based nanomaterials have gained interest because of their redox activity, antioxidant effects, anti-inflammatory properties, and regenerative potential. **Methods:** A systematic review was conducted to identify preclinical studies evaluating CeO₂-based materials for periodontitis prevention or treatment. Studies investigating free CeO₂ nanoparticles or CeO₂-integrated multifunctional systems were included. Findings were narratively synthesised according to material characteristics, antimicrobial and antioxidant effects, inflammatory modulation, and regenerative outcomes. **Results:** Evidence was limited to in vitro and animal studies. CeO₂-based materials consistently showed reactive oxygen species scavenging through reversible Ce³⁺/Ce⁴⁺ cycling, reducing oxidative stress and suppressing inflammatory signalling, particularly ROS/NF-κB. Several studies reported decreased pro-inflammatory mediators, reduced alveolar bone resorption, and improved periodontal healing. Multifunctional systems combining CeO₂ with drug delivery platforms, photodynamic approaches, or antibacterial composites enhanced biofilm control and therapeutic efficacy. CeO₂-containing membranes and scaffolds also supported osteogenesis and bone regeneration. However, marked heterogeneity in formulations, models, and outcome measures was observed, and no clinical studies were identified. **Conclusion:** CeO₂-based nanomaterials show promise as multifunctional adjuncts for periodontitis management, but clinical translation remains limited by heterogeneity, lack of standardised protocols, and absence of human evidence.

Keywords: Cerium Oxide, Periodontitis, Systematic Review

Presenter ID: PP50

Socio-Demographic Predictors of OPMD Referral Compliance in Kuala Lumpur and Putrajaya

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Background: Oral Potentially Malignant Disorders (OPMDs) can progress to invasive oral squamous cell carcinoma. Early management is vital, as late-stage diagnosis significantly increases morbidity and mortality. While screening identifies high-risk individuals, reducing mortality depends on a seamless primary-to-specialist referral pathway. 'Referral leakage' often undermines screening efforts and increases mortality risk. **Objectives:** This study aimed to evaluate referral compliance, patient socio-demographics, and predictors of compliance within an OPMD screening program at government dental clinics in Kuala Lumpur and Putrajaya. **Methods:** We retrospectively analysed 2018–2022 OPMD screening data, defining compliance as attendance at specialist appointments. Descriptive statistics, multiple logistic regression, and marginal analysis were employed to identify associations between socio-demographic variables and compliance. **Results:** Patients' median age was 49 (IQR 35–61) years; 61.3% (n=168) were male. Although 98.5% (n=270) accepted referral, only 62.6% (n=169) attended specialist appointments [median time to consultation: 5 (IQR 1–13) days]. Multiple regression identified patient sex, district and their interaction as significant predictors of compliance. Marginal difference estimation showed that females have 12.1% (95% CI: 1.0%, 23.3%) higher compliance rate, and patients from Putrajaya demonstrated 29 – 32% higher compliance rate compared to other districts. Of 87 biopsies performed, histopathology confirmed Invasive Squamous Cell Carcinoma in 20.7% and Epithelial Dysplasia in 13.8% of cases. **Conclusion:** A nearly 40% attrition rate highlights a critical referral pathway bottleneck. These findings underscore the need for targeted patient navigation and digital tracking systems to bridge the gap between screening and specialist intervention.

Keywords: Oral Potentially Malignant Disorders (OPMD), Oral Cancer, Mouth Neoplasms, Referral and Consultation, Patient Compliance, Socioeconomic Factors

Presenter ID: PP51

Hyposalivation and Candida Colonization in Thai Older Adults with Type 2 Diabetes Mellitus

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Background: Type 2 Diabetes mellitus (T2DM) is prevalent in older adults, leading to impaired immunity and reduced salivary flow, and predisposes them to oral infections, especially candidiasis. This study aimed to investigate the association of hyposalivation and *Candida* colonization in older adults with T2DM. **Methodology:** This cross-sectional study enrolled 45 Thai older adults (≥ 60 years old) diagnosed with T2DM. Stimulated and unstimulated salivary flow rate (SSFR and USFR) and pH were measured. SSFR < 0.7 ml/min or USFR < 0.1 ml/min were classified as hyposalivation. Stimulated saliva was cultured on Sabouraud-dextrose and Chromogenic agar. Data were analyzed using independent t-tests and Mann-Whitney U tests. **Results:** Hyposalivation was observed in 24 participants (53.3%). The median (interquartile range, IQR) of USFR was 0.09 (0.06-0.13) ml/min, and SSFR was 0.51 (0.43-0.69) ml/min in the hyposalivation group. These were 0.26 (0.22-0.40) and 1.40 (1.00-1.72) ml/min, respectively, in the normal-salivation group. The mean pH of unstimulated saliva in hyposalivation group was significantly lower than in normal-salivation group (7.04 ± 0.65 vs. 7.49 ± 0.49 , $p = 0.014$), while stimulated salivary pH was not significantly different ($p = 0.169$). *Candida* colonization was detected in 22 patients (91.7%) in the hyposalivation group compared to 14 patients (66.7%) in the normal-salivation group ($p = 0.06$). Interestingly, patients with hyposalivation had significantly higher levels of *Candida* colonization than normal-salivation group (2.81 ± 1.04 vs. 2.0 ± 1.3 logCFU/ml, $p = 0.02$). **Conclusion:** Hyposalivation was associated with lower unstimulated salivary pH and higher level of *Candida* colonization among older adults with T2DM, emphasizing the importance of oral care and regular monitoring to prevent oral complications in these patients.

Keywords: Candida, Diabetes Mellitus, Hyposalivation, Older Adults, Salivary Ph

Presenter ID: PP52

Oral Health Status of 16-Year-Old Students in Kota Samarahan and Asajaya Districts, Sarawak

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Background: Dental caries remains a significant public health concern in Malaysia. This study evaluated the prevalence and severity of dental caries among 16-year-old students in the Kota Samarahan and Asajaya districts of Sarawak to optimize targeted preventive strategies. **Methods:** A secondary data analysis was conducted encompassing 479 16-year-old students from nine secondary schools across both districts. Data were extracted from the Ministry of Health's Health Information Management System. Caries experience was evaluated using the World Health Organization's Decayed, Missing, Filled Teeth (DMFT) index. **Results:** Caries prevalence was notably higher in rural Asajaya (74.3%) than in Kota Samarahan (59.9%), both surpassing the national average of 43.1%. The mean DMFT was 3.16 (SD=3.15) in Asajaya and 2.24 (SD=2.99) in Kota Samarahan, which are both well above the national level of 1.31. Active decay (the D component) affected 65.79% of adolescents in Asajaya and 37.0% in Kota Samarahan. Although a large proportion of students received restorative treatment (filled teeth accounted for 57.9% in Asajaya and 49.2% in Kota Samarahan), significant untreated decay remains. **Conclusion:** A high burden of dental caries exists among adolescents in these districts, especially in rural Asajaya. The data demonstrates that restorative treatment alone is insufficient. Expanding preventive measures—such as community water fluoridation, school-based fluoride programs, and targeted oral health education—is essential to mitigate the disease burden.

Keywords: Caries, Prevention

Presenter ID: PP53

Development of A Caries-Risk Predictive Model Incorporating Salivary Biomarkers among Malaysian Young Adults

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Background: Caries risk assessment is fundamental for preventive and personalised dental care. Saliva has emerged as a promising non-invasive diagnostic medium for biomarker-based risk prediction. However, evidence among Malaysian young adults remains limited. This study aimed to develop a more comprehensive caries risk prediction by integrating clinical, behavioural, and salivary biomarker variables among Malaysian young adults. **Methods:** A cross-sectional study was conducted among 64 participants aged 20–21 years. Caries risk was categorised into low, moderate, and high according to the ICCMS-CRA framework. Unstimulated saliva samples were analysed for soluble CD14 (sCD14), statherin, and alpha-amylase using an enzyme-linked immunosorbent assay. Potential clinical, behavioural, and salivary biomarker predictors were analysed using multivariable ordinal logistic regression to confirm the significant factors of caries risk. A nomogram was constructed from the ordinal final model for individualised risk estimation. Model assumptions and discriminatory ability were evaluated. **Results:** Restoration status (OR=3.10), early caries lesions (OR=10.45), and salivary sCD14 concentration (OR=3.70) were identified as significant predictors of higher caries risk categories. The nomogram was developed as a visual tool to estimate individuals' caries risk probability based on the coefficient values of significant predictors in the final prediction model. The proportional odds assumption was satisfied with no evidence of multicollinearity. The model demonstrated good discrimination for distinguishing individuals in different caries risk categories (low, moderate, and high). **Conclusions:** An ordinal prediction model and nomogram integrating clinical indicators and salivary biomarkers showed potential for improved caries risk stratification and provided more comprehensive, personalised caries risk assessment among young adults.

Keywords: Dental Caries, Risk Prediction, Salivary Biomarkers, sCD14, Ordinal Logistic Regression, Young Adults

Presenter ID: PP54

The Effectiveness of Tooth Brushing Programme Among 10-Year-Old Schoolchildren in Kelantan, Malaysia

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Background: The Effective Tooth Brushing Programme (*Program Memberus Gigi Berkesan* - BEGIN) was introduced to improve oral cleanliness among schoolchildren. This study evaluated oral cleanliness among 10-year-old schoolchildren participating in the BEGIN and conventional Daily Tooth Brushing (DTB) programmes. **Methods:** A quasi-experimental study was conducted among 391 schoolchildren aged 10 years from government primary schools in Kelantan, Malaysia. Participants were assigned to the BEGIN group (n = 196) and the DTB group (n = 195). Oral cleanliness was assessed using O'Leary Plaque Index at baseline, after 1 day, 2 weeks, and 4 weeks of intervention. Data were analysed using repeated measures ANOVA with significance set at $P < 0.05$. **Results:** Baseline plaque scores were comparable between the BEGIN [mean (SD) = 25.6 (15.87)] and DTB [mean (SD) = 23.7 (14.28)], $P = 0.066$. A significant effect-of-time on plaque score was found within both BEGIN (mean plaque score after 4 weeks = 27.2; 95% CI = 24.87, 29.63; $F = 10.106$; $P < 0.001$) and DTB groups (mean plaque score after 4 weeks = 28.1; 95% CI = 25.69, 30.56; $F = 13.923$; $P < 0.001$). A significant time-group interaction was observed ($F = 4.334$; $P = 0.006$), indicating different plaque score trends over time between the two programmes. **Conclusion:** The BEGIN programme provided better control of oral cleanliness than the conventional DTB programme, as reflected by its greater ability to minimise the worsening of plaque scores throughout the study period.

Keywords: Oral Cleanliness, Dental Plaque, Schoolchildren

Presenter ID: PP58

Knowledge and Management of Molar Incisor Hypomineralisation among Sabah Dental Practitioners and Nurses

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Background: Hypomineralisation (MIH) is a developmental enamel defect affecting first permanent molars and often incisors. It may complicate dental treatment due to hypersensitivity and enamel breakdown. This study assessed the knowledge, awareness, and management of MIH among dental practitioners and dental therapists in Sabah, Malaysia. **Methods:** A cross-sectional questionnaire-based study was conducted among dental practitioners and dental therapists in Sabah. Data from 181 respondents regarding demographic characteristics, knowledge, awareness, and management approaches related to MIH were analysed using descriptive and bivariate analyses. Statistical significance was set at $p < 0.05$. **Results (Preliminary):** Of the 181 respondents, 90 were dental practitioners and 91 were dental therapists. Overall, 100 (55.2%) encountered MIH in clinical practice on a yearly basis, and 93 (51.4%) perceived it as a moderately serious condition. Genetic factors (80.7%) were the most commonly identified aetiological factor. Fluoride application was the preferred initial management approach (71.8%). Most respondents reported confidence in diagnosing MIH, with no significant difference between professional groups ($\chi^2 = 5.772$, $p = 0.123$). Despite high awareness of MIH, 148 respondents (81.8%) reported receiving insufficient information on the condition, including 80 dental practitioners (88.9%) and 68 dental therapists (74.7%). Nevertheless, 175 respondents (96.7%) expressed interest in further clinical training. **Conclusion:** Although awareness of MIH was high, deficiencies in knowledge and management preparedness remains. Targeted continuing professional education may improve evidence-based diagnosis and management of MIH.

Keywords: Molar Incisor Hypomineralization (MIH), Children, Pediatric Dentistry, Sabah

E-POSTER PRESENTATION

Presenter ID: EP1

Disability Oral Health Policies in South Korea, Malaysia, and Viet Nam: A Comparative Analysis

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Background: People with disabilities face disproportionate barriers to dental care access, including physical inaccessibility, caregiver dependency, and systemic gaps in service delivery, contributing to poorer oral health outcomes compared to the general population. **Methods:** This policy study compares national oral health policies and insurance systems for people with disabilities across South Korea, Malaysia, and Viet Nam. Sources included official reports from Korea's Ministry of Health and Welfare and HIRA, Malaysia's Ministry of Health Oral Health Programme, the Viet Nam National Oral Health Survey, and WHO Regional and Country Oral Health Profiles. Dental care delivery systems, insurance mechanisms, preventive programs, and caregiver education policies were systematically reviewed and compared. **Results:** South Korea mandates regional disability dental hospitals in each province, and its National Health Insurance covers annual preventive services and oral health education for people with disabilities at minimal or no cost. Malaysia's Special Needs Dentistry program offers comprehensive hospital-based care and is recognized as a regional leader in special care dentistry. Viet Nam is advancing oral health policy development through its recent National Oral Health Survey (2024–25), aiming to establish evidence-based frameworks aligned with the WHO Global Oral Health Action Plan 2030. **Conclusion:** All three countries demonstrate growing commitment to disability-inclusive oral health, yet disparities in accessibility and preventive care persist. This comparative analysis identifies transferable policy strategies that may inform more equitable oral health systems across the Asian region.

Keywords: Disability, Oral Health Policy, Insurance Systems, Special Needs Dentistry, Southeast Asia, Comparative Analysis

Presenter ID: EP2

Comparative Analysis of Prevention-Centered Oral Health Policies in South Korea, UK, USA, Japan, and Thailand

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Background: Oral diseases are the most prevalent diseases globally, and expenditure for oral health services is increasing significantly. Recognized as an essential component of overall health, prevention and health-focused national oral health policies are on the rise, shifting from a treatment-centered paradigm. **Methods:** This narrative comparative policy analysis study compares national oral health policies and insurance systems in South Korea, UK, USA, Japan, and Thailand. Sources included national insurance coverage schemes and interviews with public health experts to identify similarities and differences in preventive policy components. **Results:** The analysis revealed distinct prevention-focused strategies across the five nations. South Korea utilizes National Health Insurance for school-based fluoride programs, while Scotland's Childsmile (UK) serves as a globally cited model for pediatric prevention. Despite inconsistent coverage in the USA due to Medicaid reliance, digital reward systems were found to effectively foster preventive compliance. Japan's 8020 Campaign promotes lifelong tooth retention, and Thailand integrates antenatal dental visits as a primary care entry point. Synthesizing these international models, this study proposes two primary policy strategies for South Korea: 1) implementing a performance-based incentive system centered on prevention outcomes, and 2) establishing an integrated maternal-child oral health welfare package. These interventions are identified as critical for enhancing voluntary care utilization and reducing long-term healthcare burdens. **Conclusion:** Prevention-focused national oral health policies showed promising health outcomes and effectively mitigate oral health disparities across the life cycle.

Keywords: Oral Health, Health Policy, Preventive Oral Health

Presenter ID: EP4

The Role of Patient Safety Knowledge in Nurses' Compliance with Patient Safety Incident Reporting

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Background: Patient safety incident reporting is a fundamental component of healthcare quality improvement and the prevention of recurrent adverse events in hospitals. Despite the implementation of incident reporting systems, nurses' compliance with reporting procedures remains suboptimal. Patient safety knowledge is considered a key determinant influencing nurses' adherence to incident reporting practices. This study aimed to examine the role of patient safety knowledge in nurses' compliance with patient safety incident reporting among inpatient nurses at a hospital in Kediri, Indonesia. **Methods:** A quantitative cross-sectional study was conducted involving 83 nurses selected through proportionate stratified random sampling. Data were collected using a structured self-administered questionnaire. The association between patient safety knowledge and nurses' compliance with incident reporting was analyzed using ordinal logistic regression with a significance level of 5%. **Results:** Most participants demonstrated good patient safety knowledge (84.3%) and were compliant with patient safety incident reporting procedures (55.4%). Ordinal logistic regression analysis revealed that patient safety knowledge was a significant predictor of nurses' compliance with incident reporting ($\beta = 5.645$, $p = 0.018$). The Nagelkerke R^2 value of 0.104 indicated that patient safety knowledge accounted for 10.4% of the variance in nurses' compliance. **Conclusions:** Patient safety knowledge plays a significant role in improving nurses' compliance with patient safety incident reporting. Strengthening patient safety education, implementing continuous training programs, and fostering a non-punitive reporting culture are essential strategies to enhance reporting compliance and promote a sustainable patient safety culture within hospitals.

Keywords: Patient Safety, Patient Safety Knowledge, Incident Reporting, Nurses' Compliance, Hospital, Patient Safety Culture

Presenter ID: EP7

Clinical Characteristics of Unerupted Permanent Teeth in a Vietnamese Birth Cohort from a Dioxin-Contaminated Area

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Background: Tooth eruption is an essential process in craniofacial development. Although environmental factors may influence dental development, little is known about the relationship between perinatal dioxin exposure and unerupted permanent teeth. This study aimed to describe the prevalence and tooth-specific distribution of unerupted permanent teeth among Vietnamese adolescents from a dioxin-contaminated area. **Methods:** This cross-sectional study included 133 Vietnamese adolescents (boys: 77, girls: 56) from an ongoing birth cohort established in Da Nang, Vietnam. Two participants who had undergone orthodontic treatment were excluded. Unerupted teeth were assessed by clinical examination. The prevalence of unerupted teeth was compared by sex, residential area, perinatal dioxin exposure, and early-term birth. Tooth-specific distribution was also evaluated. Categorical variables were compared using Pearson's chi-square test, and the distribution of the number of unerupted teeth was assessed using the Cochran–Armitage trend test. **Results:** Unerupted teeth were identified in 11 of 131 participants (8.4%). No statistically significant associations were observed between the presence of unerupted teeth and sex, residential area, perinatal dioxin exposure, or early-term birth. The mandibular lateral incisors were the most frequently affected teeth, particularly among boys. **Conclusion:** No apparent effect of perinatal dioxin exposure on unerupted permanent teeth was observed. The mandibular lateral incisors were the most frequently affected teeth, particularly among boys.

Keywords: Tooth Eruption, Unerupted Teeth, Perinatal Dioxin Exposure, Vietnam, Adolescents

Presenter ID: EP8

Oral Health Knowledge, Attitudes, and Practices Associated With Demographics Among Long Lapok Longhouse Residents, Sarawak

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Background: Oral health significantly impacts daily life and systemic well-being. This study aimed to assess oral health knowledge, attitudes, and practices (KAP) among the Long Lapok longhouse community and identify their relationships with sociodemographic characteristics. **Methods:** A cross-sectional study was conducted among 63 adult permanent residents of longhouse Rh Agnes in Long Lapok, Sarawak. Data were collected via face-to-face interviews using a 44-item validated questionnaire. Statistical analysis utilized non-parametric tests, including the Spearman's correlation to evaluate age, and the Mann-Whitney U and Kruskal-Wallis H tests to evaluate categorical sociodemographic associations. **Results:** The response rate was 100% (n=63). Participants comprised 57.1% males and 42.9% females, with a median age of 39.0 years (IQR = 20.5). Utilizing the median thresholds, 55.6% of participants demonstrated 'good' knowledge and 55.6% demonstrated 'positive' attitudes. Baseline hygiene practices were strong: 96.8% brushed teeth and 98.4% used fluoride toothpaste ≥ 2 times daily. However, daily consumption of sugary drinks (55.6%) and foods (50.8%) remained highly prevalent. Age demonstrated a statistically significant correlation with poorer oral health practices ($r_s = 0.282$, $p = 0.025$). Other sociodemographic factors, including gender, education, and occupation, showed no significant associations with KAP scores ($p > 0.05$). **Conclusion:** The Long Lapok community exhibits strong baseline hygiene practices, but oral health knowledge and attitudes remain divided, and daily sugar consumption is prevalent. Because increasing age significantly correlates with poorer oral health practices, future community-based health promotion programs must be carefully targeted to address dietary habits and specifically focus on older adults.

Keywords: Oral Health, Knowledge, Attitudes, and Practices (KAP), Rural Population, Health Promotion, Sarawak

Presenter ID: EP9

Changing Patterns of Dental Caries among Penan Children in Central Sarawak: A Retrospective Analysis

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Background: Despite the long-standing implementation of school oral health services in Malaysia, dental caries remains a significant oral health concern among school children. This study aimed to evaluate changes in caries prevalence, caries experience and caries treatment needs over six years among a cohort of Penan primary school children in Belaga District, Kapit Division, Sarawak. **Methods:** This longitudinal study followed 23 Penan primary school children in Kapit Division from Primary 1 in 2019 to Primary 6 in 2024. They were examined annually by trained Dental Therapists using the World Health Organization DMFT index via the Ministry of Health Malaysia (MOH) incremental dental care. **Results:** The prevalence of dental caries in the cohort increased from 26.1% in 2019 (Primary 1) to 87.0% in 2024 (Form 6), representing more than a three-fold rise. This increase over the six years was statistically significant (McNemar test, $p < 0.001$). The change of median DMFT from 0.0 (IQR = 1.00) to 2.0 (IQR = 3.00) was statistically significant (Wilcoxon signed-rank test, $p < 0.001$). However, the change in median D from 0.0 (IQR = 1.00) to 0.0 (IQR = 2.00) was not statistically significant (Wilcoxon signed-rank test, $p > 0.01$). **Conclusion:** A significant increase in dental caries was observed among the same cohort of Penan school children in Central Sarawak over six years, despite provision of oral health care annually. These findings underscore the need to strengthen and refine preventive strategies, with greater emphasis on targeted and culturally appropriate interventions.

Keywords: Dental Caries, Caries Prevalence, Caries Experience, School Children, Penan, Belaga, Sarawak

Presenter ID: EP10

Effects of a Vitamin Derivative and Sodium Chloride Dentifrice on Halitosis

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Background: Functional dentifrices containing bioactive ingredients may contribute to improved oral hygiene and oral health. This study evaluated the clinical efficacy of a dentifrice containing vitamin derivatives and sodium chloride on plaque control, halitosis, and periodontal health. **Methods:** A 12-week randomized, double-blind, controlled clinical trial was conducted in 62 healthy adults. Participants were assigned to either a test dentifrice containing sodium chloride, tocopheryl acetate, pyridoxine hydrochloride, and sodium fluoride or a control dentifrice without active ingredients. Subjects brushed three times daily for 12 weeks. Clinical assessments were performed at baseline and after 4, 8, and 12 weeks. Outcome measures included the Patient Hygiene Performance (PHP) index, organoleptic halitosis assessment using a Visual Analogue Scale (VAS), BB Checker measurements, and the Community Periodontal Index (CPI). **Results:** The test dentifrice demonstrated significantly greater improvements than the control dentifrice from week 8 onward. At week 12, the PHP index was significantly lower in the experimental group than in the control group (1.54 ± 0.49 vs. 2.14 ± 0.60 , $p < 0.001$). BB Checker values were also significantly reduced (49.1 ± 13.0 vs. 60.4 ± 14.8 , $p = 0.004$), and organoleptic halitosis scores showed greater improvement in the experimental group (2.78 ± 1.05 vs. 4.07 ± 1.90 , $p = 0.003$). CPI scores improved significantly from baseline within the experimental group, although between-group differences were not significant. **Conclusion:** The dentifrice containing vitamin derivatives and sodium chloride significantly improved plaque control and halitosis while contributing to improved periodontal health. These findings suggest that the formulation may be a useful adjunct for maintaining oral hygiene and oral health.

Keywords: Dentifrice, Halitosis, Sodium Chloride, Vitamin Derivatives, Plaque Control, Periodontal Health

Presenter ID: EP11

Clinical Evaluation of a Dentifrice Containing Tricalcium Phosphate and Sodium Metaphosphate

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Background: Multifunctional dentifrices containing remineralizing and bioactive ingredients have been developed to improve oral health. This study evaluated the clinical efficacy of a dentifrice containing tricalcium phosphate, sodium fluoride, sodium metaphosphate, tocopheryl acetate, and potassium nitrate on halitosis, dentin hypersensitivity, gingival health, periodontal status, tooth color, and dental plaque. **Methods:** A 12-week randomized controlled clinical trial was conducted in 50 healthy adults. Participants were assigned to either an experimental group using the multifunctional dentifrice ($n=25$) or a control group using a vehicle-based dentifrice ($n=25$). Subjects brushed three times daily for 12 weeks. Clinical assessments were performed at baseline and after 4, 8, and 12 weeks. **Results:** The experimental group showed significantly greater improvements in halitosis from week 4 and dentin hypersensitivity from week 8 compared with the control group ($p < 0.05$). Gingival inflammation and dental plaque indices decreased earlier and to a greater extent in the experimental group. Tooth color improved significantly at week 8 ($p < 0.05$). Although some periodontal indices did not differ significantly between groups, consistent improvements from baseline were observed in the experimental group. **Conclusion:** The multifunctional dentifrice demonstrated beneficial effects on halitosis, dentin hypersensitivity, gingival health, plaque control, and tooth color. These findings suggest that it may be a useful option for comprehensive oral health management.

Keywords: Tricalcium Phosphate, Sodium Metaphosphate, Dentifrice, Dentin Hypersensitivity, Halitosis, Plaque Control

Presenter ID: EP12

Clinical Evaluation of a Centella Asiatica Dentifrice for Oral Health Improvement

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Background: Functional dentifrices containing bioactive ingredients may improve oral hygiene and periodontal health. This study evaluated the clinical efficacy and antibacterial activity of a dentifrice containing standardized Centella asiatica extract on gingival health, plaque control, oral malodor, and periodontal status. **Methods:** A 12-week randomized, double-blind, parallel-controlled clinical trial was conducted in 56 healthy adults. Participants were assigned to either a test dentifrice containing standardized Centella asiatica extract or a control dentifrice. Clinical assessments were performed at baseline and after 4, 8, and 12 weeks, including the Papillary-Marginal-Attachment (PMA) index, Lobene Gingival Index (GI), Patient Hygiene Performance (PHP) index, Community Periodontal Index (CPI), BB Checker values, and Visual Analogue Scale (VAS) scores. Antibacterial and bactericidal activities against Porphyromonas gingivalis were also evaluated in vitro. **Results:** The test dentifrice demonstrated significantly greater improvements than the control dentifrice from week 8 onward ($p < 0.05$). At week 12, significant reductions were observed in the PMA index ($p < 0.001$), GI ($p = 0.009$), and PHP index ($p < 0.001$). Oral malodor was significantly reduced as measured by BB Checker ($p = 0.004$) and VAS ($p = 0.003$). No adverse events were reported. In vitro, the dentifrice completely inhibited P. gingivalis growth at 3.90 mg/mL and achieved >99.99% bactericidal activity within 3 minutes. **Conclusion:** The dentifrice containing standardized Centella asiatica extract significantly improved gingival health, plaque control, and oral malodor while demonstrating strong antibacterial activity. These findings suggest that the dentifrice may be a useful preventive oral care product for maintaining oral and periodontal health.

Keywords: Centella Asiatica Extract, Dentifrice, Gingivitis, Dental Plaque, Oral Malodor, Porphyromonas Gingivalis

Presenter ID: EP13

A Peek Into Preschoolers' Toothbrushing Behaviours: Parental Questionnaire and Video Observation

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Background: Parents are the primary facilitators of oral hygiene among preschool children. However, parental reports may not reflect children's actual toothbrushing behaviours. This study compared parental self-reported toothbrushing practices with video-observed toothbrushing behaviours among Malaysian preschool children. **Methods:** A cross-sectional study was conducted among 145 parent-child dyads from selected preschools in Terengganu. Parents completed a structured questionnaire on their child's oral hygiene practices, dietary habits, and dental attendance. Children's toothbrushing behaviours were subsequently assessed using standardized video recordings. Descriptive statistics were used to summarise parental-reported and observed behaviours, while associations with oral health outcomes were analysed using appropriate statistical tests. **Results:** All parents reported that their children brushed their teeth, with 70.3% reporting brushing at least twice daily and 80.7% providing daily supervision. Most parents reported using toothpaste (99.3%), including fluoridated toothpaste (81.4%). Video assessment confirmed that 82.8% of children used fluoridated toothpaste. However, a significantly higher proportion used non-fluoridated toothpaste than reported by parents (17.2% vs 11.0%, $p = 0.030$). Despite favourable parental reports, video observation revealed inadequate toothbrushing behaviours. Two-thirds of children (66.2%) brushed for less than one minute, 89.0% used predominantly horizontal brushing strokes, and all demonstrated a non-systematic brushing pattern. Most children also rinsed their mouths multiple times after brushing (57.2%), potentially reducing fluoride retention. **Conclusion:** Despite favourable parental reports, observed toothbrushing behaviours highlight the need for continuous parental supervision to ensure effective plaque control.

Keywords: Oral Cleanliness, Preschool Children, Toothbrushing, Video Observation

Presenter ID: EP14

Tongue Coating Microbiota and Machine Learning-Based Identification of Halitosis in Young Adults

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Background: Halitosis is closely associated with tongue coating microbiota and volatile sulfur compounds (VSCs). This study investigated the prevalence and associated factors of halitosis in young adults and explored microbial biomarkers for auxiliary diagnosis. **Methods:** A cross-sectional study was conducted among 1,377 university students using questionnaires and organoleptic test (OLT). Tongue coating samples from 149 participants underwent 16S rRNA sequencing, and 60 samples received shotgun metagenomic sequencing. Microbial diversity, differential taxa, metabolic pathways, and correlations with VSCs were analyzed. Machine learning models were developed using microbial features. **Results:** The prevalence of organoleptic halitosis was 33.9%. Smoking, poor oral hygiene, lack of tongue cleaning, and high tongue coating score were significantly associated with halitosis. Halitosis subjects showed significantly higher VSC concentrations and altered tongue coating microbiota with increased alpha diversity and distinct community structure. *Alloprevotella*, *Peptococcus*, and *Solobacterium* were enriched in the halitosis group, whereas *Rothia* and *Streptococcus* were enriched in controls. A random forest model based on microbial genera showed good diagnostic performance (AUC = 0.94). Metagenomic analysis further identified enrichment of sulfur amino acid metabolism pathways in the halitosis group. **Conclusion:** Halitosis in young adults is associated with dysbiosis of tongue coating microbiota and enhanced sulfur-related metabolism. Tongue coating microbial biomarkers combined with machine learning may support auxiliary diagnosis and targeted prevention of halitosis.

Keywords: Halitosis, Tongue Coating Microbiota, Volatile Sulfur Compounds, Machine Learning, Young Adults

Presenter ID: EP16

Factors Associated with Salivary IgA Variation in Healthy Young Adults

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Background: Secretory immunoglobulin A (IgA) is a key component of mucosal immunity and is abundant in the saliva. Although various behavioral, physiological, and local factors are thought to influence secretion, these mechanisms remain incompletely understood. This study aims to investigate factors related to salivary IgA levels in healthy young adults. **Methods:** Unstimulated whole saliva was collected from 50 healthy young adults (men: 50%, mean age: 23.9 ± 2.8 years). Salivary IgA concentrations were measured using enzyme-linked immunosorbent assay (ELISA), and lactate dehydrogenase activity (LDH) was evaluated using a colorimetric assay. Body mass index and lifestyle factors were assessed via questionnaire. Diurnal variation was examined in the five participants with the highest and lowest salivary IgA levels. Oral microbiota were analyzed using next-generation sequencing, and statistical analyses were performed to evaluate associations. **Results:** Salivary IgA concentrations showed substantial inter-individual variability among participants, ranging from 64.2 to 394.4 pg/mL (mean ± SD: 174.5 ± 69.5 pg/mL). However no significant associations were observed between IgA levels and physiological or lifestyle factors. In contrast, *Streptococcus* and *Lachnoanaerobaculum orale* were relatively more abundant in the high-IgA group. A significant difference in alpha diversity, assessed using the Shannon index, was also observed between the high- and low-IgA groups. IgA levels increased from morning to midday and decreased thereafter. **Conclusion:** Salivary IgA levels showed marked inter-individual and diurnal variation. Oral microbiota, including *Streptococcus* and *Lachnoanaerobaculum orale*, may be associated with salivary IgA variation in healthy young adults.

Keywords: Saliva, IgA, Microbiota

Presenter ID: EP17

Cariogenic Diet and Toothbrushing Frequency Influence *S. mutans* Colonization Despite Fluoride Mouth Rinse Exposure

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Background: Fluoride mouth rinse (FMR) programs are implemented to reduce caries risk; however, their effectiveness may depend on behavioral factors such as cariogenic diet and toothbrushing habits. This study examined whether cariogenic food consumption and toothbrushing frequency were associated with *Streptococcus mutans* colonization, and whether these associations differed according to FMR exposure. **Methods:** This study involved 94 students. Ordinal logistic regression was performed with *S. mutans* colonization as the ordinal dependent variable. Cariogenic food consumption and toothbrushing frequency were analyzed as behavioral predictors. FMR status and interaction terms between each predictor and FMR status were included to assess whether FMR modified these associations. **Results:** Cariogenic food consumption was significantly associated with higher *S. mutans* colonization, with each one-unit increase increasing the odds of being in a higher colonization category by approximately 11% (OR = 1.11; $p < 0.001$). In contrast, higher toothbrushing frequency was significantly associated with lower *S. mutans* colonization, reducing the odds of higher colonization by approximately 58% for each one-level increase in brushing frequency (OR = 0.42; $p = 0.048$). However, the interaction between cariogenic consumption and FMR status was not significant ($p = 0.461$), and the interaction between toothbrushing frequency and FMR status was also not significant ($p = 0.519$). **Conclusion:** *S. mutans* colonization was more consistently associated with cariogenic dietary exposure and toothbrushing frequency than with FMR status. These findings suggest that FMR programs may have limited observable effects when cariogenic consumption and inadequate toothbrushing behavior remain uncontrolled.

Keywords: Streptococcus Mutans; Fluoride Mouth Rinse; Cariogenic Diet; Toothbrushing Frequency; Oral Health Behavior

Presenter ID: EP18

Management of White Spot Lesions by Resin Infiltration: A Case Series

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Background: White spot lesions (WSLs) manifest as subsurface demineralization beneath intact enamel, presenting as distinct chromatic alterations on the tooth surface. These lesions pose significant concerns regarding aesthetics and patient self-esteem. Contemporary management modalities include remineralization therapies such as casein phosphopeptide–amorphous calcium phosphate (CPP-ACP) pastes and topical fluorides microabrasion utilizing an 18% hydrochloric acid/pumice slurry, and resin infiltration. As a minimally invasive aesthetic intervention, resin infiltration demonstrates considerable efficacy; however, its clinical outcome is subject to multiple modulating factors. **Case Description:** This report delineates three clinical cases of WSLs managed via resin infiltration across varying stages of lesion progression. We further analyze the critical determinants influencing treatment efficacy. **Clinical lesson:** Resin infiltration constitutes a clinically efficacious therapeutic modality specifically for incipient carious lesions graded as ICDAS codes 2–3. Although resin infiltration offers a viable treatment pathway for WSLs, optimal aesthetic outcomes are not universally guaranteed. Clinical success necessitates stringent patient selection, strict adherence to standardized clinical protocols, and systematic post-operative follow-up.

Keywords: Resin Infiltration, White Spot Lesions, ICDAS

Presenter ID: EP21

A Scoping Review on the Application of Bioactive Glass in Endodontics

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Background: This scoping review aimed to map the available evidence on the applications and effectiveness of bioactive glass (BG) in endodontics. **Methods:** The review was conducted according to the PRISMA-ScR guidelines. Four electronic databases (PubMed, Scopus, Web of Science, and Cochrane Library) were searched up to March 2026. Studies evaluating BG in endodontic applications, regardless of study design, were included. Data on study characteristics, BG formulations, applications, outcomes, and key findings were extracted and narratively synthesized. **Results:** Sixty-five studies were included, predominantly in vitro investigations, with limited clinical evidence comprising three randomized controlled trials, two clinical studies, one case series, and five animal studies. BG was most frequently investigated in root canal filling materials (n=32), followed by intracanal medicaments or irrigants (n=19), root repair or root-end filling materials (n=9), and pulpotomy or regenerative procedures (n=5). BG demonstrated promising antimicrobial activity, particularly against *Enterococcus faecalis*, especially when combined with chlorhexidine or calcium hydroxide. Commercial BG-based sealers, including GuttaFlow Bioseal and Nishika BG Sealer, showed favourable sealing ability, dentinal tubule penetration, and postoperative outcomes. Novel BG formulations incorporating metallic ions or modified glass compositions also demonstrated encouraging physicochemical and bioactive properties. However, findings remained heterogeneous and largely laboratory-based, limiting clinical translation and preventing definitive conclusions regarding long-term superiority over conventional endodontic materials in routine clinical practice. **Conclusion:** BG demonstrates significant potential in endodontics due to its antimicrobial and mineralization-supporting properties. Nevertheless, standardized, high-quality clinical trials are required to establish long-term clinical effectiveness and superiority over conventional materials.

Keywords: Bioactive Glass, Dental Pulp, Endodontics, Root Canal, Root Canal Sealer, Literature Review

Presenter ID: EP22

Lentian Modulates the Galectin-3/NLRP3 Axis to Ameliorate Periodontitis: An In Vitro and In Vivo Study

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Background: Periodontitis is a chronic inflammatory disease driven by plaque biofilm, often leading to irreversible tissue destruction. Natural product-based therapeutic strategies are gaining attention. Lentian (LNT), a β -glucan from *Lentinula edodes*, exhibits anti-inflammatory properties, yet its role in periodontitis remains unclear. **Methods:** An in vitro periodontitis model was established using Pg-LPS-stimulated human periodontal ligament fibroblasts (hPDLFs). Oxidative stress (SOD/MDA), pyroptosis (LDH/IL-1 β /IL-18), and related mRNA/protein levels were assessed. Network pharmacology, RNA-seq, molecular docking, and NLRP3 agonist (Nigericin) rescue experiments were performed. In vivo, a ligature-induced rat periodontitis model was used to evaluate LNT efficacy via Micro-CT, histology, TRAP staining, and immunohistochemistry. **Results:** LNT (≤ 400 μ g/mL) promoted hPDLFs proliferation, reduced oxidative damage, and inhibited pyroptosis. Network pharmacology and docking suggested LNT binds Galectin-3. LNT significantly suppressed NLRP3, Caspase-1, and GSDMD expression, effects partially reversed by Nigericin, confirming NLRP3 as a key downstream target. In vivo, LNT reduced alveolar bone loss, osteoclast numbers, and tissue inflammation, while downregulating Galectin-3 and NLRP3. **Conclusion:** LNT ameliorates experimental periodontitis by inhibiting the Galectin-3/NLRP3-mediated pyroptosis pathway, positioning it as a promising natural candidate for periodontal therapy.

Keywords: Lentian; Periodontitis; Galectin-3; NLRP3 inflammasome; Pyroptosis

Presenter ID: EP23

A Novel Biomimetic Mineralization Strategy Using Tyrosine-Polylysine-Calcium Silicate Complex for Preventive Dentistry

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Background: Enamel demineralization and dentinal hypersensitivity pose major challenges in preventive dentistry. This study investigated a novel biomimetic mineralization strategy using a complex of tyrosine, polylysine, and calcium silicate in an aqueous toothpaste formulation for dual anti-caries and anti-hypersensitivity management. **Methods:** Anti-caries efficacy was evaluated using an in vitro pH-cycling demineralization-remineralization model with bovine enamel, followed by surface microhardness recovery. Anti-hypersensitivity potential was assessed using a dentinal tubule occlusion model and scanning electron microscopy over 1 to 2 days. The synergistic effects of the complex with fluorides were also investigated. **Results:** The optimized tyrosine/polylysine complex at a1:1 mass ratio exhibited significant synergistic enhancement when combined with calcium silicate and fluoride. The experimental toothpaste promoted substantial surface microhardness recovery and biomimetic repair of structural enamel defects. In addition, the formulation achieved complete dentinal tubule occlusion within 1 day, significantly outperforming control groups. **Conclusion:** The combination of tyrosine, polylysine, and calcium silicate inhibited enamel demineralization, promoted remineralization, and occludes dentinal tubules. This biomimetic toothpaste may offer a promising therapeutic approach for comprehensive anti-caries and anti-hypersensitivity management in preventive dentistry.

Keywords: Enamel; Tyrosine; Polylysine; Biomimetic Mineralization

Presenter ID: EP24

Association Between Mean Chewing Count per Bite During Onigiri Consumption and Postprandial Salivary IgA Ratio

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Background: Chewing behavior may influence oral and mucosal immune responses; however, the association between chewing behavior during food consumption and postprandial salivary immunoglobulin A (IgA) response remains unclear. This study examined whether a higher mean chewing count per bite during onigiri consumption was associated with changes in salivary IgA after eating. **Methods:** Thirty-seven healthy adult men were classified into two groups according to mean chewing count per bite during onigiri consumption: <30 chews/bite (n = 18) and ≥30 chews/bite (n = 19). Saliva samples were collected before eating and at 0–5, 10–15, and 25–30 min after eating. Salivary IgA concentration, saliva volume, and IgA secretion amount were measured. Post-chewing changes were evaluated as ratios relative to the pre-chewing value, and between-group comparisons were performed using the Mann–Whitney U test. **Results:** Participants with ≥30 chews/bite showed significantly higher salivary IgA ratios at 10–15 min and 25–30 min after eating than those with <30 chews/bite (10–15 min: 74.0 [64.3–102.3] vs 56.2 [49.5–73.5], P = 0.005; 25–30 min: 118.5 [90.1–139.5] vs 91.2 [57.0–104.5], P = 0.022). Differences in saliva volume were not significant, and IgA secretion amount showed only a nonsignificant increasing trend. **Conclusion:** A higher mean chewing count per bite during onigiri consumption was associated with a greater postprandial salivary IgA response.

Keywords: Chewing Behavior, Salivary Immunoglobulin A, Oral Immune Response

Presenter ID: EP25

Severe Early Childhood Caries During The Growth Period and Neurodevelopment Trajectories - A Murine Model Study

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Background: Our preliminary cross-sectional survey on childhood caries in mountainous regions revealed a potential correlation between severe early-childhood caries and delayed physical growth as well as intellectual development. However, the causal relationship and underlying mechanisms remain poorly understood. **Methods:** To investigate whether caries during the critical growth period adversely affects neurodevelopment, we established a murine model of comorbid caries and neurodevelopmental disorders. Single-cell RNA sequencing was performed on brain tissues to identify differentially expressed genes (DEGs) between the disease and control groups, followed by bioinformatic pathway enrichment analysis to dissect the molecular cascades involved. **Results:** Compared with controls, the disease group exhibited a marked increase in the number of pericytes in the cerebral cortex, concomitant with upregulated expression of Cyclophilin A (CyPA). Mechanistically, this elevation may trigger the CyPA-NF- κ B-MMP-9 signaling axis, leading to neuroinflammation and subsequent neural injury, as evidenced by altered expression of synaptic plasticity-related genes. **Conclusion:** Our findings provide evidence that developmental caries may compromise neurodevelopmental trajectories, potentially through pericyte-mediated inflammatory pathways. These results highlight the necessity of incorporating caries prevention into early-life neuroprotective strategies.

Keywords: Severe Early Childhood Caries, Neurodevelopment, Pericytes, Cyclophilin A, Caries Prevention

Presenter ID: EP26

Dental Health Knowledge and Interest Among Dietetic Students: A Questionnaire-Based Study

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Background: An increasing interest exists in the relationship between dentistry and nutrition with the ageing of the population. However, the dental knowledge and interest of dietetic students before dental health lectures are unclear. This study investigated the status of dental health knowledge and interest among dietetic students. **Methods:** A questionnaire survey was conducted on 71 dietetic students in Japan before dental health lectures regarding their personal and others' dental and oral difficulties, the information they know about dentistry (school education, mass media, dental clinic and others) and the dental information they should know when they become dietitians. Morphological analysis was performed to extract the nouns only, and the number of times a word appeared (> 4 times) was indicated. The number of none responses was also tabulated. **Results:** No match existed between the dental information provided and dental problems encountered, with the exception of tooth decay. The most frequent output was tooth brushing in school education (39 times), tooth decay in mass media (9 times), tooth in dental clinics (21 times) and tooth in others (8 times). Tooth brushing was mentioned four and 14 times in mass media and dental clinics, respectively, as well as in school education. Two respondents (2.8%) answered none to the question 'The dental information they should know when they become dietitians'. **Conclusions:** It is crucial to provide additional information other than tooth brushing. Dental education integrated into dietetic training is of great interest to students and may yield a high educational effect.

Keywords: Dental Information, Dietician, School Education, Mass Media, Dental Clinics, Tooth Brushing

Presenter ID: EP28

Questionnaire Validation about Parental Knowledge on Fluoridated Toothpaste for Pre-school Children in Kota Kinabalu, Sabah

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Background: Without water fluoridation in Kota Kinabalu, Sabah, preschool children rely on fluoridated toothpaste as their sole source of daily fluoride exposure. Parental knowledge and selection of fluoridated toothpaste remains poorly understood. A contextually relevant and methodologically sound questionnaire aims for reliable assessment. **Methods:** A content validation study was conducted through expert review. The questionnaire, comprising two domains and eight items, was developed in both English and Malay based on established guidelines and relevant literature. Seven subjects-matter experts, four in dental public health and three in paediatric dentistry, independently rated each item for clarity and relevance using a 4-point Likert scale. The Scale-level Content Validity Index (S-CVI/Ave) was calculated. Face validity was then assessed among parents of preschool children to evaluate the questionnaire's clarity, comprehensibility, and acceptability. **Results:** All questionnaire items demonstrated excellent content validity. The Malay version achieved an S-CVI/Ave of 0.96 for clarity and 0.98 for relevancy, while the English version achieved an S-CVI/Ave of 0.95 for clarity and 0.98 for relevancy, indicating complete agreement among experts regarding item relevance. Face validation findings indicated that the questionnaire was clear, easy to understand, and culturally appropriate. **Conclusion:** The questionnaire demonstrated excellent content validity and satisfactory face validity, supporting its suitability for assessing parental knowledge and choice of fluoridated toothpaste for pre-school children. Further studies are recommended to establish construct validity and reliability prior to large-scale use.

Keywords: Parental Knowledge, Fluoridated Toothpaste, Validation, Questionnaire, Pre-school Children

Presenter ID: EP29

Bridging Periodontal and Systemic Health through Community Empowerment: Outcomes of the *Senyum Sihat* MADANI Initiative

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Background: Integrated community screening may identify undetected periodontal and metabolic health risks. This study assessed screening findings and programme acceptability among adults attending the *Senyum Sihat* MADANI initiative. **Methods:** Cross-sectional dental and medical screenings were conducted at seven community sites in Negeri Sembilan from July to November 2025. A total of 439 adults received periodontal screening using the Basic Periodontal Examination (BPE). At sites providing comprehensive metabolic screening, random blood sugar, blood pressure and body mass index were recorded. Oral health education was provided, and participant feedback was collected. Descriptive statistics summarised the disease burden, while Spearman correlation examined the relationship between periodontitis and diabetes mellitus using available paired data. **Results:** Periodontal disease was detected in 52.5% of participants; 17.2% recorded BPE Codes 3 or 4, indicating a need for comprehensive periodontal assessment. Among participants receiving metabolic screening, 11.4% had random blood sugar above 11.1 mmol/L, 63.6% had hypertensive-range readings and 20.5% met the obesity threshold of BMI at least 27.5 kg/m². No statistically significant correlation was observed between periodontitis and diabetes mellitus (Spearman's rho = -0.055, p = 0.810). Programme acceptability was high, with 72.9% recommending biannual implementation. **Conclusion:** The initiative identified substantial periodontal and metabolic health risks among participating adults, supporting the value of integrated community screening. University–community partnerships may facilitate early identification, appropriate referral and targeted health education. Further evaluation is needed to establish longer-term effects on health outcomes.

Keywords: Periodontal Disease, Metabolic Syndrome, Community Outreach, Oral Health Promotion.

Presenter ID: EP30

Evaluating Interceptive Orthodontic Referrals by Dental Therapists: Preliminary Findings from a Pilot Competency Assessment

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Background: In Malaysia, dental therapists (DTs) provide early screening and referral of school children to dental officers (DOs) for orthodontic assessment. Therefore, DT's competency is vital for timely malocclusion identification and management. This pilot study evaluated the competency of Labuan DTs in early orthodontic screening and interceptive referral identification. **Methods:** This study was carried out using a study questionnaire that underwent a three-step validation process. Participants assessed thirteen clinical cases ranging from normal occlusion to severe malocclusion using study models and extraoral photographs with brief descriptions. Thirteen DTs were identified, and 12 responded (response rate: 92.3%). Responses were scored against a gold standard established by an orthodontist, and passing scores were determined objectively using the simplified Yes-No Angoff method. **Results:** Passing rates varied across domains: incisor relationship (25%), skeletal pattern (50%), and referral need determination (100%), with 25% meeting the overall competency threshold. Years of service correlated weakly with overall performance ($r=-0.329$) and referral needs ($r=0.191$). Time since last education session correlated moderately with overall performance ($r=-0.489$) and very weakly with referral needs ($r=-0.094$). No associations were statistically significant ($p>0.05$). **Conclusion:** This study showed that DTs in Labuan possess adequate skills for preliminary orthodontic screening, yet their competency regarding incisor and skeletal classification requires improvement. These findings highlight areas for targeted continuing professional development to strengthen early screening skills, improve recognition of malocclusion, and support appropriate referral to the DO for further assessment.

Keywords: Dental therapists, Interceptive orthodontics, Orthodontic referral.

Presenter ID: EP31

The Integrated Care Pathway for Oral Healthcare Service Provision for Individuals with Special Healthcare Needs (ICP-OHSN) Attending Primary Dental Clinic at Universiti Sains Malaysia Specialist Hospital - Development and Evaluation.

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Background: Individuals with special healthcare needs (ISHCN) face greater unmet dental needs due to limited access and fragmented care, perpetuating health inequities. An Integrated Care Pathway (ICP) bridges primary and specialist care, standardising evidence-based multidisciplinary management. As care pathways remain underutilised in Malaysia, this study developed an oral healthcare ICP for ISHCN (ICP-OHSN) at Universiti Sains Malaysia Specialist Hospital (HPUSM). **Methods:** Development began with an exploratory phase - reviewing guidelines, practices, and staff consultations. Three multidisciplinary teams designed the pathway via a six-step framework (team formation, information gathering, drafting, expert input, finalisation, pilot implementation), integrating NICE, the Seattle Care Pathway, and Malaysian guidelines. During a cross-sectional evaluation, primary care dental officers applied the ICP-OHSN to 36 ISHCN patients. Feasibility was evaluated via clinical variance tracking, and clinician satisfaction was measured using a validated 21-item scale. **Results:** A structured, goal-based ICP-OHSN was developed, reflecting multidisciplinary consensus tailored to the local context. Spanning patient entry to discharge, it incorporates a triage assessment, 13 core activities, and 37 standardised procedures. Variance analysis demonstrated high adherence; justifiable deviations occurred primarily in prescription issuance (69.4%) and emergency treatment (63.9%). Clinicians reported a high mean satisfaction score of 83.31 (SD=12.66) out of 105, with highest ratings in process identity and effect perception confirming seamless integration. **Conclusion:** The ICP-OHSN provides a robust, evidence-informed framework standardising ISHCN oral healthcare. Implementing this pathway bridges service gaps, enhances care coordination, and ultimately improves oral health outcomes. Yielding high clinical satisfaction, it offers a scalable blueprint for equitable healthcare systems.

Presenter ID: EP32

Early Functional Intervention and Objective Assessment of Orofacial Muscle Function in an 8-Year-Old Child: A Case Report

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Background: Oral functional abnormalities in childhood may affect oral health, craniofacial development, and future orthodontic needs. Early identification and functional intervention may therefore contribute to preventive oral healthcare, although objective assessment of functional change remains limited. **Case Description:** An 8-year-old child with oral functional abnormalities and an upper airway-related condition underwent adenotonsillectomy, followed by early clear aligner treatment combined with lip muscle training, tongue muscle training, breathing exercises, and a home-based functional program. Orofacial muscle function was objectively assessed using real-time force–time recording. Maximum lip endurance time increased from 10 to 13 s, while tongue-tip maximum force increased from 5.2 to 6.0 N. Changes were also observed in craniofacial, dental, and airway parameters. Upper lip length increased from 32.5 ± 1.2 to 34.1 ± 1.3 mm. Dental spacing decreased from 28.3 ± 0.9 to 27.1 ± 1.0 mm, indicating space closure during treatment. Airway cross-sectional area increased from 198.6 ± 14.7 to 215.42 ± 15.2 mm². **Clinical Lesson:** Objective measurement of lip endurance and tongue strength may complement conventional clinical observation when monitoring early functional intervention. Combining quantitative orofacial muscle assessment with upper lip length, dental space closure, and airway measurements may provide a multidimensional framework for preventive and interceptive pediatric oral care.

Presenter ID: EP34

Capacity Building for Continuous Improvement in Delivering Kotak Initiative - Kedah Experience

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Background: This review outlines the nature, clinical scope, and staff attendance of workforce capacity-building initiatives for the KOTAK tobacco cessation program within the Kedah State Oral Health Division from 2023 to 2025. **Methods:** Programmatic documentation and training activity records on tobacco cessation were systematically reviewed. Aggregated secondary data were comparatively analyzed to evaluate training distribution across diverse personnel categories, specialized instruction modules, and critical healthcare delivery metrics such as screening rates in primary care settings. **Results:** Statewide participation was high, with 480 dental officers and therapists completing the mandatory KOTAK Module for baseline proficiency. Advanced professional development (2023–2025) certified 101 personnel in CSCSP, 57 in Motivational Interviewing (MI), and 60 via the Dental Public Health Specialist Unit (DPHSU) Referral Protocol Workshop to standardize high-risk patient care pathways, meeting the universal coverage benchmark. Thirteen comprehensive graduates managed 351 cumulative intervention cases. **Conclusion:** Strategic capacity building successfully reinforced the systematic clinical delivery of KOTAK, ensuring reliable screening and targeted support. Specialized MI workshops significantly upgraded clinician confidence in navigating complex case management, correlating directly with projected gains in cessation outcomes. Continuous, data-backed oversight and ongoing professional training remain crucial to preserve clinical excellence, effectively empowering dental personnel as frontline leaders for comprehensive patient behavioral modification KOTAK, tobacco cessation, capacity building, CSCSP, smoking intervention in primary care.

Keywords: Parental Knowledge, Fluoridated Toothpaste, Validation, Questionnaire, Pre-school Children

Presenter ID: EP35

AI-Powered Automatic Tooth Segmentation from CBCT for The Fabrication of Dental Splints

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Background: Dental splints are widely used in the prevention and treatment of bruxism and temporomandibular disorders, orthodontics, and prosthodontic treatments. Current methods of splint production rely on impression models or intra-oral scans. This cross-sectional study investigates the feasibility of AI-powered tooth segmentation from CBCT for surgical splint design. **Methods:** Plaster models were digitized to serve as the reference group, with an offset value of 0.2mm (3D scan-0.2mm). AI-segmented dental arches were used to design splints with varying offset values (CBCT-0.25mm, CBCT-0.5mm, CBCT-0.75mm). 3D-printed splints underwent preclinical assessments of obstruction, stability, and interfacial space (N = 15). **Results:** In the obstruction test, 3D scan-0.2mm splints (median: 3; interquartile range (IQR): 2-3) scored higher than CBCT-0.25mm (median: 2; IQR: 1-3; p=0.048), but comparable to CBCT-0.5mm (median: 3; IQR: 3-3; p=0.20) and CBCT-0.75mm (median: 3; IQR: 3-3; p=0.32). In the stability test, there was no significant difference between 3D scan-0.2mm (median: 3; IQR: 2-4), CBCT-0.25mm (median: 2; IQR: 1-3), CBCT-0.5mm (median: 3; IQR: 3-3), and CBCT-0.75mm (median: 3; IQR: 2-3). 2D and 3D interfacial space analyses revealed that 3D scan-0.2mm splints had significantly smaller gaps (p<0.001) than splints from AI-segmentation. **Conclusion:** It is feasible to design splints using AI-powered tooth segmentation at offset values of 0.5mm and 0.75mm with acceptable performance in prevention and treatment in oral and maxillofacial surgery, orthodontics, and prosthodontics.

Keywords: Cone-Beam Computed Tomography, Artificial Intelligence, Dental Splints, Spiral CT

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A Mixed-Methods Exploration of Dental Anxiety Risk Factors and Coping Strategies to Address Dental Anxiety

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Background: Dental anxiety remains a common psychological problem affecting all age groups and continues to pose significant challenges to dental care delivery worldwide. A high prevalence of dental anxiety has been reported, but there is a lack of studies that integrate an in-depth qualitative exploration of dental anxiety with a quantitative evaluation of educational interventions. This study aimed to explore the risk factors and coping strategies related to dental anxiety among adolescents and to evaluate the effectiveness of a pop-up book-based intervention in reducing dental anxiety. **Methods:** A mixed-methods study with a sequential exploratory design was applied in this study. The qualitative phase used in-depth interviews with 30 adolescents in Kediri, Indonesia. These interviews explored experiences, perceptions, and psychosocial factors associated with dental anxiety using Precede-Proceed theory. The quantitative phase employed a pre-experimental pretest-posttest design involving 127 adolescents who received a pop-up book-based intervention. Dental anxiety levels were measured before and after the interventions using a standardized anxiety scale. Statistical analysis was performed using Wilcoxon signed-rank tests and Mann-Whitney tests, with p<0.05. Meanwhile, the qualitative data were coded by NVivo. **Results:** Qualitative analysis revealed that several major factors influenced dental anxiety. These included previous traumatic dental experiences, anticipation of pain, lack of knowledge of dental procedures, environmental conditions of the dental clinic, and family influence. Quantitative findings showed that the pop-up book-based intervention significantly reduced dental anxiety levels after the intervention (p<0.05). **Conclusion:** Dental anxiety among adolescents is determined by multifactorial psychosocial influences. Pop-up book-based education is effective in reducing dental anxiety.

Keywords: Adolescents, Dental Anxiety, Mixed-methods, Pop-up Book, Risk Factor